Magic-Induced Computational Separation in Entanglement Theory

Andi Gu,[1] Salvatore F.E. Oliviero,[2] and Lorenzo Leone[3,*]

[1] *Department of Physics, Harvard University, 17 Oxford Street, Cambridge, Massachusetts 02138, USA*
[2] *NEST, Scuola Normale Superiore and Istituto Nanoscienze, Consiglio Nazionale delle Ricerche, Piazza dei Cavalieri 7, IT-56126 Pisa, Italy*
[3] *Dahlem Center for Complex Quantum Systems, Freie Universität Berlin, 14195 Berlin, Germany*

(Received 19 April 2024; revised 24 November 2024; accepted 5 March 2025; published 5 May 2025)

Entanglement serves as a foundational pillar in quantum information theory, delineating the boundary between what is classical and what is quantum. The common assumption is that a higher degree of entanglement corresponds to a greater degree of "quantumness." However, this folk belief is challenged by the fact that classically simulable operations, such as Clifford circuits, can create highly entangled states. The simulability of these states raises the question: What are the differences between "low-magic" entanglement and "high-magic" entanglement? To understand this interplay between entanglement and magic, we take an operational approach by studying tasks such as entanglement estimation, distillation, and dilution. We uncover a separation of Hilbert space into two distinct regimes: the entanglement-dominated (ED) phase, where entanglement surpasses magic, and the magic-dominated (MD) phase, where magic dominates entanglement. This separation induces a computational phase transition: entanglement-related tasks are efficiently solvable in the ED phase, but become intractable in the MD phase. Our results find applications in diverse areas such as quantum error correction, many-body physics, and the study of quantum chaos, providing a unifying framework for understanding the behavior of quantum systems. We also offer theoretical explanations for previous numerical observations, highlighting the broad implications of the ED-MD distinction across various subfields of physics.

DOI: 10.1103/PRXQuantum.6.020324

I. INTRODUCTION

Entanglement has long stood as a central concept in quantum information, serving as a key differentiating feature between classical and quantum theories. This perspective was both enriched and challenged by the introduction of the stabilizer formalism, which identified a subset of quantum states, known as stabilizer states (and their associated circuits, termed Clifford circuits), which could be simulated efficiently using classical resources [1]. These classically simulable states could be highly entangled, and hence did not conform to the conventional understanding that entanglement could be the sole metric determining the degree to which a state is truly quantum. Building on this understanding, a new metric, referred to as "magic," or nonstabilizerness, was introduced to quantify the amount of non-Clifford resources necessary to prepare a given quantum state, providing a more nuanced measure of a state's quantumness. In fact, magic has itself been shown to be intimately connected to a number of fundamentally quantum phenomena [2–8].

However, there is little work on the interaction of these two measures of quantumness. A previous work showed that entanglement could be calculated exactly for stabilizer states [9], and more recent works have shown that the degree of magic in a state is linked to the "entanglement response" of the state [10]. In this work, we further explore the intersection of these two notions and find them deeply connected. This exploration leads to implications that extend far beyond the realm of quantum information theory. Our framework provides valuable insights into a wide array of physical phenomena, from the robustness of topological entanglement entropy in quantum error-correcting codes to the behavior of entanglement in many-body systems. Moreover, our efficient entanglement estimation protocols find applications in the study of scrambling dynamics and the characterization of quantum chaos. By bridging the gap between these seemingly disparate fields, our work highlights the fundamental role of the entanglement-magic interplay in understanding the complex behavior of quantum systems.

*Contact author: lorenzo.leone001@umb.edu

Published by the American Physical Society under the terms of the Creative Commons Attribution 4.0 International license. Further distribution of this work must maintain attribution to the author(s) and the published article's title, journal citation, and DOI.

There are many ways to understand entanglement. The approach we take in this work is *operational*: we argue that studying the role of magic in entanglement estimation and manipulation tasks leads to a clear understanding of their relationship as a whole. However, a simple operational approach does not yet reveal the full picture. A sequence of recent works [11,12] has found that operational characterizations of entanglement can change significantly under the constraint of computational limitations. Taking inspiration from this, we enhance our operational approach by studying entanglement tasks under a *computational lens*. This approach allows us to uncover a marked divide between quantum states, defined by the relationship between magic and entanglement. Specifically, we find that the Hilbert space can be divided into two distinct phases: the entanglement-dominated (ED) phase and the magic-dominated (MD) phase. These phases correspond roughly to states whose entanglement significantly surpasses magic and states in which magic dominates the entanglement. The boundary between the two phases is delineated by a computational separation induced by the competition between the two resources. That is, for a number of entanglement detection and manipulation problems, there are sample- and time-efficient quantum algorithms that solve these problems for ED states. Conversely, we show that these entanglement manipulation tasks are provably computationally intractable for the MD phase.

Our first step in understanding the ED-MD transition is in the setting of entanglement estimability. We find that, in the ED phase, we can efficiently estimate the entanglement entropy with an asymptotically (in n) vanishing error, even for volume-law states. Conversely, we show that in the MD phase, entanglement estimation is inefficient beyond logarithmic entanglement (below this, the swap test allows for efficient estimation). Finding clear evidence of an ED-MD computational separation for entanglement estimation, we then study entanglement distillation. We show that within the class of ED states, we can always efficiently find a polynomial-depth circuit that distills almost all of the entanglement into Bell pairs. We also prove the converse, which says that in the MD phase, it is impossible to find such efficient and optimal distillation protocols. Similarly, in the context of entanglement dilution, we demonstrate that we can always identify an efficient dilution protocol that utilizes an optimal number of Bell pairs to prepare ED states. In contrast, for MD states, we rule out the possibility of an efficient dilution protocol that consumes anything close to an optimal number of Bell pairs. Taken together, these findings provide substantial evidence supporting computational separation between ED and MD states. This leads us to conclude that there is a stark difference in the entanglement structure between these two phases of states (as illustrated by the ED-MD phase diagram in Fig. 1). Within the ED phase, entanglement is always structured in such a way that it can be manipulated efficiently and (almost) reversibly. In contrast, for general states in the MD phase, there is limited to no structure in the entanglement, making the entanglement manipulation inefficient and irreversible.

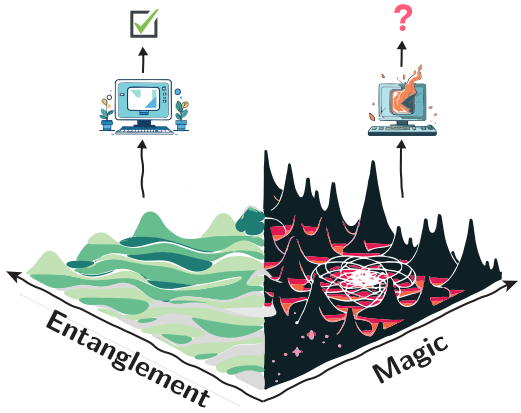


FIG. 1. The sharp distinction between entanglement-dominated and magic-dominated states. The landscape visualizes the entanglement structure of states: in the entanglement-dominated phase, entanglement is highly structured and easily manipulable, while in the magic-dominated phase, entanglement can be scrambled in such a complex way that it is completely intractable to measure or manipulate.

Given the simple entanglement estimation and manipulation protocols for ED states, one might wonder whether entanglement-dominated states are merely an esoteric theoretical construction. Practically speaking, where can we expect to find ED states? Although the majority of states within the Hilbert space are magic dominated, we show that ED states appear naturally in a wide range of physical settings, from quantum error-correcting codes to many-body systems. Moreover, it is well known that most states in Hilbert space also cannot be prepared in polynomial time, and hence are irrelevant in practice [13–15]. In contrast, there is a large class of well-known unitary operations, which almost always produce ED states, namely circuits dominated by Clifford gates. In particular, Clifford-dominated circuits with $t = o(n)$ non-Clifford gates [importantly, this far exceeds the classical simulability limit $O(\log n)$] produce ED states with overwhelming probability. This means that, in contrast with t-doped state simulation algorithms, which can only operate efficiently under the assumption that t scales logarithmically in n [16,17], our efficiency results regarding ED states hold far beyond this classical simulation limit. However, we emphasize that Clifford-dominated circuits only generate a tiny fraction of all ED states; in fact, most ED states are constructed from far more than just $O(n)$ T gates.

As applications of our theory, we demonstrate the robustness of entanglement witnesses and topological entanglement entropy for ED states, highlighting their potential for practical quantum error correction and the

study of many-body physics. We also develop an efficient testing algorithm that can classify states within the ED and MD phases. To conclude, we highlight the relevance of our findings in many-body physics by demonstrating the robustness of topological entanglement entropy in three-dimensional (3D) topological models such as the X-cube model and Haah's code. We also find connections between stabilizer code Hamiltonians and ED states, which extend the applicability of our results to quantum error-correcting codes. Lastly, we provide rigorous grounds for previous numerical observations in the literature of Clifford-dominated circuits, such as entanglement-cooling and phase transition in hybrid quantum circuits.

A. Overview

This section provides a nontechnical summary of our key findings. As highlighted in the introduction, this paper explores the interplay between two fundamental resources that separate quantum computers from classical computers: entanglement and magic. A starting point for our work is the observation that tasks related to entanglement, such as entanglement measurement, witnessing, or manipulation (e.g., EPR-pair extraction, dilution), are sample and time efficient for stabilizer states. We investigate how introducing *magic*, through non-Clifford gates in state preparation, affects the entanglement manipulability (hence entanglement structure) of stabilizer states.

We find two classes of states where entanglement-related tasks are either efficiently solvable or provably hard. This stark operational difference divides the Hilbert space into the *entanglement-dominated* phase and the *magic-dominated* phase. As previously mentioned, these phases roughly correspond to scenarios where entanglement significantly surpasses magic, leading to stabilizer properties dominating, and cases where magic dominates entanglement. For a more precise definition of the two phases, we use resource-theoretic monotones—a natural choice, given the operational approach we take. Throughout the paper, we consider n-qubit states, which are bipartite $A|B$ with n_A (respectively, n_B) the number of qubits in A (respectively, B) with $n_A + n_B = n$. Given the Pauli group $\mathbb{P}_n$, a state is stabilized by $P \in \mathbb{P}_n$ if and only if $P|\psi\rangle = |\psi\rangle$. We use the *stabilizer nullity* [4] as a measure of magic—related to the number of Pauli stabilizers—which has a clear operational meaning in terms of magic-state distillation. On the other hand, we quantify bipartite entanglement in $A|B$ with the *entanglement entropy*, which has a clear operational meaning for entanglement distillation and dilution [18].

Definition 1 (Entanglement and magic-dominated phases). Let $|\psi\rangle$ be a state with $2^{n-\nu}$ Pauli stabilizers; we say that this state has stabilizer nullity ν [4]. Let $A|B$ a bipartition with n_A and n_B qubits, and let $S_1(\psi_A)$ be the von Neumann entropy of $\psi_A := \mathrm{tr}_B |\psi\rangle\langle\psi|$. We say $|\psi\rangle$ is entanglement dominated if $S_1(\psi_A) = \omega(\nu)$, and it is magic dominated if $S_1(\psi_A) = O(\nu)$. We note that this definition depends on the bipartition $A|B$. This choice of bipartition will typically be implicit; whenever it is not clear from the context, we will specify which bipartition we are defining as entanglement dominated or magic dominated with respect to. Furthermore, we generally assume $n_A, n_B = \omega(1)$, otherwise all states would be trivially magic dominated.

We will study the measurability and manipulability of entanglement for these two classes of states. For entanglement manipulation, there are two tasks: entanglement distillation and dilution. The goal of entanglement distillation is to use LOCC (local operations and classical communication) to transform a state ψ into as many Bell pairs M_+ as possible. The number M_+ is the *distillable entanglement* of ψ. For the reverse of entanglement distillation, namely entanglement dilution, the number of Bell pairs M_- required to prepare ψ via LOCC is known as its *entanglement cost*. Having outlined the relevant entanglement tasks, we use the table below to summarize the key findings of this work. Throughout the paper, we often make use of Bachmann-Landau notation for asymptotics [19].

Theorem 1 (Efficient entanglement measurement and manipulation for ED states [Theorems 4, 6, and 8 and Lemmas 13 to 15]). Let $|\psi\rangle$ be an unknown entanglement-dominated state across the bipartition $A|B$ with $n_A, n_B = \omega(\log n)$. There exist sample- and time-efficient quantum algorithms for each of the following tasks.

Protocol	Entanglement dominated	Magic dominated
An efficient state-agnostic protocol, which produces an estimate $\tilde{S}(\psi_A)$ of the true entanglement $S_1(\psi_A)$.	$\frac{\lvert S_1(\psi_A) - \tilde{S}_1(\psi_A)\rvert}{S_1(\psi_A)} = o(1)$	$\frac{\lvert S_1(\psi_A) - \tilde{S}_1(\psi_A)\rvert}{S_1(\psi_A)} = \omega(1)$
An efficient state-agnostic protocol, which distills M_+ Bell pairs from the state ψ.	$M_+/S_1(\psi_A) = 1 - o(1)$	$M_+/S_1(\psi_A) = o(1)$
An efficient state-agnostic LOCC protocol, which uses M_- Bell pairs to prepare a state ψ across $A\|B$, where either A or B has local access to polynomial copies of ψ.	$M_-/S_1(\psi_A) = 1 + o(1)$	$M_-/S_1(\psi_A) = \omega(1)$

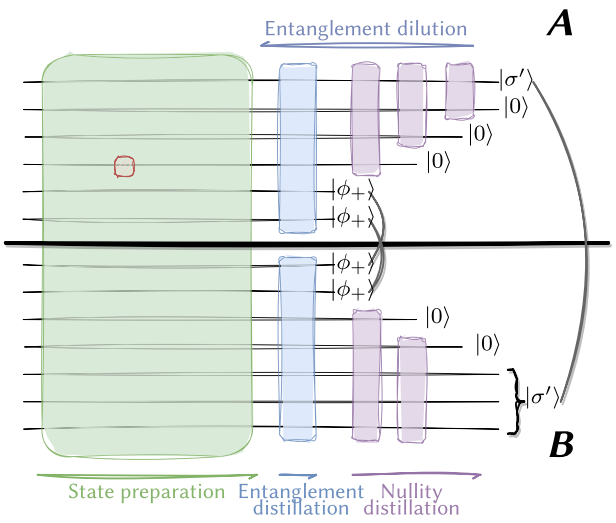


FIG. 2. Schematic of our entanglement distillation protocol (left to right) and dilution protocol (right to left).

(1) *Entanglement estimation:* Estimates $S_1(\psi_A)$ up to an $o(1)$ relative error.
(2) *Entanglement distillation:* Distills M_+ Bell pairs from ψ using LOCC operations, with $M_+/S_1(\psi_A) = 1 - o(1)$. Unlike more conventional protocols [20], this protocol requires only a single copy of the input state and makes no error: it is a *one-shot, zero-error protocol*.
(3) *Entanglement dilution:* Prepares ψ across the bipartition $A|B$ using LOCC, M_- Bell pairs, where $M_-/S_1(\psi_A) = 1 + o(1)$ and a number of bits of classical communication $N_{\mathrm{CC}} = o(S_1(\psi_A))$.

The previous theorem demonstrates that entanglement-related tasks, such as entanglement estimation, distillation, and dilution, can be performed efficiently for states in the entanglement-dominated (ED) phase. These positive results highlight the tractability of manipulating and characterizing entanglement in ED states, even for those with high entanglement entropy (i.e., volume-law states). The efficiency of these tasks in the ED phase can be attributed to the structure of entanglement in these states, which allows for effective compression and manipulation using stabilizer operations. In contrast, the following theorem establishes the hardness of these same entanglement-related tasks for states in the magic-dominated (MD) phase. It shows that, in general, estimating the entanglement entropy or performing entanglement distillation and dilution for MD states cannot be done efficiently, even when allowing for a constant relative error. This computational intractability arises from the complex and unstructured nature of entanglement for general MD states.

Theorem 2 (Hardness of entanglement estimation and manipulation for MD states [Theorem 10]). Take any bipartition $A|B$ that satisfies $n_A, n_B = \omega(\log n)$ and let $|\psi\rangle$ be a pure state with $S_1(\psi_A) = \omega(\log n)$. There is no efficient protocol that can estimate $S_1(\psi_A)$ to within a constant relative error for arbitrary MD states. Furthermore, the distillable entanglement of general MD states is $M_+/S_1(\psi_A) = o(1)$ while the entanglement cost is $M_-/S_1(\psi_A) = \omega(1)$.

We qualify this claim for readers not familiar with hardness proofs. In (almost) every hardness theorem, there are two crucial points to be aware of. The first part of any hardness theorem is the class of problems for which the problem is claimed to be hard. In Theorem 10, the only information we are given is that the state is magic dominated, which means we are forced to consider very general algorithms that work for the broad class of MD states. With more information at hand, in principle, alternative strategies could be developed, tailored to specific classes of MD states. The theorem merely states that there is no efficient strategy for entanglement-related tasks that works in full generality on *all* MD states (in contrast to our results for ED states). This brings us to the second point. Hardness proofs refer to *worst-case hardness*, meaning that among MD states, we guarantee that there is at least one (sub)class of states for which no efficient strategy can perform well. Indeed, the proof of this is constructive: we concretely identify subclasses of states for which this is true (see Sec. IV). In principle, there are probably many more classes spread around the larger set of MD states for which there are no such efficient strategies. This is best understood with a canonical example: it is well-known that finding the ground state of a general classical system is NP-hard. However, does this mean that finding the ground state of *every* classical Hamiltonian is hard? Of course not. However, there are important classical physical systems, such as spin glasses, for which finding the ground state is provably hard. This constitutes a class of systems for which the hardness proof holds true. Given the two axes representing magic and entanglement, respectively, we show that the entanglement-related tasks analyzed above are provably efficient and agnostic for *every* ED state. On the other hand, for a wide range of magic, we construct a family of MD states for which entanglement-related tasks are hard. Given that, what is the true meaning of the computational separation found in this work? It means that, in the ED phase, we are guaranteed the existence of efficient and agnostic protocols, while in the MD phase, efficient and agnostic protocols do not exist; hence, tailored protocols for specific classes should be adopted. However, at the same time, there exist classes of states within the MD phase for which no efficient protocols exist at all.

1. Irreversibility of efficient entanglement manipulation

The computational separation between entanglement- and magic-dominated phases can also be understood through the lens of entanglement irreversibility. In the absence of restrictions on the efficiency of entanglement manipulation protocols, it is well known [15] that bipartite pure-state entanglement theory is reversible: given a bipartite state $|\psi\rangle$ with entanglement entropy $S_1(\psi_A)$, one can extract $M_+ = S_1(\psi_A)$ Bell pairs via LOCC; conversely, to dilute a bipartite state $|\psi\rangle$ across the $A|B$ partition, only $M_- = S_1(\psi_A)$ Bell pairs are required.

When LOCC protocols are constrained to be efficient (i.e., one has access to only polynomially many input copies of $|\psi\rangle$), entanglement theory loses its reversibility. As shown in Ref. [11], the maximum extractable number of Bell pairs from an *uknown* state is (quasi)exponentially reduced to $M_+ = O\left(\log^2 S_1(\psi_A)\right)$. Conversely, as we demonstrate in Corollary 5, any efficient LOCC protocol that dilutes a *unknown* bipartite state across $A|B$ must use a (quasi)exponentially increased number of Bell pairs $M_- = \Omega\left(\exp(S_1(\psi_A)/2)\right)$.

From these two results, it follows that the reversibility ratio of bipartite pure-state agnostic entanglement manipulation,

$$\frac{M_+}{M_-} = O\left(e^{-\frac{S_1(\psi_A)}{2}}\right), \tag{1}$$

asymptotically vanishes in terms of the entanglement entropy $S_1(\psi_A)$. Although this result holds for general bipartite states, one might wonder whether reversibility can be restored for specific classes of states. The magic-induced computational separation proven in this paper answers this question. As a consequence of Theorems 1 and 2, Corollary 6 demonstrates that reversibility of entanglement manipulation differentiates the two phases:

$$\frac{M_+}{M_-} = \begin{cases} 1 - o(1), & \text{for ED states,} \\ o(1), & \text{for MD states,} \end{cases} \tag{2}$$

In the entanglement-dominated phase, entanglement manipulation remains reversible, while the magic-dominated phase is precisely where the general irreversibility of bipartite entanglement manipulation emerges. Specifically, for states with maximal magic $\nu = \Omega(n)$, the worst-case reversibility ratio M_+/M_- is given by Eq. (1) for an in-depth discussion of possibilities and limitations of entanglement manipulation with limited computational resources, see Ref. [21].

2. Multipartite entanglement distillation

Besides bipartite entanglement distillation, there is also the challenge of multipartite entanglement distillation. In this scenario, if k parties share an entangled state, their goal is to distill some target k-partite entangled state (e.g., a GHZ state) using LOCC operations. Interestingly, it has been established that this task is unachievable for the vast majority of states [22,23]. However, identifying a generalization of the ED phase in the k-partite setting, in Theorem 7 we show that we can *deterministically* distill many copies of a k-partite GHZ state from ED states using an efficient LOCC protocol.

3. Entanglement witnessing and robustness

While precisely measuring entanglement can be a challenging and noise-sensitive task, the less ambitious goal of merely witnessing entanglement can be easier and more noise resilient [24,25]. We will address this task in Sec. III B. The purpose of a witness is to experimentally validate the presence of genuine entanglement in an imperfectly prepared version of the target state. We define an entanglement witness for ED states that can be measured with $O(1)$ sample complexity. With Theorem 9, we strengthen this result by defining a similar witness to multipartite entanglement. This witness verifies entanglement across k parties—that is, it rules out the possibility of the state being unentangled across any of the k given partitions. As a corollary of this, we find that entanglement for ED states can be far more robust than the entanglement of generic states; see Corollary 2. The reason for this is as follows. The Fannes inequality roughly says that $|S_1(\rho_A) - S_1(\psi_A)| \lesssim n_A T$, where T is the trace distance between ψ and its noisy version ρ. If ψ were a generic state with $S_1(\psi_A) \sim \sqrt{n_A}$, then we would generally need $T < 1/\sqrt{n_A}$ to guarantee that ρ were not separable across $A|B$. On the other hand, if ψ was in the ED phase, we show that we could tolerate up to $T \lesssim 1 - 2^{-\Omega(\sqrt{n_A})}$, showing that the entanglement within the ED phase is robust.

4. Phase classification and testing

In light of the clear distinction between ED-MD phases, one might ask whether it is possible, given query access to an unknown state $|\psi\rangle$, to determine the phase in which it resides. We formalize this task as a property testing problem and show that the separation between ED-MD phases can indeed be efficiently tested. More precisely, in Theorem 13, we present a polynomial-time algorithm that can discriminate whether $|\psi\rangle$ is an ED state or it is ϵ-far from any state in the entanglement-dominated phase and, as such, lies in the MD phase.

5. Applications to physics

We conclude by discussing the implications of our results in the context of many-body physics. First, leveraging the result of Ref. [26], we stem a potential application of the techniques developed in our work. In Sec. VI A, we show how ED states appear in the context of many-body

physics. We demonstrate the robustness of ground-state entanglement for such models and study the robustness of topological entanglement entropy to perturbations in models such as the X-cube model or Haah's code. Taking advantage of the fact that the topological entanglement entropy in these models scales extensively, we show that this topological entanglement persists under any perturbation of the Hamiltonian that has a subextensive [i.e., $o(n)$] number of terms, *regardless the strength of the perturbation*. While topological entanglement entropy is traditionally understood to be robust against weak and local perturbations [27], we find that for models with extensive topological entropy, this robustness extends far beyond such cases. To determine whether this characteristic is solely due to the extensivity of the topological entropy, we numerically study the robustness of topological entanglement in models where the topological entropy is an order-one constant, e.g., the toric code [28]. Through extensive numerical analysis (up to approximately 100 qubits), we find that for such models, the topological entropy is not robust against the same type of perturbations and decays exponentially with the strength of the perturbative term. We thus conclude that this kind of robustness, highlighted by our techniques, is a unique feature of models (such as 3D topological models) with extensive topological entanglement. These findings shed new light on 3D topological order, which remains mainly elusive to this day [29]. We then extend these studies to the settings of quantum quench, finding similar conclusions. Furthermore, in Sec. VI D, we present a method for quantifying the Lyapunov exponent—the rate at which correlations propagate in unitary dynamics—within random quantum circuits. In particular, our algorithm is efficient for Clifford + T circuits with up to a sublinear number of T gates. Notably, the sublinear number of non-Clifford gates does not prevent these circuits from exhibiting randomness properties, which are rigorously quantified by k-designs (ensembles of circuits that mimic completely random unitaries up to the kth moment of the distribution). Indeed, in the seminal paper [30], it was shown that even a constant number of non-Clifford gates sparsely embedded in large Clifford circuits can provide approximate k-designs up to $k \sim \sqrt{n}$. Specifically, for a number of non-Clifford gates growing strictly superlogarithmically, such circuits become indistinguishable from Haar random unitaries. Our algorithm leverages the efficiency of estimating entanglement in the ED phase. Although the Lyapunov exponent is prominently studied in the context of many-body quantum chaos, it has also been extensively investigated in random quantum circuits [31], which are traditionally considered to provide a good approximation to chaotic dynamics [32]. The ability to efficiently measure the Lyapunov exponent in Clifford-dominated circuits enhances the study of the emergence of chaotic dynamics in such circuits.

6. Understanding phenomenology with ED-MD phases

Leveraging the theoretical tools developed in this work, in Sec. VII, we analytically and rigorously elucidate certain numerical observations in the literature. Specifically, we offer a straightforward proof demonstrating that entanglement in states constructed from Clifford gates, supplemented by, at most, a linear number of non-Clifford gates, can be efficiently and agnostically cooled down. This result was previously observed numerically in Ref. [33]. Moreover, we delve into the study of the magic-entanglement transition in hybrid quantum circuits, complementing the analysis from Ref. [34]. In this context, we rigorously establish the absence of ED states in these circuits, confirming a numerical observation of Ref. [34].

7. Pseudocode

Finally, we provide pseudocode for many of the protocols we have just described. First, we show how to *exactly* compute the 2-Rényi entanglement entropy of a state with stabilizer nullity ν using a classical algorithm whose runtime is $O(4^{\nu}n^3)$. We then show how to efficiently monitor the stabilizer entanglement produced by a t-doped Clifford circuit using an efficient classical algorithm, and finally show how to efficiently witness entanglement for ED states. These procedures can be of independent interest, so we dedicate a separate section to these pseudocodes in Sec. VIII.

B. Related works

This work aims to contribute to the growing body of research within quantum resource theory that investigates how quantum resources interact with each other. Although our work presents a detailed exploration of the relationship between magic and entanglement, similar ideas have been previously explored to some extent, providing a foundation for our investigation. This section discusses and contextualizes our results in light of these earlier contributions.

Recently, there has been a growing interest in understanding how the efficiency of quantum algorithms impacts the feasibility of quantum resource protocols, particularly in the context of entanglement [11] and magic [35]. To understand these developments, we refer to a seminal work [20], which showed that the number of Bell pairs that can be distilled per copy of bipartite state ψ_{AB} is asymptotically equal to the von Neumann entanglement $S_1(\psi_A)$ of the state. Furthermore, it was shown that this rate is optimal. Despite the apparent finality of this result, there are several drawbacks associated with the protocol used to achieve this. First, since distillation techniques make use of the idea of typical sets, it requires *many* copies of the input state—for even simple cases, up to 10^6 copies may be required before the desired concentration begins to kick in Ref. [36]. Second, it relies on knowledge of the input state (i.e., it is not input state agnostic). Finally,

there are no guarantees on the computational efficiency of the protocol. In fact, the recently introduced concept of quantum pseudoentanglement demonstrates rigorously that this protocol cannot be efficient in general [11,35], a problem elaborated by the framework of *computational entanglement theory* [12]. Our work presents an entanglement distillation protocol that circumvents all three of these problems for arbitrary entanglement-dominated states: it works just as well on one copy of the state as on many copies, it is agnostic to the input state, and is computationally efficient. The performance trade-off for these gains is minimal: we are able to distill almost all the entanglement out of entanglement-dominated states. Furthermore, we show in a converse result that this is the best one can hope for. No efficient, input state-agnostic and optimal protocol can exist for states that are magic dominated. Additionally, we generalize these findings to entanglement dilution, proving a magic-induced computational separation in entanglement cost. We offer the first no-go results for entanglement dilution using pseudorandom states. While a computational separation in entanglement cost has been found in Ref. [12], this work only ruled out a one-way LOCC dilution protocol. In contrast, our results hold for general LOCC channels with an arbitrary number of communication rounds.

Beyond the context of quantum resource theories, our results also have strong connections with the quantum information literature on t-doped stabilizer states. This class of states is of broad interest because they are widely believed to be the simplest class of states that cannot be classically simulated [for $t = \omega(\log n)$], but can also be prepared on early fault-tolerant quantum hardware. We outline the essential features of these t-doped stabilizer states that have been found already in past works. We will also describe how our work relates to these previous results and present directions for future investigation. Unsurprisingly, the first place where t-doped stabilizer states emerged was in the context of classical simulation of quantum circuits. The Gottesman-Knill theorem [1] showed that Clifford circuits can be efficiently represented with $O(n^2)$ bits and classically simulated with runtime $O(n^3)$. The introduction of t non-Clifford gates makes the simulation exponentially harder in t. This fact was first explored in Ref. [16], and then improved several times, e.g., in Refs. [17,37]. Despite improvements, the most advanced classical algorithm for simulating t-doped Clifford circuits still exhibits exponential scaling in t, which has been shown to be optimal [38]. Therefore, circuits (and the states they generate) that use $t = O(\log n)$ nonstabilizer gates can be efficiently classically simulated. However, once $t = \omega(\log n)$, any classical simulation algorithm requires a superpolynomial amount of resources.

Doped Clifford circuits have been proven to be useful for quantum information processing tasks. In particular, they appear in the context of unitary k-designs, which are ensembles of unitaries that reproduce the first k moments of the uniform measure over the unitary group. The Clifford group is a unitary 3-design [39] and does not constitute a unitary k-design for any $k \geq 4$ [40]. Therefore, it is natural to ask whether injecting t non-Clifford gates can produce higher k-designs. Although Refs. [41,42] showed that $t = \Theta(n)$ is required for an exact k-design, in Ref. [30], it was shown that for $t = O(k^4)$ (independent of n), a t-doped Clifford circuit successfully produces an *approximate* k-design. Since approximate k-designs suffice for the vast majority of quantum information processing tasks, this is just one demonstration of the usefulness of t-doped Clifford circuits.

Lastly, t-doped Clifford circuits and t-doped stabilizer states have recently attracted attention in the context of quantum learning theory. The objective of quantum learning is to obtain a classical description of an unknown state or unitary transformation while being restricted to a particular access model, a process known as *tomography*. Although the task of learning an arbitrary state ψ or unitary operator U has an exponential sample complexity in general, it is widely understood that these learning tasks can become tractable by restricting oneself to learning particular classes of circuits or states. To name just a few examples of this, Ref. [43] recently showed that $O(1)$ depth quantum circuits can be learned efficiently, and Ref. [44] showed that matrix product states can also be learned with polynomial resource requirements. This pattern holds true for the class of t-doped Clifford circuits [45,46] and t-doped stabilizer states [47–50]: each of these works, using distinct methods and access models, has shown that t-doped circuits and states can be learned using $\mathrm{poly}(n, 2^t)$ sample complexity and $\mathrm{poly}(n, 2^t)$ classical postprocessing time. However, this means that learning t-doped Clifford circuits and t-doped stabilizer states becomes infeasible whenever $t = \omega(\log n)$, exhibiting the same threshold as classical simulation algorithms. In addition to suggesting that quantum learning theory has a strong relation to the classical simulability of quantum mechanics [44,48,51], this fact raises the question: Are there information processing tasks that are both relevant and efficient beyond the seemingly ubiquitous threshold of $t = \omega(\log n)$?

A concurrent work [52] made progress along these lines. While their work focuses on bounding entanglement entropy (their Theorem 3.1 resembles our Lemma 8), our paper focuses on the operational and computational aspects of the entanglement-magic interplay. Similarly, in Refs. [45,46], it was demonstrated that quantum information scrambled by a t-doped Clifford circuit could be *unscrambled* and recovered using a carefully constructed Clifford circuit whenever $t < n$. This extends far beyond the classical simulability barrier. This result, in addition to providing insight into information scrambling, is an indication that certain information processing tasks can still exhibit efficiency beyond the classical simulation threshold

$t = \omega(\log n)$. In other words, while the task of learning a full description of such t-doped stabilizer states might be hard, restricting the objective from learning a full description of the state to simply learning certain characteristics of the state can render the problem feasible even beyond the classical simulability barrier. A particular characteristic of quantum states that occupies a central role in quantum information is its *entanglement*. In this direction, a recent work [53] shows how even for highly entangled states, estimating the entanglement entropy of states possessing Clifford symmetries can be significantly improved using techniques developed in Refs. [45,46]. Here, we present a detailed analysis of learning and manipulating the entanglement of t-doped stabilizer states, aiming to extend beyond the conventional limits imposed by classical simulability barriers.

C. Future directions

This paper thoroughly describes the interplay between two resource theories: the resource theory of magic and the resource theory of entanglement. We discuss how magic influences the efficiency of entanglement tasks, presenting both positive and negative results. This presents new research directions and ideas, some of which we list below.

(1) The ED phase includes states that contain up to $t = o(\exp n)$ non-Clifford gates, yet we are able to show that entanglement manipulation, detection, and estimation are efficient. This fact raises the question of whether this class of states is efficiently classically simulable; while we believe that is not the case, as we can always embed intractable problems, none of the known classical-simulation algorithms, e.g., [37] are specifically designed for entanglement-dominated states. We also need not demand full classical simulability, asking instead the slightly weaker question of whether there are other quantum information-related tasks that can be performed efficiently for ED states.

(2) We explored the synergy between the resource theory of entanglement and the resource theory of magic in the context of entanglement theory. We characterized how magic affects entanglement theory in positive and negative directions. However, the other way around is also possible. For instance, the analog of entanglement distillation in magic resource theory is magic-state distillation, in which one aims to distill a maximal number of copies of the magic state $|B\rangle \propto |0\rangle + e^{-i\pi/4}|1\rangle$ using only Clifford operations. Similarly to entanglement distillation, it is known that at most $O(\mathcal{M}(\rho))$ copies of $|B\rangle$ can be distilled from ρ, where $\mathcal{M}(\cdot)$ is an arbitrary magic monotone [54]. It is thus interesting to develop a theory, akin to the one developed here, that explores the flip side of this interplay: how does entanglement affect the efficiency of magic-state distillation? While it is clear that the optimal rate of magic-state distillation can be achieved for product states, how is this rate affected by entanglement? Is it perhaps true that the ED phase is characterized by efficient entanglement distillation, while the MD phase is characterized by efficient magic-state distillation? Answering these questions opens an exciting avenue for future research.

(3) In the paper, the distinction between ED and MD phases is made using the (smoothed) stabilizer nullity as the measure of magic, and the von Neumann entanglement entropy to quantify entanglement. However, we expect that this separation is more general and independent of the specific measures employed. An exciting future direction is to explore the generality of our conclusions by using different measures of entanglement and magic, and possibly deriving conclusions that are independent of the particular measures chosen.

(4) The ED phase is characterized by those states for which the entanglement is dominated by the stabilizer properties of the state and, as such, can be easily manipulated. This concept is akin to "Gaussification," a process where continuous variable states are analyzed by studying their Gaussian component. Particularly relevant are Refs. [55–58], where the authors study continuous variable entanglement under this lens. It is, therefore, natural to ask whether a theory similar to the one presented in this work can also be developed in the context of bosons and fermions, analyzing the interplay between the resource theory of fermionic (respectively, bosonic) non-Gaussianity.

(5) A naturally arising question prompted by our work is the possibility of a computationally efficient and agnostic distillation protocol for matrix product states (MPSs). Our no-go results for the MD phase are applicable only under the condition that the von Neumann entropy scales strictly faster than logarithmically with the local system size. The inadequacy of these results in capturing MPSs suggests the feasibility of an efficient and agnostic entanglement distillation protocol within this regime. One plausible approach involves the efficient learning of the state, with a sample efficiency suitable for polynomially large bond dimensions [59,60]. Subsequently, an analysis of the computational cost to identify the optimal distillation protocol.

II. STABILIZER NULLITY AND STABILIZER ENTANGLEMENT

We provide a glossary of notation for convenience (see Table I).

TABLE I. Table containing the notation used throughout the text.

Symbol	Definition
$\mathrm{SEP}(\mathscr{H}_A;\mathscr{H}_B)$	The set of separable states with respect to the bipartition $A\|B$
$\mathbb{P}_n$	The Pauli group over n qubits
G	The Abelian group of stabilizers for a state ψ
G_A	The Abelian subgroup of stabilizers for ψ that act nontrivially only on subsystem A
S	A generating set for the full set of stabilizers G; always has cardinality $\log_2\|G\|$
S_A	A generating set for G_A
ψ_A	The partial trace of the state ψ respect to the subsystem B in the bipartion $A\|B$
LOCC	Local operations and classical communication protocols [61]
$\mathcal{E}(\psi;A\|B)$	The stabilizer entanglement of ψ across the bipartition $A\|B$ (Definition 7)
$S_\alpha(\rho)$	The α-Rényi entropy of a state ρ (Definition 6)
M_+	The number of Bell pairs distillable across a bipartition
M_-	The number of Bell pairs consumed to dilute a state across a bipartition
t	The number of non-Clifford gates used to prepare a t-doped stabilizer state
l	The maximum number of qubits any non-Clifford gate acts on
ν	The stabilizer nullity of a state; it is defined by $\|S\| = n - \nu$, and satisfies the bound $\nu \leq 2lt$
Π	The stabilizer portion of a t-doped stabilizer state (Theorem 3)

Stabilizer states are capable of capturing a key aspect of "quantumness" that other classically simulable states (such as matrix product states) are unable to: they can exhibit any degree of entanglement, ranging from complete separability to full volume-law entanglement. Moreover, an early work showed that the entanglement of stabilizer states can be computed exactly in $O(n^3)$ time by studying the structure of the generators [9]. This work is representative in that it recognized how two important characteristics of stabilizer states (i.e., compact classical representation and the ability to exhibit high entanglement) could be combined in powerful ways to better understand the nature of entanglement. Indeed, this combination has made the stabilizer formalism a useful tool for a wide array of applications ranging from quantum error correcting codes to condensed-matter physics. However, entanglement is not all there is to quantum computation: early work showed that Clifford circuits (the unitaries that generate stabilizer states) are probably not even universal for purely *classical* computation [16]. This motivated the introduction of quantum magic as a means of quantifying the degree to which a state could be approximated using stabilizer states. In other words, magic measures the amount of non-Clifford resources required to prepare a particular state. Then, by definition, entanglement is insufficient for true quantum computational advantage: one must have a nontrivial degree of quantum magic as well.

A stabilizer state $|\psi\rangle$ is a state such that there is a size 2^n Abelian subgroup of the Pauli group $\mathbb{P}_n$ whose elements P all satisfy $P|\psi\rangle = |\psi\rangle$. This subgroup, known as the stabilizer group G, is finitely generated by n Pauli operators $S = \{g_j \mid g_j \in \mathbb{P}_n\}$. Then, stabilizer states can always be written as

$$|\psi\rangle\langle\psi| = \prod_{i=1}^{n}\left(\frac{\mathbb{1}+g_j}{2}\right). \tag{3}$$

The compact representation of stabilizer states in terms of just n generators immediately lends itself to their efficient simulability [1]. We note that the generators g_j are not unique—that is, there are many possible choices of generators S such that $G = \langle S\rangle$. This can be understood using the tableau representation of the stabilizer group, first introduced in Ref. [16], which takes advantage of the isomorphism between the Pauli group and $\mathbb{F}_2^{2n}$. This isomorphism is as follows: any generator $g_j \in \mathbb{P}_n$ can be represented as a length $2n$ binary string $\vec{s}_j = x_{j,1}x_{j,2}\ldots x_{j,n}z_{j,1}z_{j,2}\ldots z_{j,n}$, such that $g_j \propto \prod_{i=1}^{n}(X_i)^{x_{j,i}}(Z_i)^{z_{j,i}}$ (where the proportionality is up to a factor i^l for some integer l). The product of any two Pauli operators $g_j g_k$ corresponds to a binary string that is equal to $\vec{s}_j \oplus \vec{s}_k$, where $\oplus$ is addition modulo 2. We then see that if we choose a particular set of generators g_j with binary string representations as defined above, we can write a tableau representation $\mathcal{T}_S \in \mathbb{F}_2^{|S|\times 2n}$ of the stabilizer group

$$\mathcal{T}_S = \left[\begin{array}{cccc|cccc} x_{1,1} & x_{1,2} & \cdots & x_{1,n} & z_{1,1} & z_{1,2} & \cdots & z_{1,n} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,n} & z_{2,1} & z_{2,2} & \cdots & z_{2,n} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ x_{|S|,1} & x_{|S|,2} & \cdots & x_{|S|,n} & z_{|S|,1} & z_{|S|,2} & \cdots & z_{|S|,n} \end{array}\right], \tag{4}$$

where each row corresponds to a generator g_j. Although we note that $|S| = n$ by definition for all pure stabilizer states, we leave Eq. (4) in this general form to allow for the possibility that the stabilizer group is not full rank (i.e., $|S| < n$). Since the entire stabilizer group is given by rowsp $(\mathcal{T}_S)$, the tableau can be changed using any sequence of elementary row operations (i.e., swapping rows or adding rows to other rows), as these operations do not change the row space of a matrix. We will find it

useful to define the local stabilizer group G_A (G_B), which is the subgroup of stabilizers that act as the identity on the subsystem B (A). A tableau representation of the local stabilizer group G_A can be found efficiently as follows. We define a linear map $\mathcal{P}_B : g_A \otimes g_B \to \mathbb{1} \otimes g_B$, which projects out the A component of a given stabilizer. Acting with $\mathcal{P}_B$ on $\mathcal{T}_S$ sets all $x_{j,i} = z_{j,i} = 0$ for all $j = 1, \ldots, |S|$ and any $i \in A$, and leaves the rest of the matrix elements unchanged. Then, $G_A = \ker(\mathcal{P}_B(\mathcal{T}_S)^T)$, so finding a basis for the rows of $\ker(\mathcal{P}_B(\mathcal{T}_S)^T)$ suffices to find a tableau representation for the generators of G_A, which we call S_A. The procedure for finding S_B follows identically.

A. Stabilizer nullity and ν-compressible states

Closely related to stabilizer states are the Clifford unitaries, which are the group $\mathcal{C}_n$ of unitaries that normalize the Pauli group. That is, $\mathcal{C}_n := \{U \mid U\mathbb{P}_n U^\dagger = \mathbb{P}_n\}$. Since Clifford unitaries map stabilizer states to stabilizer states (they act by mapping each of the generators $g_j \to U g_j U^\dagger \in \mathbb{P}_n$), Clifford circuits, initialized with a stabilizer state (such as $|0\rangle^{\otimes n}$) are classically simulable. The canonical gateset for Clifford circuits consists of the Hadamard gate H, the phase gate S and the CNOT gate. Yet, as discussed previously, Clifford gates alone are insufficient to realize universal quantum computation: one needs to include just one non-Clifford gate to realize full universality. Injection of non-Clifford resources in the state-preparation results in a property known as *magic*. There are many choices to reasonably quantify magic [2,4,6,54,62,63]. In this work, we utilize the *stabilizer nullity*, introduced in Ref. [4]. In this work, this proves to be the most natural choice.

Definition 2 (Stabilizer nullity). Consider a pure state $|\psi\rangle$. If $|\psi\rangle$ is stabilized by $n - \nu$ algebraically independent Pauli operators P (i.e., $P|\psi\rangle = |\psi\rangle$), then the state has stabilizer nullity ν. Equivalently, the state $|\psi\rangle$ has a stabilizer group, hereby denoted as G, of cardinality $|G| = 2^{n-\nu}$. This notion can be easily extended to mixed states ρ. If ρ commutes with $n - \nu$ algebraically independent Pauli projectors [i.e. $((\mathbb{1} + P)/2)\rho((\mathbb{1} + P)/2) = \rho$], then we say the state has stabilizer nullity ν.

Given our earlier claim that these states might be understood as perturbations of exact stabilizer states, one might expect that these states exhibit a structure similar to that of stabilizer states. We show that indeed we can write states with stabilizer nullity ν using an algebraic structure that closely mirrors that of Eq. (3).

Theorem 3 (Algebraic structure of states with stabilizer nullity ν [47]). Any pure state $|\psi\rangle\langle\psi|$ with stabilizer nullity ν can be written

$$|\psi\rangle\langle\psi| = \underbrace{\frac{1}{2^\nu}\left(\mathbb{1} + \sum_{i=1}^{k} \mathrm{tr}(h_i\psi)h_i\right)}_{H} \underbrace{\left(\prod_{j=1}^{n-\nu}\left(\frac{\mathbb{1} + g_j}{2}\right)\right)}_{\Pi}; \quad k + 1 \le 2^{2\nu}, \tag{5}$$

where $S := \{g_j \in \mathbb{P}_n \mid j = 1, \ldots, |S|\}$ is a generating set for the stabilizer group of $|\psi\rangle$. The "bad generators" $h_i \in \mathbb{P}_n$ of $|\psi\rangle$, which do not necessarily obey any special structure, beyond the fact that $\mathrm{tr}(h_i\psi) \neq 0$. Π is a rank-2^ν projector, and H is a Hermitian operator whose operator norm is bounded: $\|H\|_\infty \le 2^\nu$. Finally, Π and H commute.

Proof. Let the stabilizer group G associated with the state $|\psi\rangle\langle\psi|$ have cardinality $|G| = 2^{n-\nu}$. For every h_i such that $\mathrm{tr}(h_i|\psi\rangle\langle\psi|) \neq 0$, it follows that every Pauli operator in the coset $h_i G$ also has expectation value $\mathrm{tr}(h_i|\psi\rangle\langle\psi|)$, as $\mathrm{tr}(h_i P_j|\psi\rangle\langle\psi|) = \mathrm{tr}(h_i|\psi\rangle\langle\psi|)$ for any stabilizer P_j, as $P_j|\psi\rangle = |\psi\rangle$. First, from the above reasoning, the set of Pauli operators, which have nonzero support on $|\psi\rangle$, namely $\mathrm{supp}(\psi) = \{P \in \mathbb{P}_n \mid \mathrm{tr}(P\psi) \neq 0\}$, can be partitioned into disjoint cosets $h_i G$ for $i = 0, \ldots, k$ where we use the convention that $h_0 = \mathbb{1}$. Second, since the $\alpha = 0$ stabilizer Rényi entropy M_0, defined as $\log|\mathrm{supp}(\psi)| - \log n$, we immediately have $|\mathrm{supp}(\psi)| \le 2^{n+\nu}$, is upper bounded by ν [5]. These two facts allow us to conclude that there are at most $|\mathrm{supp}(\psi)|/|G| \le 2^{n+\nu}/2^{n-\nu} = 2^{2\nu}$ coset representatives h_i. That is, $k + 1 \le 2^{2\nu}$. In summary, we can write

$$\begin{aligned}|\psi\rangle\langle\psi| &= \frac{1}{2^n}\sum_{i=0}^{k}\mathrm{tr}(h_i\psi)\sum_{P_j\in G} h_i P_j \\ &= \frac{1}{2^\nu}\left(\mathbb{1} + \sum_{i=1}^{k}\mathrm{tr}(h_i\psi)h_i\right)\left(\prod_{j=1}^{n-\nu}\left(\frac{\mathbb{1} + g_j}{2}\right)\right).\end{aligned} \tag{6}$$

■

Remark 1. The stabilizer component of the state Π is a valid state on its own, and more importantly, it is a (mixed) stabilizer state. In Ref. [64], it was shown that state Π has a very special relationship with $|\psi\rangle\langle\psi|$. First, it is the closest stabilizer state to $|\psi\rangle\langle\psi|$, as measured by the relative entropy $D(\Pi \parallel \psi) := -\mathrm{tr}\Pi\log\psi - S_1(\Pi)$. Furthermore, Π can be realized by "averaging" many copies of $|\psi\rangle\langle\psi|$ using an operation known as the quantum convolution.

The structure established by Theorem 3 shows that states with nullity ν can be *algebraically* separated into a stabilizer component Π and a nonstabilizer component H,

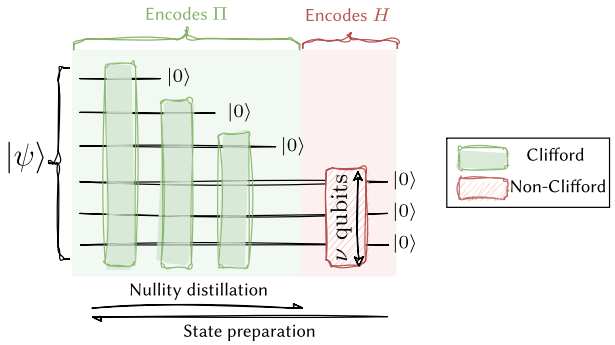


FIG. 3. We visualize the algebraic structure of ν-compressible states. Any ν-compressible state can be prepared using a general unitary acting on ν qubits, which depends only on H, and some global Clifford unitary, which depends only on Π.

where each component has a number of exponential terms in $n-\nu$ and ν, respectively. This suggests the possibility of decomposing the components into a tensor-product structure, specifically, that the operator $|\psi\rangle\langle\psi|$ can be factored into two separate *states*, one corresponding to H and one for Π. The following algorithm establishes this to be possible. We term this protocol "nullity distillation," because it essentially distills the nonstabilizer portion of a state H onto a number of qubits equal to the stabilizer nullity (see Fig. 3). This protocol constructs a Clifford unitary that takes as input a n-qubit state ρ with $|S|$ stabilizer generators and outputs a state $|0\rangle\langle 0|^{\otimes|S|}\otimes\rho''$, where ρ'' is a state on $n-|S|$ qubits. The idea is simple: we first identify a Clifford unitary that maps the first generator to Z_1, after which the state must be $|0\rangle\langle 0|\otimes\rho'$—this is because if a state is stabilized by Z_1, it must have $\mathrm{tr}(\rho_1 Z_1)=1 \implies \rho_1=(\mathbb{1}+Z_1)/2 \implies \rho=|0\rangle\langle 0|\otimes\rho'$, where ρ_1 is the reduced density matrix of ρ on qubit 1. We then find a Clifford (acting on the remaining $n-1$ qubits) that maps the second generator to Z_2, and so on, until we have a state $|0\rangle\langle 0|^{\otimes|S|}\otimes\rho''$. In Sec. VI C, we will see how the nullity distillation circuit can be interpreted as the *decoding circuit* for a stabilizer error-correcting code.

ALGORITHM 1. Nullity distillation.

Input: A state ρ with stabilizer generators $S=\{g_1,g_2,\ldots,g_{|S|}\}$
Output: A state $U\rho U^\dagger=|0\rangle\langle 0|^{\otimes|S|}\otimes\rho'$
1: **function** NullityDistillation(ρ)
2: $U\leftarrow\mathbb{1}$
3: **for** $i\leftarrow 1,\ldots,|S|$ **do**
4: $V\leftarrow$ any Clifford acting on qubits $\{i,i+1,\ldots,n\}$ such that $Vg_iV^\dagger=Z_i$
5: $U\leftarrow U\circ V$
6: **end for**
7: **return** $U\rho U^\dagger$
8: **end function**

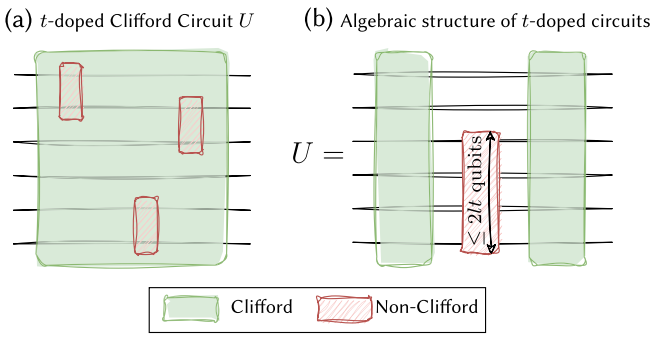


FIG. 4. In (a), we show a t-doped Clifford circuit, where the green background indicates a "substrate" of Clifford gates, which are implicitly assumed to dominate the circuit compared to the doped non-Clifford gates. In this circuit, we have set in particular $t=3$ (the number of non-Clifford gates) and $l=2$ (the number of qubits each non-Clifford acts on). In (b), we show an equivalent representation of the t-doped circuit shown in (a): it can always be rewritten as a Clifford C_1, with a general (non-Clifford) unitary acting on $\leq 2lt$ qubits, followed by another Clifford C_2 [45,46].

These considerations motivate us to denote states with stabilizer nullity ν as *ν-compressible states*, consistently reminding the reader that nullity can be distilled away using a simple Clifford operation.

Definition 3 (ν-compressible states). Let $|\psi\rangle$ be a pure quantum state. We say $|\psi\rangle$ is ν *compressible* if it can be prepared by the application of a unitary U acting on ν qubits followed by a Clifford circuit C. Any state $|\psi\rangle$ is ν compressible if and only if it has stabilizer nullity ν. The equivalence between stabilizer nullity ν states and ν-compressible states is established by Algorithm 1.

A well-known class of ν-compressible states is the set of t-doped stabilizer states. These are states created by the action of Clifford circuits containing a number t of l-qubit non-Clifford gates. Let us formally define them below.

Definition 4 (t-doped stabilizer states). An n-qubit state is a pure t-doped stabilizer state if $|\psi\rangle=U|0\rangle^{\otimes n}$, where a t-doped Clifford circuit U is a unitary operator containing Clifford gates and at most t non-Clifford l-qubit gates with $l=O(1)$ (see Fig. 4 for a visualization). A mixed t-doped stabilizer state is any state ρ for which there exists a purification $|\psi\rangle$ such that $|\psi\rangle$ is a pure t-doped stabilizer state, and the purifying system has at most $\lceil\log\mathrm{rank}\rho\rceil$ qubits. For brevity, we will often refer to t-doped stabilizer states simply as t-doped states.

Lemma 1 (t-doped stabilizer states are ν-compressible states). Let $|\psi\rangle$ be a t-doped stabilizer states. Then $|\psi\rangle$ is ν compressible with $\nu\leq 2lt$.

Proof. This has been shown in Ref. [65] (see also Refs. [45,46]). The core idea of the proof can be summarized

as follows: the application of any l-qubit non-Clifford gate removes at most $2l$ stabilizer generators from a state, so after applying t of these gates, we have at most $\nu \leq 2lt$. ■

However, the class of ν-compressible states is far from being limited to t-doped stabilizer states. For instance, unitaries that encode completely general H (see Fig. 3) can have up to exponential depth in ν, hence up to $\exp(\nu)$ non-Clifford gates. This shows that not only are ν-compressible states strictly more general than t-doped stabilizer states, they even include classes of states that are not efficiently preparable.

1. Smoothed stabilizer nullity

In Definition 2, we introduced the stabilizer nullity ν as the number of Pauli operators that fail to stabilize the state $|\psi\rangle$. As explained in Sec. II A, the stabilizer nullity is closely related to the number of non-Clifford gates required to prepare the state and serves as an *effective* count of such gates in the preparation process. However, an attentive reader might notice that this definition lacks robustness: even a small perturbation to the state can cause a substantial increase in the stabilizer nullity and, consequently, in the perceived magic of the state. However, this issue is a limitation of the definition itself, as magic, as a resource measure, is robust against small perturbations. To address this, we introduce a more robust notion of stabilizer nullity.

Definition 5 (Smoothed stabilizer nullity). Let $|\psi\rangle$ be a pure quantum state, and $\varepsilon > 0$. Then its ε-stabilizer nullity is defined as

$$\nu_\varepsilon(\psi) := \min\{\nu(\phi) \,:\, \| \,|\psi\rangle\langle\psi| - |\phi\rangle\langle\phi| \,\|_1 \leq \varepsilon\}. \quad (7)$$

In other words, we define the smoothed stabilizer nullity as the minimum stabilizer nullity within the ε ball surrounding the state. Note that $\nu_{\varepsilon=0}$ corresponds to the stabilizer nullity in Definition 2. To draw a helpful analogy, the robust definition of stabilizer nullity is similar to the robust definition of gate complexity: without an ε-ball tolerance, gate complexity is generally unbounded, whereas with an $\varepsilon > 0$ tolerance, it is at most exponential in the number of qubits [13].

In Sec. V, as emphasized throughout the paper, we present an efficient quantum algorithm designed to measure the stabilizer nullity. Importantly, given a target precision ε, the algorithm outputs the minimum stabilizer nullity among the states within the ε ball surrounding the state of interest. Thus, the algorithm effectively measures the smoothed version of the stabilizer nullity, as defined in Definition 5. Given the relevance of the smoothed stabilizer nullity, we proceed to outline its key properties.

First, any state with ε-stabilizer nullity ν_ε is within ε in trace distance of a state that admits a decomposition as described in Theorem 3. This proximity enables straightforward manipulation of such states, as will be demonstrated throughout the paper. Moreover, we can prove the following lemma.

Lemma 2. Given a pure quantum state $|\psi\rangle$ attaining $|\mathrm{tr}(P\psi)| \geq 1 - \varepsilon'$ for $P \in S$, where S contains algebraically independent Pauli operators, then ψ has $\nu_\varepsilon \leq n - |S|$ for every $\varepsilon \leq \sqrt{|S|\varepsilon'}$.

Proof. First of all, from Fact 2.21 of Ref. [48], we know that S contains mutually commuting operators necessarily if $\varepsilon' < 1/2$. Define $\phi_P = \mathrm{sign}(\mathrm{tr}(P\psi))$ for $P \in S$ and the projector $\Pi = \prod_{P\in S}(\mathbb{1} + \phi_P P)/2$. Define the projected state $\psi_\varepsilon := \Pi\psi\Pi/\mathrm{tr}(\Pi\psi)$. Notice that ψ_ε has stabilizer nullity $n - |S|$, being stabilized by $P \in S$. The gentle measurement lemma [36] implies

$$\|\psi - \psi_\varepsilon\| \leq \sqrt{\sum_{P\in S} \mathrm{tr}(P\psi)} \leq \sqrt{(n-\nu)\varepsilon'}. \quad (8)$$

Therefore, for any $\varepsilon < \sqrt{(n-\nu)\varepsilon'}$, ψ_ε lies in the ε ball of ψ. This concludes the proof. ■

Lemma 2 provides an operational interpretation of the ε-stabilizer nullity, similar to the interpretation of the stabilizer nullity $\nu_{\varepsilon=0}$. Specifically, it establishes a close connection with the number of Pauli operators that quasistabilize the state, meaning those with expectation values greater than or equal to $1/2$.

Building on the analogy with the robust version of gate complexity, in Lemma 1, we demonstrated that states constructed using at most t non-Clifford gates that are l local have a stabilizer nullity bounded by $2lt$. However, as previously noted, the construction of a specific state $|\psi\rangle$ exactly requires an infinite sequence of gates from a discrete set (e.g., Clifford + T), rendering the bound in Lemma 1 ineffective. Instead, states that can be approximately prepared, with an error up to ε in trace distance, using t l-local non-Clifford gates (approximate T count), have an ε-stabilizer nullity bounded by $2lt$. This leads to the following generalization of Lemma 1.

Corollary 1 (Approximate t-doped states are approximately compressible). Let $|\psi\rangle$ be a state that can be prepared with t l-local non-Clifford gates up to an accuracy ε in trace distance. Then, the ε-stabilizer nullity satisfies $\nu_\varepsilon(|\psi\rangle) \leq 2lt$.

Proof. Since there exists a state $|\phi\rangle$, ε close in trace distance, that can be prepared using at most $2lt$ non-Clifford gates, it follows that $\nu(|\phi\rangle) \leq 2lt$. By Definition 5, this implies that the ε-stabilizer nullity is bounded above by $2lt$. ■

Beyond capturing magic by bounding the effective number of non-Clifford gates required for state preparation (Lemma 1), the stabilizer nullity quantifies magic in a precise sense: it is a monotone within the resource theory of magic [54], as was proven in Ref. [4]. A monotone for the resource theory of magic is zero for stabilizer states and does not increase under stabilizer operations, which are a combination of the following elementary operations: Clifford unitaries, measurements in the computational basis, partial trace, addition of ancillary systems, and combinations thereof conditioned on measurement outcomes (see Ref. [66]). In resource theory, monotones are crucial for bounding conversion rates between resource states [67]. Specifically, Ref. [4] established several resource-theoretic bounds using stabilizer nullity as a monotone in magic-state resource theory. In this framework, the ε-stabilizer nullity extends these bounds to a robust setting, as demonstrated in the following lemma.

Lemma 3. Let $|\psi_1\rangle$ and $|\psi_2\rangle$ be two resource states with ε-stabilizer nullities $\nu_\varepsilon(\psi_1)$ and $\nu_\varepsilon(\psi_2)$, respectively. If $\nu_\varepsilon(\psi_1) < \nu_\varepsilon(\psi_2)$, then no stabilizer protocol can distill $|\psi_2\rangle$ from $|\psi_1\rangle$.

Proof. Let $|\phi_1\rangle$ be the optimal state as per Definition 5, satisfying $\|\psi_1 - \phi_1\|_1 \leq \varepsilon$ and $\nu(\phi_1) = \nu_\varepsilon(\psi_1)$. Assume, for contradiction, that there exists a stabilizer protocol $\mathcal{E}$ that, when applied to ψ_1, produces ψ_2, i.e., $\mathcal{E}(\psi_1) = \psi_2$. Applying the same protocol to ϕ_1 would yield a state ε close to ψ_2:

$$\|\mathcal{E}(\phi_1) - \psi_2\|_1 = \|\mathcal{E}(\phi_1) - \mathcal{E}(\psi_1)\|_1 \leq \|\phi_1 - \psi_1\|_1 \leq \varepsilon, \tag{9}$$

where we have used the data-processing inequality [15]. Equation (9) implies that $\nu(|\psi_2\rangle) \leq \nu(|\psi_1\rangle)$, contradicting the assumption $\nu_\varepsilon(|\psi_1\rangle) < \nu_\varepsilon(|\psi_2\rangle)$. ■

In Sec. V, we describe how the stabilizer nullity of an unknown, yet pure, n-qubit state $|\psi\rangle$ can be efficiently measured using an algorithm with a resource complexity of $\text{poly}(n, \varepsilon^{-1})$, where ε defines the ε ball around $|\psi\rangle$. Importantly, the algorithm outputs ν_ε, as measuring ν with $\varepsilon = 0$ would require infinite computational resources. Thus, we see that the robust notion of stabilizer nullity is indeed the most natural definition to adopt in this manuscript. For simplicity, however, in the remainder of the paper, we will work with the stabilizer nullity as defined in Definition 2, i.e., $\nu_{\varepsilon=0}$. Nonetheless, all of our conclusions can be readily extended to ν_ε with $\varepsilon > 0$ by allowing an ε error in trace distance, which renders our conclusions robust to perturbations. Where relevant, we will also indicate how our results can be expressed in terms of ν_ε. In Sec. IV E, we provide a robust formulation of Definition 1 for the entanglement- and magic-dominated phases through Definition 5. Additionally, we extend Theorems 1 and 2, establishing the robustness of the computational separation between these two phases.

2. Simulability of ν-compressible states

t-doped stabilizer states are ubiquitous in the quantum information literature (and as we will see in Sec. VI, they appear in physics as well). An aspect of these states that has been closely studied is their classical simulability; existing algorithms generally achieve a time (and space) complexity $\text{poly}(n, 2^{c \cdot t})$ for some constant c [37,68]. In fact, it has been shown that no simulation algorithm can achieve subexponential runtime in terms of T-gate count [38]. This result holds for single-qubit non-Clifford gates, but more broadly, it is easy to see that simulation of states that have stabilizer nullity up to ν must have runtimes that scale exponentially in ν. Algorithm 1 makes this clear: the set of general ν-qubit states is in a one-to-many correspondence with states that have stabilizer nullity ν, so a general simulation algorithm for states with stabilizer nullity ν would immediately imply a simulation algorithm for general ν-qubit states. Since such a general simulation algorithm is widely believed to require exponential (in ν) classical runtime, it follows that simulating arbitrary states with stabilizer nullity $\nu = \omega(\log n)$ cannot be classically efficient.

Example 1. A universal model of quantum computation is given by the Clifford + T model. In particular, any state can be expressed as, $|\psi\rangle \propto \langle 0^t|(C_{n+t}|0^n\rangle \otimes |T\rangle^{\otimes t})$, where $|T\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and C_{n+t} is a Clifford circuit acting on $n + t$ qubits. It is widely believed that when $t = \omega(\log n)$, this class of states lies beyond the regime of classical simulability. From Lemma 1, we know that $\nu \leq t$. Therefore, whenever $|\psi\rangle$ exhibits high entanglement ($S_1 = \omega(t)$) across an extensive bipartition of qubits $A|B$, the state resides in the entanglement-dominated phase.

B. Distribution and occurrence of entanglement-dominated states

Before exploring the operational distinctions between entanglement- and magic-dominated phases, a natural question to ask is where entanglement-dominated states can be found in practice. Recall that the two phases were defined in Definition 1 in terms of stabilizer nullity and entanglement entropy. However, there are many different measures available for entanglement. Here, we focus on the well-known α-Rényi entropy, which for $\alpha = 1$ reduces to the entanglement entropy $S_1(\psi_A)$ employed in Definition 1.

Definition 6 (α-Rényi entanglement entropy). For a pure bipartite state $|\psi\rangle_{AB}$, the α-Rényi entanglement entropy

across the bipartition $A|B$ is

$$S_\alpha(\psi_A) = \frac{1}{1-\alpha}\log\big(\mathrm{tr}(\psi_A^\alpha)\big) = S_\alpha(\psi_B). \tag{10}$$

For the special cases $\alpha \in \{0, 1, \infty\}$, the Rényi entropy can be defined as the limit of S_α as $\alpha \to 0$, 1 (which corresponds to the familiar von Neumann entropy), or ∞. Explicitly, for these three cases

$$S_\alpha(\psi_A) = \begin{cases} \log \mathrm{rank}\psi_A & \text{for } \alpha = 0 \\ -\mathrm{tr}(\psi_A \log \psi_A) & \text{for } \alpha = 1 \\ -\log \mathrm{Pur}(\psi_A) & \text{for } \alpha = 2 \\ -\log \|\psi_A\|_\infty & \text{for } \alpha = \infty. \end{cases} \tag{11}$$

where $\mathrm{Pur}(\psi_A) := \mathrm{tr}(\psi_A^2)$. An important fact about the Rényi entanglement entropies is that they are monotonically decreasing in α, which means that $S_\alpha(\psi_A) \geq S_\beta(\psi_A)$ for any $\alpha \leq \beta$.

Despite the fact that the entanglement entropy $S_1(\psi_A)$ is the unique measure that fully characterizes bipartite entanglement in pure states [18], the α-Rényi entropies serve as a theoretically valuable tool, as we will clear in the following sections.

The following lemma states that the vast majority of states are MD. This may seem to throw cold water on the hope of finding any entanglement-dominated states, but for the reasons we discuss below, this is not so.

Lemma 4 (Typicality of magic-dominated states). For any bipartition $A|B$ with $n_A \leq n_B$, the states in the Haar ensemble are magic dominated with probability 1:

$$\Pr_{|\psi\rangle\sim\mathrm{Haar}}[|\psi\rangle \text{ is magic dominated}] = 1. \tag{12}$$

Proof. All but a measure zero set of states in the Haar measure have no stabilizers, so they have $\nu = n$ with probability 1. Since the entanglement of these states is at most n_A (i.e., strictly less than n), these states are all magic dominated. ■

In spite of Lemma 4, the vast majority of states in the Haar measure will also never be observed in practice, as they take exponential time to prepare. In the following, we will explore a much more practically relevant setting, where entanglement-dominated states turn out to be ubiquitous: *polynomial-depth random quantum circuits*, or more specifically, unitary k-designs. Several works [30,41,42] have shown that a very simple form of random circuit forms a unitary k-design. These circuits take the form of a sequence of global random Clifford unitaries, interleaved with local Haar random unitaries (which are, importantly, not necessarily Clifford unitaries). The measure over this random ensemble of circuits is μ_t:

$$\mu_t = \mu_{\mathrm{Cl}_n} * \mu_{\mathrm{Haar}_l} * \mu_{\mathrm{Cl}_n} * \mu_{\mathrm{Haar}_l} * \underbrace{\cdots}_{t\,\text{times}} * \mu_{\mathrm{Cl}_n} * \mu_{\mathrm{Haar}_l} * \mu_{\mathrm{Cl}_n}, \tag{13}$$

where $*$ denotes the convolution between measures, μ_{Haar_l} is the measure over l-qubit Haar random unitaries (acting on the first l qubits) with $l = O(1)$, and μ_{Cl_n} is the uniform distribution over n-qubit Clifford unitaries. Crucially, this measure generates an approximate k-design with just $t = O(\mathrm{poly}\, k)$. That is, using very few non-Clifford resources, we can quickly generate good k-designs [30]. This leads us to observe that this ensemble generates states that are almost exclusively entanglement dominated. We will use $\mu(\mathcal{S}_t)$ to denote the measure over states induced by μ_t: that is, we sample from $\mu(\mathcal{S}_t)$ by sampling a unitary $U \sim \mu_t$ and applying $U|0\rangle^{\otimes n}$.

Lemma 5 (Typicality of entanglement-dominated phase within $\mu(\mathcal{S}_t)$). For any bipartition $A|B$, most t-doped stabilizer states, with $t = o(n_A)$, are entanglement dominated. That is

$$\Pr_{|\psi\rangle\sim\mu(\mathcal{S}_t)}[|\psi\rangle \text{ is entanglement dominated}] \geq 1 - 2^{-n_A/2+1}. \tag{14}$$

Proof. If $t = o(n_A)$, a sufficient condition for $|\psi\rangle$ to be entanglement dominated is that $S_2(\psi_A) \geq n_A/2$, since $S_1(\psi_A) \geq S_2(\psi_A)$. Recall that $S_2(\psi_A) = -\log \mathrm{tr}(\psi_A^2)$, and applying the Markov inequality, we have

$$\begin{aligned} \Pr_{|\psi\rangle\sim\mu(\mathcal{S}_t)}\left[S_2(\psi_A) \geq \frac{n_A}{2}\right] &\geq 1 - \frac{\mathbb{E}_{|\psi\rangle\sim\mu(\mathcal{S}_t)}[\mathrm{Pur}(\psi_A)]}{2^{-n_A/2}} \\ &\geq 1 - 2^{-n_A/2+1}, \end{aligned} \tag{15}$$

where in the second inequality we use the fact that $\mu(\mathcal{S}_t)$ forms at least a 2-design to conclude that

$$\mathbb{E}_{\mu(\mathcal{S}_t)}[\mathrm{Pur}(\psi_A)] = \frac{2^{n_A} + 2^{n_B}}{2^n + 1} \leq \frac{2}{2^{n_A}}. \tag{16}$$

■

However, as explained in Sec. II A, t-doped stabilizer states are only one example of ν-compressible states, characterized by a stabilizer nullity $\nu \leq 2lt$. Indeed, as illustrated in Fig. 5, they form only a vanishing fraction of those. Therefore, we study the general class of ν-compressible states, and find that they too are almost all ED. To establish this result, we choose the uniform measure on ν-compressible states, defined as the convolution of the uniform measure over the Clifford group μ_{Cl_n}

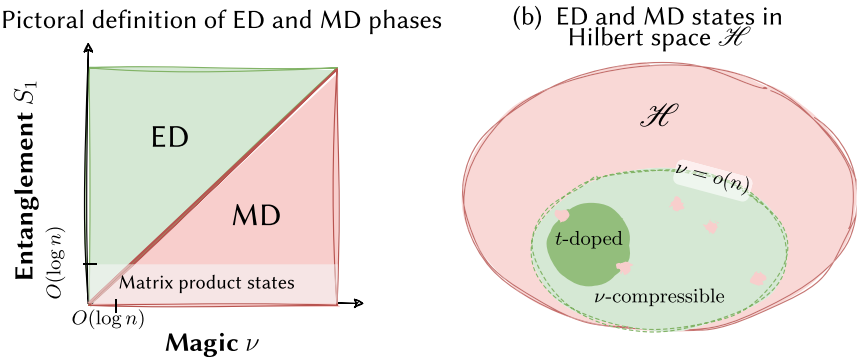


FIG. 5. In (a), we show a representation of the entanglement- and magic-dominated phase. The entanglement is quantified, given the bipartition, by the entanglement entropy, while magic is quantified by the stabilizer nullity (Definition 1). The class of matrix-product states occupies a special place in our theory, as we are unable to rule out efficient entanglement manipulation protocols for MPS, despite the fact that most of these states are magic dominated. Indeed, examples of efficient MPS-specific protocols have already been found [44]. In (b), we show the portion of Hilbert space occupied by ν-compressible states for $\nu = o(n)$. These states occupy a vanishing fraction of the full Hilbert space. Similarly, a counting argument shows that t-doped states (which are all $2lt$-compressible states) are also a vanishing fraction of all $2lt$-compressible states. However, for both of these classes of states, the entanglement-dominated phase is typical (see Lemmas 4 to 6).

and μ_{Haar_ν} representing the uniform measure on the unitary group. In other words, $\mu_\nu = \mu_{\mathrm{Cl}_n} * \mu_{\mathrm{Haar}_\nu}$. Recalling Definition 3, we have the following lemma.

Lemma 6 (Typicality of ED phase for ν-compressible states). Most ν-compressible states, drawn from the measure μ_ν with $\nu = o(n)$, are entanglement dominated with respect to exponentially many cuts $A|B$, where each half of the bipartition is size $\Omega(n)$.

Proof. Let us consider any cut $A|B$ with $n_A, n_B = \Omega(n)$. Let us assume, without loss of generality, that $n_A \leq n_B$. A sufficient condition for $|\psi\rangle$ to be entanglement dominated is that $S_2(\psi_A) \geq n_A/2$, since $S_1(\psi_A) \geq S_2(\psi_A)$. Recall that $S_2(\psi_A) = -\log \mathrm{tr}(\psi_A^2)$, and applying the Markov inequality, we have

$$\Pr_{|\psi\rangle\sim\mu_\nu}\left[S_2(\psi_A) < \frac{n_A}{2}\right] \leq \frac{\mathbb{E}_{|\psi\rangle\sim\mu(\mathcal{S}_t)}[\mathrm{Pur}(\psi_A)]}{2^{-n_A/2}} \leq 2^{-n_A/2+1}, \tag{17}$$

where in the second inequality we use the fact that μ_ν forms at least a 2-design to conclude that

$$\mathbb{E}_{\mu(\mathcal{S}_t)}[\mathrm{Pur}(\psi_A)] = \frac{2^{n_A}+2^{n_B}}{2^n+1} \leq \frac{2}{2^{n_A}}. \tag{18}$$

Consider now a set $K := \{A|B : n_A, n_B = \Omega(n)\}$ of cuts of size $|K| = o(2^{-n})$. The failure probability that a randomly sampled state $|\psi\rangle \sim \mu_\nu$ is enough entangled over all the cuts in K is bounded as

$$\Pr_{|\psi\rangle\sim\mu_\nu}\left[S_2(\psi_A) < \frac{n_A}{2} : \forall A|B \in K\right] \leq |K|2^{-\Omega(n)} = o(1). \tag{19}$$

Hence, the probability that $|\psi\rangle \sim \mu_\nu$ is ED over exponentially many cuts is $1 - o(1)$. ■

The findings of Lemmas 4 to 6 are summarized in Fig. 5(b). The majority of states in the Hilbert space are in the MD phase, but the majority of ν-compressible states are ED states. Within ν-compressible states, t-doped stabilizer states with a sublinear number of non-Clifford gates are also ED states.

We now proceed to introduce one of the central tools employed throughout this paper: the stabilizer entanglement.

C. Stabilizer entanglement

When dealing with entanglement, a fundamental question is to understand whether a state is entangled or not, and more specifically, what the degree of entanglement is. Given the fact that the Rényi entropies are defined in terms of the reduced density matrix ψ_A (Definition 6), we can use the structure of a ν-compressible state to *exactly* calculate any Rényi entropy of a generic state. Using this invariance, summarized by Eq. (5), we show below that the case of $\alpha = 2$ is particularly simple.

Lemma 7 (Exact 2-Rényi entropy). For any pure state $|\psi\rangle$, the 2-Rényi entanglement entropy across a bipartition $A|B$ is

$$S_2(\psi_A) = (n_A - |S_A|) - \log\left(\sum_{i=0}^{k} \delta_i \mathrm{tr}^2(h_i\psi)\right), \quad (20)$$

where S_A are the generators for the local stabilizer group G_A (i.e., the subgroup of the original stabilizer group that acts trivially on the B subsystem), $\delta_i \in \{0,1\}$ is an indicator variable for whether there is some stabilizer P_i of $|\psi\rangle$ such that $P_i h_i$ acts trivially on the B subsystem, and by convention $h_0 = \mathbb{1}$, $\delta_0 = 1$.

Proof. An equivalent formulation of Eq. (5) is

$$\begin{aligned} |\psi\rangle\langle\psi| &= \frac{1}{2^{n-|S|}} \sum_{i=0}^{k} \mathrm{tr}(h_i\psi) h_i \Pi \\ &= \frac{1}{2^n} \sum_{i=0}^{k} \mathrm{tr}(h_i\psi) \sum_{P_j \in G} h_i P_j . \end{aligned} \quad (21)$$

Note that if there is no stabilizer $P_j \in G$ such that $P_j h_i$ acts trivially on the B subsystem, then $\mathrm{tr}_B(h_i\Pi) = 0$. We will call the surviving indices $\mathcal{I} = \{i \mid \exists P_i \in G;\ \mathrm{tr}_B(h_i P_i) \neq 0\}$. We can write

$$\psi_A = \mathrm{tr}_B(|\psi\rangle\langle\psi|) = \frac{1}{2^{n-|S|}} \sum_{i\in\mathcal{I}} \mathrm{tr}(h_i\psi)\mathrm{tr}_B(h_i P_i \Pi), \quad (22)$$

since $P_i\Pi = \Pi$. To understand the term $\mathrm{tr}_B(h_i P_i \Pi)$, recall that $\Pi = \prod_{j=1}^{|S|}((\mathbb{1} + g_j)/2)$, so $h_i P_i \Pi$ can be expanded as a linear combination of Pauli operators of the form $h_i P_i P_j$ where $P_j \in G$. By the definition of $\mathcal{I}$, $h_i P_i$ acts as the identity on subsystem B. Therefore, when we take the partial trace over B, a term $h_i P_i P_j$ survives if and only if P_j also acts as the identity on B. These surviving stabilizers P_j are precisely those generated by S_A, which gives us $\mathrm{tr}_B(h_i P_i \Pi) = h_i P_i \Pi_A$, where $\Pi_A = \prod_{j=1}^{|S_A|}((\mathbb{1} + g_j)/2)$. In summary, defining $h_i' := h_i P_i$:

$$\begin{aligned} \psi_A &= \frac{1}{2^{n_A}} \sum_{i\in\mathcal{I}} \mathrm{tr}(h_i\psi) \sum_{P_j \in G_A} h_i' P_j \\ &= \underbrace{\frac{1}{2^{n_A-|S_A|}} \left(\mathbb{1} + \sum_{i\in\mathcal{I}} \mathrm{tr}(h_i\psi) h_i'\right)}_{H_A} \underbrace{\left(\prod_{j=1}^{|S_A|} \left(\frac{\mathbb{1} + g_j}{2}\right)\right)}_{\Pi_A}. \end{aligned} \quad (23)$$

The purity of this state is exactly

$$\begin{aligned} \mathrm{tr}(\psi_A^2) &= \frac{1}{4^{n_A}} \sum_{i\in\mathcal{I}} \mathrm{tr}^2(h_i\psi) 2^{n_A+|S_A|} \\ &= \frac{1}{2^{n_A-|S_A|}} \sum_{i=0}^{k} \delta_i \mathrm{tr}^2(h_i\psi). \end{aligned} \quad (24)$$

Since $S_2(\psi_A) = -\log \mathrm{tr}(\psi_A^2)$, this shows the desired claim. ■

Each of the linear-algebraic manipulations in Algorithm 2 can be done efficiently using the tableau formalism, including the calculation of $|S_A|$ and δ_i. The primary computational cost is the loop over $i = 0, \ldots, k$, as k can be as high as $2^{2\nu}$. The overall computational complexity of Algorithm 2 is $O(4^\nu n^3)$. Therefore, similar to low-rank stabilizer simulation algorithms, this exact calculation of the 2-Rényi entropy can be done in polynomial time for $\nu = O(\log n)$.

At this point, we recall our earlier assertion that most of the entanglement structure could be ascertained simply by looking at the stabilizers of a state. Indeed, the form of Eq. (20) is quite suggestive: it appears that there is a "stabilizer" component of the entanglement $n_A - |S_A|$, and a correction due to the nonstabilizer components of the state. The following lemma makes this intuition precise.

Lemma 8. For any pure state ψ and bipartition $A|B$, the following holds for all α:

$$n_A - |S_A| - 2\nu \leq S_\alpha(\psi_A) \leq n_A - |S_A|, \quad (25)$$

where $\psi_A = \mathrm{tr}_B(\psi)$. For the von Neumann entropy, a tighter bound

$$n_A - |S_A| - \nu \leq S_1(\psi_A) \leq n_A - |S_A| \quad (26)$$

holds. An identical bound holds for any Rényi entropy with $\alpha \leq 2$.

Proof. To prove the statement, we recall the hierarchy of the α-Rényi entropies, we have $S_\infty(\psi_A) \leq S_\alpha(\psi_A) \leq S_0(\psi_A)$. Hence, it suffices to lower bound $S_\infty(\psi_A)$ and upper bound $S_0(\psi_A)$. Referring to the algebraic structure of ψ_A in Eq. (23), the lower bound proceeds as follows: $-S_\infty(\psi_A) = \log \|\psi_A\|_\infty = \log \|H_A\Pi_A\|_\infty \leq \log \|H_A\|_\infty \leq \log\left((k+1)/(2^{n_A-|S_A|})\right) \leq -(n_A - |S_A|) + 2\nu$, so we have $S_\infty(\psi_A) \geq n_A - |S_A| - 2\nu$. For the upper bound, $S_0(\psi_A) = \log \mathrm{rank}\psi_A = \log \mathrm{rank}(\Pi_A H_A) \leq \log \mathrm{rank}\Pi_A = n_A - |S_A|$.

The proof of the second inequality follows by lower bounding $S_2(\psi_A)$ using Lemma 7. Specifically, it suffices to show that $\log\left(\sum_{i=0}^{k} \delta_i \mathrm{tr}^2(h_i\psi)\right) \leq \nu$. Observe that by applying a nullity distillation unitary C (Algorithm 1) to the state ψ, we get $|0\rangle^{\otimes(n-\nu)} \otimes |\phi\rangle$, where $|\phi\rangle$ is a ν-qubit state. The nullity distillation circuit C is constructed specifically to map the stabilizer generators of ψ to single-qubit Z operators on the first $n - \nu$ qubits. Therefore, for any operator h_i that has nonzero expectation value $\mathrm{tr}(h_i\psi) \neq 0$, $Ch_iC^\dagger$ must act as the identity on these first $n - \nu$ qubits [otherwise it would anticommute with at least

one of the stabilizers, forcing $\mathrm{tr}(h_i\psi) = 0$]. This means $Ch_iC^\dagger = \mathbb{1}_{n-\nu} \otimes P_i$ for some Pauli operator P_i acting on the remaining ν qubits. We then have

$$\begin{aligned}\sum_{i=0}^{k} \mathrm{tr}^2(h_i\psi) &= \sum_{i=0}^{k} \mathrm{tr}^2((Ch_iC^\dagger)(C\psi C^\dagger)) \\ &= \sum_{i=0}^{k} \mathrm{tr}^2((\mathbb{1}_{n-\nu} \otimes P_i)(|0\rangle\langle 0|^{\otimes(n-\nu)} \otimes |\phi\rangle\langle\phi|)) \\ &\leq \sum_{P\in\mathbb{P}_\nu} \mathrm{tr}^2(P\phi) = 2^\nu, \end{aligned} \tag{27}$$

where the final equality holds because $|\phi\rangle$ is a pure ν-qubit state. We then have

$$\begin{aligned}\log\left(\sum_{i=0}^{k} \delta_i \mathrm{tr}^2(h_i\psi)\right) &\leq \log\left(\sum_{i=0}^{k} \mathrm{tr}^2(h_i\psi)\right) \\ &= \log\left(\sum_{P\in\mathbb{P}_\nu} \mathrm{tr}^2(P\phi)\right) = \nu, \end{aligned} \tag{28}$$

as desired. ■

It may at first appear that $n_A - |S_A|$ is a fundamental quantity for computing the entanglement of a state. Indeed, Lemma 8 recovers the fact that, as shown in Ref. [9], the entanglement of a stabilizer state across a bipartition $A|B$ is given exactly by $n_A - |S_A|$. However, for general states, the quantity $n_A - |S_A|$ is not necessarily symmetric across the bipartition. That is, unlike stabilizer states, we do not necessarily have $n_A - |S_A| = n_B - |S_B|$. This can be seen by a simple example: consider $T_1 |+\rangle^{\otimes 2}$, where T_1 is the T gate acting on the first qubit. This state has only one stabilizer: X_2. Therefore, if A is the first qubit and B is the second, we will have $n_A - |S_A| = 1$ while $n_B - |S_B| = 0$. This problem motivates us to introduce a closely related quantity, termed the "stabilizer entanglement," that is manifestly symmetric across the bipartition. We will show that this stabilizer entanglement not only captures the entanglement just as well as the quantity $n_A - |S_A|$, but it also serves as a useful entanglement monotone for restricted classes of LOCC protocols, namely *stabilizer LOCC protocols*.

Definition 7 (Stabilizer entanglement). For a given bipartition $A|B$, the stabilizer entanglement $\mathcal{E}(\rho; A|B)$ is

$$\mathcal{E}(\rho; A|B) := \frac{|S_{AB}|}{2} = \frac{|S| - (|S_A| + |S_B|)}{2}, \tag{29}$$

where S_{AB} are the generators for the subgroup $G_{AB} \cong G/(G_A \cdot G_B)$ of stabilizers that are unaccounted for by the local stabilizer groups G_A and G_B. Note that ρ may be a mixed or pure state.

Theorem 4 (Bounded approximation error). For a pure state, the difference between $\mathcal{E}(\psi; A|B)$ and $S_\alpha(\psi_A)$ is $O(\nu)$. Specifically, the following holds for all α:

$$\mathcal{E}(\psi; A|B) - 3\nu/2 \leq S_\alpha(\psi_A) \leq \mathcal{E}(\psi; A|B) + \nu/2. \tag{30}$$

For the von Neumann entropy,

$$\mathcal{E}(\psi; A|B) - \nu/2 \leq S_1(\psi_A) \leq \mathcal{E}(\psi; A|B) + \nu/2. \tag{31}$$

Proof. Let us first prove Eq. (30), as Eq. (31) follows from analogous reasoning. Let us define

$$z := \frac{(n_A - |S_A|) + (n_B - |S_B|)}{2} = \frac{n - |S_A| - |S_B|}{2}. \tag{32}$$

Using the fact that $S_\alpha(\psi_A) = S_\alpha(\psi_B)$ (which follows from the fact that the state ψ is pure), applying Eq. (25), we have the following inequality:

$$z - 2\nu \leq S_\alpha(\psi_A) \leq z. \tag{33}$$

Now, since the dimension of the stabilizer group is effectively $|S| = n - \nu$, this implies that $z = \mathcal{E}(\psi; A|B) + \nu/2$, and therefore we finally have

$$\mathcal{E}(\psi; A|B) - 3\nu/2 \leq S_\alpha(\psi_A) \leq \mathcal{E}(\psi; A|B) + \nu/2. \tag{34}$$

This proves Eq. (30). The bound for von Neumann entropy in Eq. (31) holds with identical reasoning, by using the tighter bound from Eq. (26) instead of Eq. (25). This concludes the proof. ■

As previously mentioned, the stabilizer entanglement is not an entanglement monotone in general. However, we show that it qualifies as an entanglement monotone for a restricted class of LOCC operations, which we introduce below.

Definition 8 (Stabilizer LOCC protocols). Stabilizer protocols are all procedures that are comprised of the following four operations: (i) evolution by Clifford unitaries, (ii) inclusion of an arbitrary number of ancillae initialized in $|0\rangle$, (iii) discarding any subset of qubits, (iv) measurement in the computational basis, and (v) any combination of the first four operations conditioned on these measurement outcomes. Stabilizer LOCC protocols with respect to a bipartition $A|B$ are stabilizer protocols wherein the Clifford unitaries are allowed only to act locally on either A or B, but no entangling unitaries are allowed across the bipartition. Classical communication is still allowed for stabilizer LOCC protocols, so operations on either side of the bipartition can be conditioned on measurement outcomes from the other side of the bipartition. In short, stabilizer LOCC protocols are to general stabilizer protocols as LOCC protocols are to general quantum channels.

Note that this definition of stabilizer protocols allows for protocols conditioned on classical randomness, as m random bits can be generated by including m ancillae, acting with a Hadamard on each (to get $|+\rangle^{\otimes m}$), and measuring in the computational basis.

Theorem 5 (Monotonicity under stabilizer LOCC). The stabilizer entanglement $\mathcal{E}$ is nonincreasing under stabilizer LOCC operations.

Proof. It suffices to show that $|S_{AB}|$ is nonincreasing under (i) local Clifford unitaries, (ii) inclusion of m ancillae in $|0\rangle^{\otimes m}$, (iii) partial trace, and (iv) computational basis measurements.

(1) Consider a Clifford unitary $U = U_A \otimes \mathbb{1}$, which acts nontrivially only on subsystem A (the proof for a Clifford acting on subsystem B follows identically). By definition, for any generator $g_A \otimes g_B = g \in \mathbb{P}_n$, $UgU^\dagger = U_A g_A U_A^\dagger \otimes g_B$. Therefore, any generator, which acts trivially only on subsystem B must still act trivially on B after applying U_A, and the same goes for generators, which act trivially on A. This is to say, $|S_A|$ and $|S_B|$ cannot decrease, and since U is a Clifford unitary, $|S|$ remains unchanged. Therefore, $|S_{AB}|$ cannot increase.
(2) Adding m ancillae initialized in $|0\rangle^{\otimes m}$ adds m generators which are single-qubit Z operators on each of the ancilla sites. However, since these generators act locally on either side of the bipartition, this does not change the number of generators $|S_{AB}|$ that span A and B.
(3) Tracing out a subset of the qubits can act only to discard stabilizers of the state.
(4) A measurement on site Z_i in the computational basis can have one of three effects.
    (a) If Z_i is already a stabilizer of $|\psi\rangle$, there is no change.
    (b) If Z_i commutes with each of the stabilizers of $|\psi\rangle$ but is not itself a stabilizer of $|\psi\rangle$, it is added to the list of generators for the stabilizer group of the state, with a $\pm$ phase depending on the measurement outcome. This does not change the number of generators $|S_{AB}|$ that spans A and B.
    (c) If Z_i anticommutes with one of the generators of the stabilizer group for $|\psi\rangle$, the general update procedure is to pick an arbitrary anticommuting stabilizer g_j and update all the other stabilizers by multiplying them with g_j, then replace g_j with Z_i. To show that $|S_{AB}|$ does not increase, assume that we have written our generators in the form $S = S_A \cup S_B \cup S_{AB}$. There are two possibilities. First, if there is a $g_j \in S_A$ that anticommutes with Z_i, we will need to multiply some generators in S_A and S_{AB} by g_j, but none in S_B (since all generators in S_B commute with Z_i). Multiplying a generator in S_A by g_j does not change its locality, and neither does replacing g_j by Z_i, so $|S_{AB}|$ cannot increase. In the second case, all $g_j \in S_A$ commute with Z_i, so there must be some $g_j \in S_{AB}$ that anticommutes with Z_i. The update step comprises multiplying some generators in S_{AB} by g_j, and replacing g_j by Z_i—again, since the other generating sets S_A and S_B are untouched, this does not increase $|S_{AB}|$. ■

The monotonicity of the stabilizer entanglement under stabilizer LOCC protocols concludes our technical preliminaries. In the subsequent section, we leverage these tools to investigate entanglement manipulation and characterization through LOCC stabilizer protocols.

III. ENTANGLEMENT THEORY THROUGH STABILIZER PROTOCOLS

In this section, we explore entanglement manipulation and detection tasks, examining the significance of stabilizer nullity in these tasks. We focus on three core tasks: entanglement distillation, entanglement dilution, and the witnessing of multipartite entanglement, all through the unifying framework of the stabilizer entanglement. Starting with entanglement distillation, we investigate the ability to purify entangled states from noisy states with stabilizer nullity ν, crucial for quantum communication and computation. Conversely, entanglement dilution addresses the synthesis of entangled states from a set of more basic quantum resources, including shared entangled pairs and local non-Clifford resources. This discussion is framed within the constraints imposed by the nullity ν, highlighting the interplay between entanglement, magic, and the operational limits of stabilizer LOCC protocols. Finally, we move away from entanglement manipulation to the mere task of verifying the presence of entanglement, known as entanglement witnessing. We develop a framework for witnessing genuine multipartite entanglement for ν-compressible states, which we use to prove the robustness of entanglement of these states against noise and imperfections. We note that each of the algorithms we present below will assume *a priori* knowledge of the stabilizer generators S of the state. This assumption is not necessary: as we discuss in Sec. V, S is readily available in most settings.

A. Entanglement distillation and dilution

1. Entanglement distillation

Maximally entangled states are a powerful resource in quantum information theory, with many protocols

leveraging the properties of such a state. However, noise makes it harder to synthesize a maximally entangled state without errors. Entanglement distillation is based on the idea that, given multiple copies of a mixed state, it is possible to convert the entanglement in the noisy state into a certain number of pure, useful (i.e., maximally entangled) states [18,69]. The specific entanglement distillation task we focus on in this work is as follows. Two parties, A and B, are given a noisy state ρ_{AB}, and their goal is to use some LOCC protocol to distill as many two-qubit maximally entangled states (i.e., Bell pairs, commonly known as "ebits") as possible from ρ_{AB}. Given the theme of this work, we will further restrict our attention to stabilizer LOCC protocols, for which the framework of ν-compressible states gives us a rich set of tools to construct good entanglement distillation protocols. In this section, we will describe a single-shot, deterministic, and error-free entanglement distillation protocol for ν-compressible states. Importantly, our algorithm requires only one copy of the input state and does not rely on using many copies of a state and projecting into some typical set; this is motivated by practical concerns, as it has been shown that one may need up to approximately 10^6 copies [36] to achieve the necessary concentration for the asymptotically optimal rates found in Ref. [20].

To start, we state a no-go result that circumscribes the optimal performance of any stabilizer LOCC protocols for entanglement distillation.

Lemma 9 (Distillability bounds). No stabilizer LOCC can distill more than $\mathcal{E}(\psi; A|B)$ Bell pairs from a state ψ.

Proof. Assume there were some stabilizer protocol $\mathcal{S}$ that could distill $M > \mathcal{E}(\psi; A|B)$ Bell pairs from ψ. Then, $\psi' := \mathcal{S}(\psi)$ is a state with at least $2M$ stabilizers in S'_{AB}, as each Bell pair contributes two generators XX and ZZ spanning the bipartition $A|B$. However, then $\mathcal{E}(\psi'; A|B) \geq M > \mathcal{E}(\psi; A|B)$, which contradicts Theorem 5. ■

Remarkably, in Theorem 6, we show that we can almost saturate this upper bound. First, we need the following lemma.

Lemma 10. If a state ρ is stabilized by X_iX_j and Z_iZ_j, it must be of the form $\rho = |\phi_+\rangle\langle\phi_+|_{ij} \otimes \sigma$, where $|\phi_+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ and σ is some arbitrary mixed state.

Proof. Let ρ_A be the reduced density matrix of ρ on sites i and j. By assumption $\mathrm{tr}(\rho_A X_iX_j) = \mathrm{tr}(\rho_A Z_iZ_j) = 1$. There is only one density matrix that satisfies this, which is $\rho_A = |\phi_+\rangle\langle\phi_+|$. Since ρ_A is pure, it must be unentangled with the rest of the system, which implies $\rho = \rho_A \otimes \sigma = |\phi_+\rangle\langle\phi_+| \otimes \sigma$ for some mixed state σ. ■

By employing Lemma 10, we now demonstrate that, for certain classes of states, specifically the entanglement-dominated states defined in Sec. IV, we can saturate the bound given in Lemma 9.

Theorem 6 (Efficient bipartite entanglement distillation). For any bipartite (pure or mixed) state ρ, there exists a stabilizer LOCC protocol $\mathcal{S}(\rho) := (U_A \otimes U_B)\rho(U_A \otimes U_B)^\dagger$ comprised solely of two Clifford unitaries U_A and U_B such that

$$\mathcal{S}(\rho) = |\phi_+\rangle\langle\phi_+|^{\otimes M}_{A'\cup B'} \otimes \sigma_E; \tag{35}$$

where $M = \lfloor \mathcal{E}(\rho; A|B) - \nu/2 - S_0(\rho)/2 \rfloor$, A' and B' are size-M subsystems of A and B, respectively, and $E := \overline{A' \cup B'}$ is the rest of the system. Moreover, a depth $O(n)$ circuit representation of U_A and U_B can be found classically in $O(n^3)$ time.

Proof. For any bipartition $A|B$, the generators G for the stabilizer group of any state can always be brought into a "canonical form" $G = G_A \cup G_B \cup G_{AB}$. Following the analysis in Ref. [9], G_{AB} can be decomposed into pairs of generators with specific anticommutation relations. Specifically, for a stabilizer state, G_{AB} can be written as exactly $M = |G_{AB}|/2$ pairs $(g_k, \bar{g}_k)$ such that the projections of g_k and $\bar{g}_k$ onto A (or B) anticommute, while these projections commute with the projections of all other generators.

For a pure stabilizer state, Ref. [9] proves this structure stems from having exactly n generators, with no "unpaired" generators in G_{AB}. Our case requires a modified analysis since we consider both pure states ψ with stabilizer nullity ν (having $n - \nu$ generators) and mixed states ρ, whose purified states have at least $n + n^* - \nu$ generators, where $n^* = \lceil S_0(\rho) \rceil$. Using the tableau representation of the stabilizers and applying row reduction (as detailed in Ref. [9]'s Lemma 1), we can arrange the generators such that the first $2n^*$ generators act nontrivially on both the original and purifying systems, while the remaining $n - \nu - n^*$ generators act trivially on the purifying system. This structure guarantees at least $n - \nu - \lceil S_0(\rho) \rceil$ generators for ρ's stabilizer group.

Following the analysis framework of Ref. [9]'s Lemma 3, we can analyze the constraints on the generator structure. Let M be the number of paired generators $(g_k, \bar{g}_k)$ in G_{AB}, z be the number of unpaired generators, and $|S_A|$ and $|S_B|$ be the sizes of the local stabilizer groups. Then two inequalities must hold:

$$|S_A| + |S_B| + 2M + 2z \leq n \tag{36a}$$

$$|S_A| + |S_B| + 2M + z \geq n - \nu - \lceil S_0(\rho) \rceil, \tag{36b}$$

where the first follows from dimension constraints, and the second follows from counting generators. Subtracting these inequalities yields $z \leq \nu + \lceil S_0(\rho) \rceil$. Since

$\mathcal{E}(\rho;A|B) = |G_{AB}|/2 = M + z$, we can conclude there are at least $M \geq \mathcal{E}(\rho;A|B) - \nu/2 - \lceil S_0(\rho)\rceil/2$ paired generators in G_{AB}. As we show below, these paired generators directly enable entanglement distillation through local Clifford operations.

We label the paired stabilizers in S_{AB} as $(g[i], \bar{g}[i])$ for $i = 1, \ldots, M$. We will denote the restriction of $g[i]$ to subsystem A as $g_A[i]$, with analogous definitions for $g_B[i], \bar{g}_A[i], \bar{g}_B[i]$. It suffices to design our Cliffords U_A, U_B as follows. We require

$$U_A g_A[i] U_A^\dagger = X_{A_i}, \quad U_B g_B[i] U_B^\dagger = X_{B_i}, \tag{37a}$$

and for the other stabilizer in each pair, we require

$$U_A \bar{g}_A[i] U_A^\dagger = Z_{A_i}, \quad U_B \bar{g}_B[i] U_B^\dagger = Z_{B_i}, \tag{37b}$$

where A_i and B_i are the ith sites of A and B, respectively. This can always be done: the resulting stabilizers have the same commutation and anticommutation relations as the original stabilizers. That is, $g_A[i]$ anticommutes with $\bar{g}_A[i]$, but commutes with all other stabilizers. For convenience, let us abbreviate $U = U_A \otimes U_B$. Now, to see, for instance, that $X_{A_i}X_{B_i}$ stabilizes $\rho' := U\rho U^\dagger$, we observe that $X_{A_i}X_{B_i} = Ug[i]U^\dagger$. Therefore, using the fact that $g[i]\Pi = \Pi$:

$$\begin{aligned} X_{A_i}X_{B_i}\rho' &= (Ug[i]U^\dagger)(U\Pi H U^\dagger) \\ &= U(g[i]\Pi H)U^\dagger = U\Pi H U^\dagger = \rho'. \end{aligned} \tag{38}$$

A similar proof holds for $Z_{A_i}Z_{B_i}$. Finally, applying Lemma 10, we see that we have distilled a product of M Bell pairs on sites $(A_i \cup B_i)$, which are all unentangled with the rest of the system.

Finally, to see that these two unitaries U_A and U_B can be designed in $O(n^3)$ time, we use the fact that the constraints Eq. (37) can be satisfied using linear-algebraic manipulations (Gaussian elimination) using the tableau representation of the stabilizer generators. The efficiency of the two unitaries U_A and U_B follows from the fact that they are Clifford unitaries, which can all be implemented with $O(n^2/\log n)$ gates and depth $O(n)$ [16]. ■

Remark 2. Theorem 6 can be extended to a robust version using the ε-stabilizer nullity introduced in Sec. II A as follows: given a (possibly mixed) state ρ with ε-stabilizer nullity ν_ε, there exists a stabilizer protocol $\mathcal{S}$ that distills M Bell pairs, as in Eq. (35), with an error of (at most) ε in trace distance. This follows from the observation that the stabilizer protocol $\mathcal{S}$ in Theorem 6 consists only of bipartite Clifford unitaries, which preserve the trace distance between states.

For the sake of clarity, we now present an example of the application of Theorem 6 with a practical scenario.

Example 2 (Entanglement distillation via stabilizer LOCC). Consider a six-qubit state $|\psi\rangle$ with $\nu = 2$ shared between Alice (qubits 1–3) and Bob (qubits 4–6) with the following independent stabilizer generators (we omit to label the qubit where the Pauli operators act identically):

$$\begin{aligned} g_1 &= X_1X_2Z_4Y_5 \\ g_2 &= Y_1X_4Y_4 \\ g_3 &= Y_2Y_3X_5X_6 \\ g_4 &= X_2X_3Y_5Z_6. \end{aligned} \tag{39}$$

Observe that this state has $\mathcal{E}(\rho;A|B) = 2$, so Theorem 6 guarantees we can achieve $M = 1$. First, we identify which stabilizers can be paired. Although all stabilizers are nonlocal (acting nontrivially on both sides), not all can form Bell pairs. g_1 and g_2 have the correct anticommutation structure on both Alice and Bob's side to be transformed into a Bell pair, while g_3 and g_4 locally commute, hence cannot be transformed into a Bell pair. To distill the entanglement, we construct local Clifford unitaries U_A and U_B as follows. On Alice's side, U_A must transform X_1X_2 to X_1 and Y_1 to Z_1, while on Bob's side, U_B must transform Z_4Y_5 to X_4 and Y_4 to Z_4 (Fig. 6). After applying these unitaries, the first two stabilizers become

$$\begin{aligned} U_A g_1 U_A^\dagger \otimes U_B g_1 U_B^\dagger &= X_1X_4 \\ U_A g_2 U_A^\dagger \otimes U_B g_2 U_B^\dagger &= Z_1Z_4 \end{aligned} \tag{40}$$

while the remaining stabilizers transform into some new nonlocal stabilizers g_3', g_4' that cannot be paired to form additional Bell pairs. The final state takes the form

$$(U_A \otimes U_B)\,|\psi\rangle = \left|\phi^+\right\rangle_{1,4} \otimes \left|\psi'\right\rangle_{2,3,5,6}, \tag{41}$$

where $\left|\phi^+\right\rangle$ is a maximally entangled Bell pair and $\left|\psi'\right\rangle$ is a state stabilized by the transformed versions of g_3 and g_4.

This example demonstrates how our protocol can extract Bell pairs even when they are not manifest in the original stabilizer description, requiring nontrivial local Clifford operations on both sides. It also shows that not all nonlocal stabilizers can be paired to form Bell pairs—this

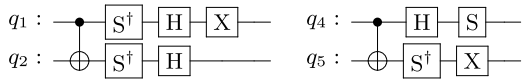


FIG. 6. The local Clifford circuits U_A (left) and U_B (right) that transform the nonlocal stabilizers (g_1, g_2) into a Bell pair, in Example 2. Note that Alice's circuit acts only on qubits 1 and 2, while Bob's circuit acts only on qubits 4 and 5. These circuits map the complicated nonlocal stabilizers to the simple form XX and ZZ across the bipartition, manifestly showing the distilled Bell pair.

is reflected in the fact that $M_+ = 1$ despite having four nonlocal stabilizers.

Theorem 6 demonstrates how the *bipartite* entanglement of a state can be distilled into Bell pairs. We can consider an even more ambitious task: distilling *multipartite* entanglement. That is, if we have a state that is split across $k > 2$ parties $A_1, A_2, \ldots, A_k$, can we distill some target k-partite entangled state using stabilizer LOCC? To begin answering this question, we first need to establish a meaningful measure of multipartite entanglement. In Ref. [9], a measure of multipartite entanglement for stabilizer states was proposed. For a k-partition $\{A_1, A_2, \ldots, A_k\}$, the authors first defined a local subgroup G_{loc}:

$$G_{\text{loc}} := \bigcup_{i=1}^{k} G_{A_i}, \tag{42}$$

which is the union of all stabilizers that act trivially on at least one of the partitions A_i. As this is an Abelian subgroup of $\mathbb{P}_n$, it has a generating set S_{loc} of size $|S_{\text{loc}}| = \log_2 |G_{\text{loc}}|$. The multipartite entanglement measure proposed by Ref. [9] was

$$\mathcal{E}(\psi; A_1|\ldots|A_k) = |S| - |S_{\text{loc}}|. \tag{43}$$

It was later shown that this entanglement measure has a convenient operational interpretation: it is exactly equal to the number of GHZ states distillable from the stabilizer state. Remarkably, we find that we can extend Eq. (43) to any state and achieve an almost identical operational interpretation. In this setting, we will assume that the input state has no stabilizers, which act entirely locally on any given partition A_i—this is called the assumption of "full local rank" in Ref. [70]. This is an easy assumption to satisfy in practice; if our state has any such local stabilizers, we can always preprocess our state by applying some local Clifford that maps these local stabilizers to a single-qubit Z. The state then separates into a tensor product $|0\rangle \otimes |\psi'\rangle$, so we trace this qubit out, as it cannot contribute to any form of entanglement distillation. We formalize these ideas in the following theorem.

Theorem 7 (Efficient multipartite entanglement distillation). Consider a k-partition $\{A_1, \ldots, A_k\}$ and a pure state $|\psi\rangle$ with stabilizers that act purely locally on any particular A_i. Then, there are k local Clifford unitaries $\{U_{A_i} \mid i = 1, \ldots, k\}$ such that $|\psi'\rangle := \bigotimes_{i=1}^{k} U_{A_i} |\psi\rangle$ is a state with at least

$$p := \mathcal{E}(\psi; A_1|\ldots|A_k) - \nu \tag{44}$$

copies of the k-partite GHZ states shared across all k parties. Furthermore, if $k = 3$ (the special case of a tripartition), $|\psi'\rangle$ can also be guaranteed to contain at least

(1) $(|S| - |S_A| - p)/2 - \nu$ shared Bell pairs between B and C,
(2) $(|S| - |S_B| - p)/2 - \nu$ shared Bell pairs between A and C, and
(3) $(|S| - |S_C| - p)/2 - \nu$ shared Bell pairs between A and B.

Proof. Let us first outline the strategy of the proof: for a given state $|\psi\rangle$, we will construct a pure (exact) stabilizer state $|\sigma\rangle$ that we term *stabilizer completion* of $|\psi\rangle$. We will then construct a set of k-local Clifford unitaries $\{U_{A_i} \mid i = 1, \ldots, k\}$ that are designed to distill multipartite entanglement from $|\sigma\rangle$. However, we will finally show that nevertheless, these unitaries work very well on $|\psi\rangle$ as well.

We first define the stabilizer completion. We take the stabilizer generators S of the state and find any Pauli P (which is not a stabilizer of $|\psi\rangle$), which simultaneously commutes with each of the generators, then add this Pauli to the stabilizer generators. This can be done efficiently using the tableau formalism: the kernel of $\mathcal{T}_S$ [see Eq. (4)] is equal to exactly all those Pauli operators, which commute with all of the generators in S; from there, it is easy to identify a Pauli, which is not in the rowspace of $\mathcal{T}_S$ (i.e., is independent of all the existing generators). This process of expanding the stabilizer generators S is repeated iteratively until we have constructed a full set of stabilizer generators T of size n (we note here that $S \subseteq T$ by construction). The new set of generators T corresponds to some pure stabilizer state $|\sigma\rangle$.

For pure stabilizer states, $\mathcal{E}(\sigma; A_1|\ldots|A_k) = |T| - |T_{\text{loc}}|$ GHZ states can be distilled from $|\sigma\rangle$ using some local Clifford unitaries $\{U_{A_1}, \ldots, U_{A_k}\}$ [70, Theorem 3]. In the process of building T up from S, we added at most $n - |S|$ stabilizers, which act trivially on at least one of the partitions A_i. That is, $|T_{\text{loc}}| \leq |S_{\text{loc}}| + (n - |S|)$. Also, since $|\sigma\rangle$ is a pure stabilizer state, $|T| = n$. Therefore, $\mathcal{E}(\sigma; A_1|\ldots|A_k) \geq n - (|S_{\text{loc}}| + (n - |S|)) = |S| - |S_{\text{loc}}| = \mathcal{E}(\psi; A_1|\ldots|A_k)$. Now, we ask what happens when we naively apply the local Clifford unitaries on the original state: $|\psi'\rangle := \bigotimes_{i=1}^{k} U_{A_i} |\psi\rangle$? These unitaries acted on the stabilizer completion $|\sigma\rangle$ by mapping k tuples of stabilizers $\{g_1, g_2, \ldots, g_k\} \subseteq T$ onto the k stabilizers for a k-partite GHZ state. However, we note that some of these stabilizers may be "phantom" stabilizers, as they may not have been part of the original set of stabilizers S (i.e., they were added to T in the stabilizer completion process). In this case, the desired GHZ state would not have been created, and some garbage may be output. However, there are at most ν "phantom" stabilizers. Therefore, at most ν GHZ states (that would have been created if $|\psi\rangle$ were equal to $|\sigma\rangle$) will not be distilled correctly. However, the remaining GHZ states are unaffected by these "defective" GHZ states, so we then arrive at the desired claim that at least $\mathcal{E}(\psi; A_1|\ldots|A_k) - \nu$ GHZ states have been distilled with these Clifford unitaries. The claim

about the simultaneous distillability of Bell pairs in the tripartite case follows identically from Ref. [70, Theorem 5], and applying our reasoning about "phantom" stabilizers (i.e., the ν "phantom" stabilizers destroy at most ν Bell pairs). ■

Hence, we conclude that ν-compressible states admit distillation protocols, either bipartite or multipartite, similar to the closely related stabilizer states in terms of Clifford operations. This is particularly appealing, as Clifford operations allow for efficient classical manipulation and can also be implemented fault tolerantly in many quantum error-correction codes, which admit *transversal* Clifford gates, such as the color code [15]. Furthermore, noting that the distillation protocol involves only multipartite Clifford unitaries, similar to Theorem 6, Theorem 7 can also be made robust using the ε-stabilizer nullity, as discussed in Remark 2.

2. Entanglement dilution

Let us now consider the converse task of entanglement distillation, known as entanglement dilution. The objective is to prepare an entangled target state $|\psi\rangle$ shared between two parties, Alice and Bob. Since this state is entangled, it cannot be prepared with LOCC alone: Alice and Bob must start with some amount of shared entanglement. Namely, we typically assume that they start with some number of Bell pairs shared between them. Since we will also restrict ourselves to stabilizer LOCC, we will also allow for some amount of magic resources in preparing the state. More precisely, we will allow one of the parties to hold a non-stabilizer state of a bounded size to start with. Given these constraints, entanglement dilution can be accomplished using a number of resources depending on the nullity ν of $|\psi\rangle$. The bare minimum amount of resources one might expect is that we require $\mathcal{E}(\psi; A|B)$ shared ebits (and, if we are unreasonably optimistic, *no* local magic states or classical communication). We will show that these optimistic expectations are only off target by $O(\nu)$.

Theorem 8 (Efficient entanglement dilution). If Bob has local access to a pure bipartite state $|\psi_{AB}\rangle$, then Alice and Bob can together prepare the state $|\psi_{AB}\rangle$ across the bipartition $A|B$ via a stabilizer LOCC protocol that uses at most

$$\begin{aligned} &\mathcal{E}(\psi; A|B) + \nu/2 \quad \text{shared ebits and} \\ &2\nu \quad \text{bits of classical communication.} \end{aligned} \tag{45}$$

Proof. Recall the output of the entanglement distillation protocol in Eq. (35):

$$\mathcal{S}(|\psi\rangle\langle\psi|) = |\phi_+\rangle\langle\phi_+|^{\otimes M}_{A'\cup B'} \otimes \sigma_E, \tag{46}$$

where $E = (A \setminus A') \cup (B \setminus B')$. We will define $A'' := A \setminus A'$ and $B'' := B \setminus B'$ for brevity. We observe that if we are able to prepare σ_E across A'' and B'', we will be able to prepare $|\psi\rangle$ by inverting $\mathcal{S}$, which comprises applying $U_A^\dagger$ and $U_B^\dagger$ locally on A and B, respectively.

First, we develop the structure of the state σ_E. σ_E must have at least $n - 2M - \nu$ stabilizers, and from Eq. (36), there are at most ν stabilizers that act nontrivially on A'' and B'' (these are the "unpaired" stabilizers in G_{AB} that were useless for entanglement dilution). Therefore, there are at least $n - M - 2\nu$ stabilizers, which act entirely locally on A'' or B''. Using nullity distillation (see Algorithm 1), we can find two Clifford unitaries $V_{A''}$ and $V_{B''}$ such that $(V_{A''} \otimes V_{B''})\,\sigma_E\,(V_{A''} \otimes V_{B''})^\dagger = |0\rangle\langle 0|^{\otimes(n-2M-2\nu)} \otimes \sigma'$, where σ' is a state spanning A and B on at most 2ν qubits.

Our entanglement dilution protocol is then as follows. Let us assume, without loss of generality, that $|\sigma'\rangle$ is supported on more qubits in B than in A (so, it acts on at most ν sites in A). Then, we first prepare $|\sigma'\rangle$ completely locally on B, by using the procedure shown in Fig. 7. Specifically, Bob uses his copy of $|\psi\rangle$ locally. Having prepared $|\psi\rangle$, Bob then executes the rest of the protocol entirely locally, which results in some number of local Bell pairs (which Bob can throw out) and some $|0\rangle$ states, which he also throws out. He is left with the state $|\sigma'\rangle$. Having prepared $|\sigma'\rangle$, B then sends A's portion of $|\sigma'\rangle$ using state teleportation, which costs at most ν ebits and 2ν bits of classical communication. Having received $|\sigma'\rangle$, and assuming A and B share M additional ebits, the nullity distillation (purple) and entanglement distillation (blue) unitaries in Fig. 7 can be inverted. With this, A and B now share $|\psi\rangle$. ■

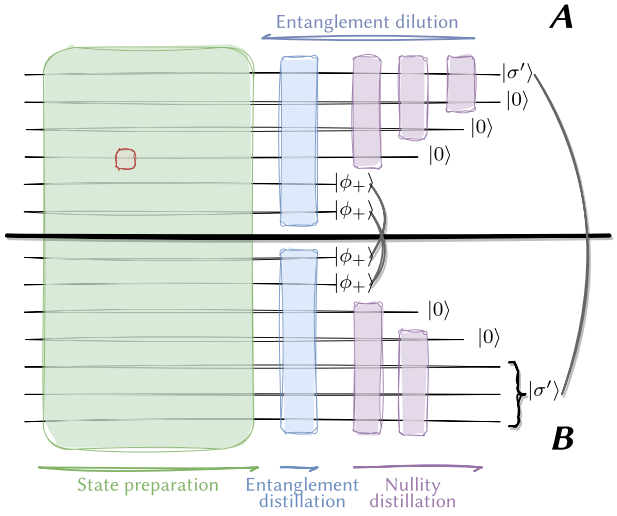


FIG. 7. State preparation (green), followed by entanglement distillation (blue), and finally "nullity distillation" (purple). The nullity distillation Clifford unitaries $V_{A''}, V_{B''}$ produce $|0\rangle$ states and a state $|\sigma'\rangle$ on at most 2ν qubits.

Remark 3. Theorem 8 can be extended to a robust version using the ε-stabilizer nullity introduced in Sec. II A as follows: given local access to a bipartite state $|\psi\rangle$ with ε-stabilizer nullity ν_ε, Alice and Bob can dilute $|\psi\rangle$ across the bipartition $A|B$ with an error of at most ε in trace distance. This result follows from the fact that the dilution protocol described in Theorem 8 employs a CPTP map to dilute the state, and by the data-processing inequality [15], the resulting state across the bipartition is at most ε-distant in trace distance from $|\psi\rangle$.

Similarly to what was done with Theorem 6, we now provide an example of Theorem 8.

Example 3 (Entanglement dilution with stabilizer LOCC). We now show how to prepare the six-qubit state $|\psi\rangle$ with $\nu = 2$ from Example 2 using entanglement dilution. Recall that $|\psi\rangle$ has stabilizer generators:

$$\begin{aligned} g_1 &= X_1X_2Z_4Y_5 \\ g_2 &= Y_1X_4Y_4 \\ g_3 &= Y_2Y_3X_5X_6 \\ g_4 &= X_2X_3Y_5Z_6. \end{aligned} \tag{47}$$

From Theorem 8, since $\mathcal{E}(\psi; A|B) = 2$ and $\nu = 2$, we know that we require three Bell pairs and 4 bits of classical communication. To see this concretely, we know that $|\psi\rangle$ can be transformed via local Clifford operations into $\left|\phi^+\right\rangle_{1,4} \otimes \left|\psi'\right\rangle_{2,3,5,6}$. Therefore, to prepare $|\psi\rangle$, Alice and Bob can do as follows:

(1) Share one Bell pair $\left|\phi^+\right\rangle$ between qubits 1 and 4.
(2) Have Bob locally prepare the state $\left|\psi'\right\rangle$ (which he can do since he holds a copy of $|\psi\rangle$) and share Alice's portion of $\left|\psi'\right\rangle$ using quantum teleportation.
(3) Then, apply $U_A^\dagger$ and $U_B^\dagger$ (as defined by Fig. 6) locally to reverse the distillation unitaries.

The protocol requires $M_- = 1$ Bell pairs for the first step, plus two Bell pairs to teleport Alice's portion of $\left|\psi'\right\rangle$ (since it acts on two qubits), for a total cost of $M_- = 3$ Bell pairs. The classical communication cost is 4 bits (two bits per qubit teleported).

This example demonstrates how entanglement dilution can be achieved by "running the distillation circuit backwards" after establishing the required entanglement resources via teleportation. Note that while the original state had only one distillable Bell pair, preparing it requires more ebits due to the need to share the state $\left|\psi'\right\rangle$ that could not be decomposed into Bell pairs.

B. Witnessing multipartite entanglement

An entanglement witness $\mathcal{W}$ is an observable defined with respect to a target state ψ. The purpose of a witness is to experimentally validate the presence of genuine entanglement in an imperfectly prepared version of the target state ψ, which we call ρ. Mathematically, an observable $\mathcal{W}$ serves as an entanglement witness with respect to a particular bipartition $A|B$ when

$$\mathrm{tr}(\mathcal{W}\rho) < 0 \implies \rho \notin \mathrm{SEP}(\mathscr{H}_A; \mathscr{H}_B), \tag{48}$$

where $\mathrm{SEP}(\mathscr{H}_A; \mathscr{H}_B)$ is the set of states that is separable with respect to the bipartition $A|B$. More precisely, $\mathrm{SEP}(\mathscr{H}_A; \mathscr{H}_B)$ is the convex hull of the set of unentangled pure states

$$\left\{|\psi_A\rangle \otimes |\psi_B\rangle \mid |\psi_A\rangle \in \mathscr{H}_A, |\psi_B\rangle \in \mathscr{H}_B\right\}. \tag{49}$$

A typical form for an entanglement witness is

$$\mathcal{W} = \alpha\mathbb{1} - |\psi\rangle\langle\psi| \quad \alpha := \max_{\rho \in \mathrm{SEP}(\mathscr{H}_A;\mathscr{H}_B)} \mathrm{tr}(\rho\,|\psi\rangle\langle\psi|). \tag{50}$$

This observable satisfies Eq. (48) by definition of α. The usefulness of this entanglement witness is measured by α: for instance, in the extreme case $\alpha = 1$, the value of $\mathrm{tr}(\mathcal{W}\rho)$ does not tell us anything about the entanglement of ρ, as $\mathrm{tr}(\mathcal{W}\rho)$ will then always be positive.

In this section, we will aim to develop an even more useful tool: a singular entanglement witness that simultaneously detects entanglement across a large number of bipartitions $\mathcal{B} = \{(A_i, B_i) \mid i = 1, \ldots, |\mathcal{B}|\}$. That is,

$$\mathrm{tr}(\mathcal{W}\rho) < 0 \implies \rho \notin \mathrm{SEP}(\mathscr{H}_{A_i}; \mathscr{H}_{B_i}) \quad \forall i = 1, \ldots, |\mathcal{B}|. \tag{51}$$

We call this a witness of genuine *multipartite* entanglement. In fact, we will construct something even stronger than this: we will define a class of entanglement witnesses that is not only able to rule out separability, but can rule out "approximate" separability. To be precise, for any fixed level of entanglement E, we define the set of E-entangled states $\mathrm{SEP}^{(E)}(\mathscr{H}_A; \mathscr{H}_B)$ as the convex hull of the set of pure states $\mathcal{X}_E := \left\{|\psi\rangle\langle\psi| \mid S_{1/2}(\psi_A) \leq E\right\}$, noting that $\mathrm{SEP}^{(E=0)}(\mathscr{H}_A; \mathscr{H}_B) = \mathrm{SEP}(\mathscr{H}_A; \mathscr{H}_B)$ recovers the original set of exactly separable states. Moreover, it holds that $\mathrm{SEP}^{(E)}(\mathscr{H}_A; \mathscr{H}_B) \supset \mathrm{SEP}(\mathscr{H}_A; \mathscr{H}_B)$. We will say that a state ρ exhibits genuine multipartite E entanglement with respect to a set of bipartitions $\mathcal{B}$ if $\rho \notin \mathrm{SEP}^{(E)}(\mathscr{H}_{A_i}; \mathscr{H}_{B_i})$ for all $i = 1, \ldots, |\mathcal{B}|$. The main result of this section will be a simple and efficient construction for a multipartite E-entanglement witness $\mathcal{W}^{(E)}$, satisfying

$$\mathrm{tr}(\mathcal{W}^{(E)}\rho) < 0 \implies \rho \notin \mathrm{SEP}^{(E)}(\mathscr{H}_{A_i}; \mathscr{H}_{B_i}) \quad \forall i = 1, \ldots, |\mathcal{B}|. \tag{52}$$

Again, we show that the stabilizer component Π of the target state ψ contains almost all the relevant information for

this multipartite entanglement detection. However, before this, we first need the following lemma.

Lemma 11. For a given bipartition $A|B$ where both halves of the system have size M, and for some level of entanglement $E \in [0, n]$,

$$\max_{\rho \in \mathrm{SEP}^{(E)}(\mathscr{H}_A;\mathscr{H}_B)} \mathrm{tr}(\rho\, |\phi_+\rangle\langle\phi_+|^{\otimes M}) \leq 2^{-M+E}, \tag{53}$$

where $|\phi_+\rangle$ is a Bell pair spanning A and B.

Proof. First, we observe that, since $\mathrm{tr}(\rho\, |\phi_+\rangle\langle\phi_+|^{\otimes M})$ is linear in ρ, we only need to maximize over states $|\psi_{AB}\rangle \in \mathcal{X}_E$, and $\mathrm{SEP}^{(E)}(\mathscr{H}_A;\mathscr{H}_B)$ is defined as the convex hull of $\mathcal{X}_E$. We first write the optimization over all states in $\mathcal{X}_E$ by doing a trivial transformation, wherein we maximize first over all reduced density matrices ψ_A that have bounded entropy, and then maximize over all possible purifications $|\psi_{AB}\rangle$ of that state.

$$\begin{aligned} &\max_{|\psi_{AB}\rangle \in \mathcal{X}_E} \left|\langle \psi_{AB}|\phi_+\rangle^{\otimes M}\right|^2 \\ &\quad = \max_{\substack{\psi_A \\ S_{1/2}(\psi_A)\leq E}} \max_{\substack{|\psi_{AB}\rangle \\ \mathrm{tr}_B(|\psi_{AB}\rangle\langle\psi_{AB}|)=\psi_A}} \left|\langle \psi_{AB}|\phi_+\rangle^{\otimes M}\right|^2. \end{aligned} \tag{54}$$

Note that according to Uhlmann's theorem [15], the inner optimization is known to be precisely equal to the fidelity:

$$\max_{\substack{|\psi_{AB}\rangle \\ \mathrm{tr}_B(|\psi_{AB}\rangle\langle\psi_{AB}|)=\psi_A}} \left|\langle \psi_{AB}|\phi_+\rangle^{\otimes M}\right|^2 = \mathcal{F}(\mathrm{tr}_B(|\phi_+\rangle\langle\phi_+|^{\otimes M}), \psi_A), \tag{55}$$

where $\mathcal{F}(\rho,\sigma) := \mathrm{tr}(\sqrt{\sqrt{\rho}\sigma\sqrt{\rho}})^2$ is the Uhlmann fidelity. Since $\mathrm{tr}_B(|\phi_+\rangle\langle\phi_+|^{\otimes M}) = 2^{-M}\mathbb{1}$,

$$\max_{|\psi_{AB}\rangle \in \mathcal{X}_E} \left|\langle \psi_{AB}|\phi_+\rangle^{\otimes M}\right|^2 = \max_{\substack{\psi_A \\ S_{1/2}(\psi_A)\leq E}} 2^{-M}\mathrm{tr}(\sqrt{\psi_A})^2. \tag{56}$$

Since $\mathrm{tr}(\sqrt{\psi_A})^2 = \exp\big(S_{1/2}(\psi_A)\big)$, the maximum possible value is at most 2^E by assumption that $S_{1/2}(\psi_A) \leq E$. Therefore, the overall maximum is bounded by 2^{-M+E}. This concludes the proof. ∎

Remark 4. The careful reader might ask whether similar bounds could be derived using the von Neumann entropy S_1 rather than the 1/2-Rényi entropy $S_{1/2}$. While S_1 is indeed the most natural choice for quantifying entanglement in pure states, our proof technique fundamentally relies on properties of the 1/2-Rényi entropy. Specifically, the key step in our proof uses Uhlmann's theorem to bound the maximum overlap between states, which directly relates to $S_{1/2}$ through the equation $\mathrm{tr}(\sqrt{\psi_A})^2 = \exp\big(S_{1/2}(\psi_A)\big)$. An analogous connection to S_1 is not known, and it remains an open question whether similar witnessing bounds could be established using S_1. Such a result would be particularly valuable as it would connect our witness more directly to the conventional theory of entanglement. However, this may require entirely new proof techniques beyond the scope of our current framework.

Theorem 9 (Multipartite E-entanglement witness). Consider a set of bipartitions $\mathcal{B} = \{(A_i, B_i) \mid i = 1, \ldots, |\mathcal{B}|\}$, a pure state ψ, and some level of entanglement E. Recall, from Theorem 3, that Π is the stabilizer projector of the target state ψ: $\Pi = \prod_{g_j \in S}((\mathbb{1} + g_j)/2)$, where S is a generating set for the stabilizer group of ψ. The following observable is a multipartite E-entanglement witness with respect to the bipartitions $\mathcal{B}$:

$$\begin{aligned} \mathcal{W}^{(E)} &= 2^{-M(\mathcal{B})+E}\mathbb{1} - \Pi \quad M(\mathcal{B}) : \\ &= \min_i \lfloor \mathcal{E}(\psi; A_i|B_i) - \nu/2 \rfloor, \end{aligned} \tag{57}$$

where $\mathcal{E}(\psi; A_i|B_i)$ is the stabilizer entanglement with respect to the bipartition $A_i|B_i$. When $E = 0$, we recover the conventional definition of a multipartite entanglement witness in Eq. (51).

Proof. Let us fix a particular bipartition $A|B$. To prove the statement, it suffices to show that

$$\max_{\rho \in \mathrm{SEP}^{(E)}(\mathscr{H}_A,\mathscr{H}_B)} \mathrm{tr}(\Pi\rho) \leq 2^{-M(\mathcal{B})+\lceil E \rceil}. \tag{58}$$

We can freely transform Π by a product of local unitaries without changing the maximum value of Eq. (58), as this is equivalent to transforming ρ by local unitaries [which leaves $\mathrm{SEP}^{(E)}(\mathscr{H}_A, \mathscr{H}_B)$ invariant], and the maximum is over all possible $\rho \in \mathrm{SEP}^{(E)}(\mathscr{H}_A, \mathscr{H}_B)$. Therefore, we use the machinery we developed in entanglement distillation, wherein we showed that Π is equivalent, under a local unitary $U_A \otimes U_B$, to a projector onto a stabilizer group with generators $S_A \cup S_B \cup S_{AB}$, see Sec. III A. Furthermore, the generators in S_{AB} contained at least $M(\mathcal{B})$ pairs $X_{A_i}X_{B_i}, Z_{A_i}, Z_{B_i}$. Denoting $\tilde{S}_{AB}$ to be the remaining unpaired generators in S_{AB}, we have

$$\begin{aligned} (U_A \otimes U_B)^\dagger \Pi (U_A \otimes U_B) = &\left[\prod_{g \in S_A \cup S_B \cup \tilde{S}_{AB}} \frac{\mathbb{1}+g}{2}\right] \\ &\times \left[\prod_{i=1}^{M(\mathcal{B})} \frac{\mathbb{1} + X_{A_i}X_{B_i}}{2}\right] \\ &\times \left[\prod_{i=1}^{M(\mathcal{B})} \frac{\mathbb{1} + Z_{A_i}Z_{B_i}}{2}\right]. \end{aligned} \tag{59}$$

Since for any two projectors Π_1 and Π_2, $\mathrm{tr}(\Pi_1\Pi_2\rho) \leq \mathrm{tr}(\Pi_2\rho)$, we simply drop the first projector (containing

terms in $S_A, S_B, \tilde{S}_{AB}$) and find

$$\begin{aligned}&\mathrm{tr}(\Pi\rho)\\ &\leq \max_{\rho} \mathrm{tr}\left(\left[\prod_{i=1}^{M(\mathcal{B})} \frac{\mathbb{1}+X_{A_i}X_{B_i}}{2}\right]\left[\prod_{i=1}^{M(\mathcal{B})} \frac{\mathbb{1}+Z_{A_i}Z_{B_i}}{2}\right]\rho\right),\end{aligned} \tag{60}$$

where the $\max_\rho$ is for $\rho \in \mathrm{SEP}^{(E)}(\mathscr{H}_A, \mathscr{H}_B)$. Denoting the first $M(\mathcal{B})$ sites of A as A' (with a similar definition for B'), we see that

$$\begin{aligned}&\left[\prod_{i=1}^{M(\mathcal{B})} \frac{\mathbb{1}+X_{A_i}X_{B_i}}{2}\right]\left[\prod_{i=1}^{M(\mathcal{B})} \frac{\mathbb{1}+Z_{A_i}Z_{B_i}}{2}\right]\\ &= |\phi_+\rangle\langle\phi_+|_{A'\cup B'}^{\otimes M(\mathcal{B})} \otimes \mathbb{1}_E,\end{aligned}$$

where ϕ_+ is a Bell pair spanning the bipartition $A|B$ and $E = \overline{(A'\cup B')}$. Putting all together, we finally obtain

$$\begin{aligned}\max_{\rho} \mathrm{tr}(\Pi\rho) &\leq \max_{\rho'\in \mathrm{SEP}^{(E)}(\mathscr{H}_{A'},\mathscr{H}_{B'})} \mathrm{tr}(|\phi_+\rangle\langle\phi_+|^{\otimes M(\mathcal{B})}\rho')\\ &\leq 2^{-M(\mathcal{B})+E},\end{aligned} \tag{61}$$

where the last bound follows from Lemma 11. This concludes the proof. ■

Corollary 2 (Robustness of multipartite entanglement). For any set of bipartitions $\mathcal{B}$ and any target pure state $|\psi\rangle\langle\psi|$, if an imperfectly prepared state ρ satisfies

$$\tfrac{1}{2}\|\rho - |\psi\rangle\langle\psi|\|_1 < 1 - 2^{-M(\mathcal{B})/2}, \tag{62}$$

then ρ exhibits genuine multipartite entanglement with respect to $\mathcal{B}$.

Proof. The condition Eq. (62) implies $\mathrm{tr}(\rho\,|\psi\rangle\langle\psi|) > 2^{-M(\mathcal{B})}$. It then suffices to show that

$$\mathrm{tr}(\rho\,|\psi\rangle\langle\psi|) > 2^{-M(\mathcal{B})} \implies \mathrm{tr}(\rho\Pi) > 2^{-M(\mathcal{B})} \tag{63}$$

as we showed that $\mathrm{tr}(\rho\Pi) \leq 2^{-M(\mathcal{B})}$ for all separable ρ. To show this, we show the contrapositive. Assume $\mathrm{tr}(\rho\Pi) \leq 2^{-M(\mathcal{B})}$. Defining $\rho' := \Pi\rho\Pi$, we see that $\mathrm{tr}(\rho') = \mathrm{tr}(\rho\Pi) \leq 2^{-M(\mathcal{B})} \implies \|\rho'\|_\infty \leq 2^{-M(\mathcal{B})}$, so $\mathrm{tr}(\rho\,|\psi\rangle\langle\psi|) = \mathrm{tr}(\rho'\,|\psi\rangle\langle\psi|) \leq \|\rho'\|_\infty \leq 2^{-M(\mathcal{B})}$. ■

Corollary 3 (Robustness of multipartite E entanglement). For any set of bipartitions $\mathcal{B}$ and any target pure state $|\psi\rangle\langle\psi|$, if an imperfectly prepared state ρ satisfies

$$\tfrac{1}{2}\|\rho - |\psi\rangle\langle\psi|\|_1 < 1 - 2^{(-M(\mathcal{B})+E)/2}, \tag{64}$$

then ρ exhibits genuine multipartite E entanglement with respect to $\mathcal{B}$.

Proof. The proof is identical to the one of Corollary 2 and follows entirely from Lemma 11 and Theorem 9. ■

Lemma 12 (Efficient measurement protocol). Assume we have the target pure state $|\psi\rangle\langle\psi|$. For any state ρ that is bounded in trace distance from $|\psi\rangle$ (i.e., $\frac{1}{2}\|\rho - |\psi\rangle\langle\psi|\|_1 \leq \epsilon$), and some maximum failure probability δ, the number of Pauli measurements N required to detect genuine multipartite entanglement with respect to the bipartitions $\mathcal{B}$ is given by

$$N = \left\lceil \frac{2\log(2/\delta)}{(1-(\epsilon+2^{-M(\mathcal{B})}))^2} \right\rceil. \tag{65}$$

Proof. Given the bound in trace distance

$$\tfrac{1}{2}\|\rho - |\psi\rangle\langle\psi|\|_1 \leq \epsilon, \tag{66}$$

we have that

$$|\mathrm{tr}(\mathcal{W}\,|\psi\rangle\langle\psi|) - \mathrm{tr}(\mathcal{W}\rho)| = |\mathrm{tr}(\Pi\,|\psi\rangle\langle\psi|) - \mathrm{tr}(\Pi\rho)| \leq \epsilon. \tag{67}$$

Since $\mathrm{tr}(\Pi\,|\psi\rangle\langle\psi|) = \mathrm{tr}(\Pi^2 H) = \mathrm{tr}(\Pi H) = \mathrm{tr}(|\psi\rangle\langle\psi|) = 1$, we must then have $\mathrm{tr}(\Pi\rho) \geq 1-\epsilon$. The required resolution Δ to which we need to measure the entanglement witness, in order to rule out biseparability across any of the bipartitions in $\mathcal{B}$, is $\Delta = 1 - (\epsilon + 2^{-M(\mathcal{B})})$.

To estimate $\mathrm{tr}(\Pi\rho)$, we use the fact that Π can be expanded as a linear combination $|S|^{-1}\sum_{P_j\in G} P_j$. We can construct an unbiased estimator $\tilde{\Pi}_j$ for $\mathrm{tr}(\Pi\rho)$ by measuring a random Pauli observable $P_j \in G$ for ρ. We note that randomly selecting $P_j \in G$ can be done by choosing a subset of generators from S, and multiplying them together. Equivalently, $P_j = \prod_{i=1}^{|S|} g_i^{x_i}$, where $x_i \sim$ Bernoulli $(1/2)$. Finally, since $-1 \leq \tilde{\Pi}_j \leq 1$, we can use Hoeffding's inequality to find that the estimator $\tilde{\Pi} := N^{-1}\sum_{j=1}^{N}\tilde{\Pi}_j$ obeys

$$\Pr\left[|\tilde{\Pi} - \mathrm{tr}(\Pi\rho)| \geq \Delta\right] \leq 2\exp\left(-\frac{N\Delta^2}{2}\right), \tag{68}$$

from which we can conclude that $N = \lceil 2\log(2/\delta)/\Delta^2\rceil$ samples suffices to limit the failure probability to δ. ■

IV. THE ENTANGLEMENT-DOMINATED PHASE VS. MAGIC-DOMINATED PHASE

We now apply all our findings to characterize the computational separation between entanglement-dominated states and magic-dominated states. To achieve this, let us examine the separation from the bottom up. The main results of the previous section (e.g., Theorems 4, 6 to 7 and 9) have a common formal similarity: for a broad

class of entanglement tasks, the stabilizer entanglement $\mathcal{E}$ can be used to replace the true entanglement entropy S_α, up to an additive correction $\propto \nu$. This simple fact proves to be useful, as everything about the stabilizer entanglement (through the generating set S_{AB}) can be computed and manipulated using a polynomial amount of classical or quantum resources. This, however, entails the drawback of introducing an error or inefficiency (for entanglement characterization or manipulation, respectively) of size $O(\nu)$. This led us to draw a distinction between the two classes, or phases, of states, ED and MD, defined in Definition 1. We find an operational distinction between these two phases. As we will show in this section, if we restrict ourselves to stabilizer protocols, we can achieve optimal performance for ED states on a number of entanglement related tasks. Conversely, there are clear counterexamples that prove efficient LOCC protocols are completely ineffective for entanglement related tasks on MD states. However, we are only able to exhibit counterexamples in the magic-dominated phase for any $\nu = \omega(\log n)$. For MD states with $\nu = O(\log n)$ or $S_1 = O(\log n)$, we are unable to rule out the possibility of efficient protocols for the simple reason that these states are both efficiently learnable and classically simulable. Indeed, it is reasonable to expect that the entanglement in these states can be manipulated efficiently by brute force inspection of the classical description of the state. To begin with, let us provide a characterization of the entanglement-dominated phase in terms of the stabilizer entanglement (see Definition 7), which proves to be instrumental in the efficient detection of ED states, as demonstrated later in Theorem 13.

Corollary 4 (Characterization of the ED phase). Let $A|B$ be a bipartition. A pure state $|\psi\rangle$ is entanglement dominated if and only if $\mathcal{E}(\psi; A|B) = \omega(\nu)$.

Proof. Recalling Definition 1, ED states are defined by the condition $S_1(\psi_A) = \omega(\nu)$. Then, from Theorem 4, it is immediate to see that an analogous condition is $\mathcal{E}(\psi; A|B) = \omega(\nu)$. ■

To appreciate the significance of the ED phase, we start with an example that illustrates the robustness of entanglement within the entanglement-dominated phase.

Example 4 (Robustness of entanglement for entanglement-dominated states). Consider a volume-law ED state $|\psi\rangle$. Let us fix a cut $A|B$ with $n_A, n_B = \Omega(n)$. By definition, this means that for some constant $1 \geq \kappa > 0$ and a large enough n, $S_1(\psi_A) \geq \kappa n$. For any (possibly mixed) state ρ, if

$$\frac{1}{2}\|\rho - |\psi\rangle\langle\psi|\,\psi\|_1 < 1 - o\left(\frac{1}{\text{poly}\, n}\right) \implies \rho \notin \text{SEP}^{(E)}(\mathscr{H}_A; \mathscr{H}_B), \quad (69)$$

for any $E = \kappa n - o(n)$. This is to say, ρ must also be a volume-law entangled state.

Proof. The proof is a straightforward consequence of Corollary 3. Since $\mathcal{E}(\psi; A|B) \geq \kappa n - \nu/2$, so $M \geq \kappa n - \nu$. Let δ be any function satisfying $\nu + \omega(\log n) < \delta < o(n)$ [since $\nu = o(n)$, this is always possible]. Then, choosing $E = \kappa n - \delta$, we then have $M - E > \omega(\log n)$. In virtue of Corollary 3, it follows that if

$$\frac{1}{2}\|\rho - |\psi\rangle\langle\psi|\,\psi\|_1 < 1 - o\left(\frac{1}{\text{poly}\, n}\right) \quad (70)$$

then $\rho \notin \text{SEP}^{(E)}(\mathscr{H}_A; \mathscr{H}_B)$ with $E = \kappa n - o(n)$, which concludes the proof. ■

In practice, this example means that for any convex decomposition of $\rho = \sum_i p_i |\phi_i\rangle\langle\phi_i|$, there exists at least one state $|\phi_i\rangle$ whose reduced density matrix $\phi_{i,A}$ satisfies $S_{1/2}(\phi_{i,A}) \geq \kappa n - o(n)$. In other words, volume-law entanglement-dominated states feature robust bipartite entanglement, as one can draw a ball of *almost maximal radius* (i.e., close to 1 up to a quickly decaying factor) around $|\psi\rangle$, inside which every state has volume-law entanglement.

In what follows, we will show that the entanglement-related tasks analyzed in this paper can be solved efficiently and are nearly optimal for any state in the ED phase. Given the desirable features of these ED states, we anticipate that one might wonder whether it is possible to detect whether a particular state is ED or not. We resolve this question later in Theorem 13, where we show that indeed we can verify whether an unknown state is in the ED phase or not.

A. Entanglement estimation in the ED phase

A common task in entanglement characterization is checking the scaling of a state's entanglement. For instance, volume-law states are characterized by an entanglement that scales with $\Theta(n_A)$. This task of *entanglement characterization* can then be formalized as follows. Fix a bipartition $A|B$, and consider an entanglement class $f(n_A)$. Given a state $|\psi\rangle$, the aim is to check whether $S_\alpha(\psi_A) = \Theta(f(n_A))$. In general, achieving this efficiently is not possible. Consider for instance setting $\alpha = 2$ and $f(n_A) \sim n_A$ (which means we are checking whether a state is volume-law entangled or not). Without any prior knowledge about the state ψ, this necessitates checking whether $\text{tr}(\psi_A^2) = \exp(-\Omega(n_A))$. Although $\text{tr}(\psi_A^2)$ can be estimated using just two copies of ψ and a simple swap test, exponentially many (in n_A) repeated measurements are necessary to resolve $\text{tr}(\psi_A^2)$ to the required level of accuracy to determine whether it is exponentially small. For anything but a small bipartition n_A, this quickly becomes infeasible. In

contrast, we show below that for entanglement-dominated states, the task of entanglement characterization is easy.

Lemma 13 (Efficient entanglement characterization). Let us fix a bipartition $A|B$, and take an entanglement class $f(n_A)$. For any ED state ψ, we can check whether $S_\alpha(\psi_A) = \Theta(f(n_A))$ using polynomially many copies of ψ and polynomial classical computing resources. Moreover, if indeed $S_\alpha(\psi_A) = cf(n_A) + o(f(n_A))$ for some c, we can estimate the coefficient of proportionality c with an asymptotically vanishing error as $n_A \to \infty$.

Proof. The proof follows by combining two facts. First, the stabilizer entanglement $\mathcal{E}(\psi; A|B)$ is measurable with polynomially many copies of ψ and polynomial classical computing resources (see Lemma 16). Second, by Theorem 4, the difference between stabilizer entanglement and entanglement entropy is at most $O(\nu)$. Since $\mathcal{E}(\psi; A|B) = \omega(\nu)$ by assumption, see Definition 1, we have that $S_\alpha(\psi_A) = \Theta(f(n_A)) \iff \mathcal{E}(\psi; A|B) = \Theta(f(n_A))$. That is, $\mathcal{E}(\psi; A|B) = \Theta(f(n_A))$ is both necessary and sufficient to conclude that $S_\alpha(\psi_A) = \Theta(f(n_A))$ as well.

To show the second claim, we again apply Theorem 4 to conclude that

$$\left| \frac{S_\alpha(\psi_A)}{f(n_A)} - \frac{\mathcal{E}(\psi; A|B)}{f(n_A)} \right| = \frac{o(f(n_A))}{f(n_A)} = o(1), \tag{71}$$

so that $\mathcal{E}(\psi; A|B)/f(n_A)$ serves as an estimate for c with asymptotically vanishing error. ■

An immediate corollary is that we can distinguish between volume-law entanglement and subvolume-law entanglement. We just set $f(n_A) = n_A$ and calculate c using the machinery of the proof of Lemma 13. When $\nu = o(n_A)$, we will have that $c = \Theta(1)$ for volume-law states and $c = o(1)$ for subvolume-law states. This sharp divide allows us to determine whether a state is volume law or not.

A simpler entanglement characterization task is witnessing entanglement. In the ED phase, the entanglement witnessing becomes effective and efficient. This efficiency stems from Theorem 9, and it can be applied to the following tasks: bipartite entanglement witnessing, bipartite E-entanglement witnessing, multipartite entanglement witnessing, and multipartite E-entanglement witnessing. In all these tasks, as shown in Lemma 12, the entanglement can be witnessed by measuring the projector Π associated with the stabilizer group G of the state $|\psi\rangle$. Indeed, for some state ρ that is an imperfectly prepared version of the state $|\psi\rangle$, it is sufficient to measure the overlap between Π and the state ρ. If this overlap is larger than the number $2^{-M(\mathcal{B})+E}$, then we can conclude the state ρ must have entanglement at least $S_{1/2} \geq E$. Here, $M(\mathcal{B}) := \min_i[\mathcal{E}(\psi; A_i|B_i) - \nu/2]$ encodes the entanglement of $|\psi\rangle$ on the (possibly multipartite) partition $\mathcal{B} = \{(A_i, B_i) \mid i = 1, \ldots, |\mathcal{B}|\}$ and E is the level of entanglement of the set of states within $\mathrm{SEP}^{(E)}$ defined in Sec. III B.

In the ED phase, the entanglement witness is useful, and it is worth giving an explicit example below. Although we focus on the case of bipartite entanglement, the generalization to the multipartite case is straightforward.

Example 5. Consider a bipartition $A|B$, and let $|\psi\rangle$ be an ED state. By Definition 1, one thus has $\mathcal{E}(\psi; A|B) = \omega(\nu)$ and therefore $M(\mathcal{B}) = \mathcal{E}(\psi; A|B)(1 - o(1)) \geq \mathcal{E}(\psi; A|B)/2$. Since estimating the overlap $\mathrm{tr}(\Pi\rho)$ can be done efficiently according to Lemma 12, one can efficiently witness any level of entanglement E up to a threshold determined by the stabilizer entanglement $\mathcal{E}(\psi; A|B)$. In formula,

$$\mathrm{tr}(\Pi\rho) > 2^{E - \mathcal{E}(\psi; A|B)/2} \implies \rho \notin \mathrm{SEP}^{(E)}(\mathscr{H}_A; \mathscr{H}_B). \tag{72}$$

In other words, if the stabilizer entanglement $\mathcal{E}(\psi; A|B)$ is known, depending on the measured value of $\mathrm{tr}(\Pi\rho)$, one can rule out the possibility that ρ has entanglement (as measured by $S_{1/2}$) less than E. The multipartite case follows easily by applying identical reasoning with $\min_i \mathcal{E}(\psi; A_i|B_i)$, that is, the minimum stabilizer entanglement across each cut $A_i|B_i$.

Despite our positive results for entanglement witnessing in the ED phase, unlike other entanglement-related tasks, we cannot prove any general hardness results for the MD phase. With that being said, note that in the MD phase, the entanglement witness presented in Sec. III B becomes completely ineffective: $\mathcal{M}(\mathcal{B}) \leq 0$ by definition of MD states. For any level of entanglement E, the relation $\mathrm{tr}(\Pi\rho) \leq 1$ is always trivially satisfied and does not carry any useful information.

B. Entanglement manipulation in the entanglement-dominated phase

Similar to entanglement characterization, entanglement manipulation is effective for ED states.

Lemma 14 (Optimal distillation). For any state $|\psi\rangle$ in the ED phase, there exists a one-shot and zero-error stabilizer LOCC protocol that distills a number M_+ of Bell pairs that satisfies

$$\frac{M_+}{S_1(\psi_A)} = 1 - o(1). \tag{73}$$

Proof. We have already shown in Theorem 6 that there always exists a stabilizer protocol that distills $M_+ = \lfloor \mathcal{E}(\psi; A|B) - \nu/2 \rfloor$ Bell pairs from any state. Using Theorem 4, we see that $M_+ \geq \lfloor S_1(\psi_A) - 2\nu \rfloor \geq S_1(\psi_A) -$

$2\nu - 1$. Since $\nu/S_1(\psi_A) = o(1)$ by definition for ED states, we have

$$\frac{M_+}{S_1(\psi_A)} = 1 - o(1) \tag{74}$$

in the ED phase. ■

One can show a simple converse claim for Lemma 14. Namely, there exist MD states $|\psi\rangle$ for which no stabilizer protocols can distill *any* Bell pairs. This begins to suggest a computational phase transition between ED and MD states, but still relies on the restriction to *stabilizer* LOCC protocols. Later, in Theorem 10, we show a stronger converse result that holds for *all* LOCC protocols. For now, the proof idea for the simpler claim is as follows: if the circuit used to prepare $|\psi\rangle$ uses non-Clifford resources, which destroy all the stabilizers in S_{AB}, no entanglement can be distilled via stabilizer LOCC. Concretely, consider a circuit where a Haar random unitary acts on $\min(n_A, n_B) < s < n$ qubits on both sides of the bipartition. This is an ensemble consisting almost entirely of MD states, as $\nu = \Theta(s)$ for all but a measure-zero set of states in this ensemble, while the entanglement is also $\Theta(s)$ with probability $1 - o(1)$. Also, $|S_{AB}| = 0$ for all but a measure-zero set of states ψ in this ensemble, while $S_1(\psi_A) \geq 1$ asymptotically almost surely. This is all to say, that there must exist some MD state $|\psi\rangle$ in the ensemble with $|S_{AB}| = 0$ but $S_1(\psi_A) \geq 1$. Since no stabilizer protocol can distill more than $|S_{AB}|/2$ Bell pairs from $|\psi\rangle$ (Lemma 9), there are no distillable Bell pairs from $|\psi\rangle$.

The following lemma examines the performance of efficient stabilizer protocols in the entanglement dilution task for ED states.

Lemma 15 (Optimal dilution). For any state $|\psi\rangle$ in the ED phase, there exists a stabilizer LOCC for dilution that achieves

$$\frac{M_-}{S_1(\psi_A)} = 1 + o(1), \tag{75}$$

where M_- is the required number of Bell pairs. Furthermore, the required number of bits of classical communication N_{CC} and local magic resource states N_{magic} both satisfy

$$\frac{N_{\mathrm{CC}}}{S_1(\psi_A)}, \quad \frac{N_{\mathrm{magic}}}{S_1(\psi_A)} = o(1). \tag{76}$$

Proof. Similarly to Theorem 6, the proof follows from Theorem 8 and the definition of ED states (Definition 1). ■

C. Hardness results in the magic-dominated phase

While in previous sections, we showed the feasibility of several entanglement tasks in the ED phase, this section shows the converse: namely, entanglement tasks become hard in the MD phase. Specifically, we show that there is no sample-efficient algorithm for entanglement estimation in the MD phase, and furthermore that there is no distillation protocol that can distill even a constant fraction of the entanglement into Bell pairs. Remarkably, while for the ED phase, one-shot and zero-error stabilizer LOCC protocols have near-optimal performance, below we rule out any sample-efficient LOCC protocol for the MD phase, even those that are allowed multiple rounds of classical communication or multiple copies of the input state. These findings suggest an information-theoretic separation between the two phases, which we further speculate upon in Sec. IV D. We note that states with $\nu = O(\log n)$ can be learned efficiently—see Refs. [47,48]—and therefore the computation of the entanglement entropy for these states is sample efficient. This means that our hardness results, while they rule out entanglement manipulation protocols that work well on *all* MD states, and more specifically for those with stabilizer nullity ν for any $\nu = \omega(\log n)$, do not rule out efficient protocols, which are specifically tailored to work on the class of MD states with $\nu = O(\log n)$. Indeed, a sample-efficient distillation protocol for states with $\nu = O(\log n)$ involves (1) learning a tomographic description of the state, and (2) applying the distillation protocol introduced by Bennett in Ref. [20], which assumes knowledge of the Schmidt basis. With these clarifications in place, we now proceed to formalize our hardness result.

Theorem 10 (Hardness of entanglement tasks for the magic-dominated phase). Let $A|B$ be any bipartition satisfying $n_A, n_B = \omega(\log n)$. Let $|\psi\rangle$ a magic-dominated state satisfying $S_1(\psi_A) = \omega(\log n)$. Then any sample-efficient protocol

(1) estimates $S_1(\psi_A)$ to within a relative error at least $\omega(1)$;
(2) distills at most a vanishing fraction $M_+/S_1(\psi_A) = o(1)$ of Bell pairs;
(3) dilutes the state $|\psi\rangle$ across the bipartition $A|B$, using at least $M_-/S_1(\psi_A) = \omega(1)$ Bell pairs.

Moreover, these no-go results hold even if the protocols are guaranteed *a priori* that the input MD states all obey $\nu = \Theta(f(n))$, for any $f(n) = \omega(\log n)$.

Proof. Let us set $\nu = \Theta(\log^c(n))$ for $c > 1$. Define the following families of states:

$$\mathcal{E}_{\mathrm{Haar}(\nu)} = \left\{ |\phi_{\mathrm{Haar}}\rangle \otimes |0\rangle^{\otimes(n-\nu)} \mid |\phi_{\mathrm{Haar}}\rangle \in \mathrm{Haar}_\nu \right\} \tag{77}$$

where Haar_ν is the ν-qubit Haar ensemble. Similarly, we define

$$\mathcal{E}_{f,S,\nu} = \left\{ \left|\phi_{f,S}\right\rangle \otimes |0\rangle^{\otimes(n-\nu)} \mid \left|\phi_{f,S}\right\rangle \in \mathcal{E}_{f,S}(\nu) \right\}, \tag{78}$$

where $\mathcal{E}_{f,S}(\nu)$ is the ensemble of subset phase states defined on ν qubits, with f being a random boolean function and S being a random subset of $\{0,1\}^\nu$ [11]. Let us set $|S| = \Theta(\exp \log^{c'}(n))$ with $1 < c' < c$.

Let us assume, without loss of generality, that our bipartition $A|B$ has A being the first $\nu/2$ qubits and B being the rest. Note that $|\psi\rangle \in \mathcal{E}_{\mathrm{Haar}(\nu)}$ has $S_1(\psi_A) = \Theta(\nu) = \Theta(\log^c(n))$ up to negligible failure probability, while $|\psi\rangle \in \mathcal{E}_{f,S,\nu}$ obeys $S_1(\psi_A) = \Theta(\log|S|) = \Theta(\log^{c'}(n))$ up to negligible failure probability. We also note that the states in both ensembles $\mathcal{E}_{\mathrm{Haar}(\nu)}$ and $\mathcal{E}_{f,S,\nu}$ are MD—that is, these states obey $S_1(\psi_A) = O(\nu)$ up to a negligible failure probability [35]. Finally, denoting $\Phi_{\mathcal{E}}^{(m)} := \mathbb{E}_{|\psi\rangle\sim\mathcal{E}} |\psi\rangle\langle\psi|^{\otimes m}$ with $\mathcal{E} \in \{\mathcal{E}_{f,S,\nu}, \mathcal{E}_{\mathrm{Haar}(\nu)}\}$, as shown in Ref. [11], we have

$$\left\| \Phi_{\mathcal{E}_{f,S,\nu}} - \Phi_{\mathcal{E}_{\mathrm{Haar}(\nu)}} \right\|_1 \leq o\left(\frac{1}{\mathrm{poly}\, n}\right) \quad \forall m = O(\mathrm{poly}\, n), \tag{79}$$

where we used that $|S| = \Theta(\exp\{\mathrm{poly}\}\log n)$. This is all to say that the two ensembles are statistically indistinguishable, meaning that any quantum algorithm that uses a polynomial number of copies of $|\psi\rangle$ cannot distinguish whether a given state belongs to $\mathcal{E}_{\mathrm{Haar}(\nu)}$ or $\mathcal{E}_{f,S,\nu}$.

(1) We first show that there cannot be any sample efficient algorithm estimating $S_1(\psi_A)$ up to a relative error $O(1)$. Assume, for the sake of contradiction, that there does exist such an algorithm. However, this would enable one to distinguish between states drawn from the two ensembles $\mathcal{E}_{f,S,\nu}$ and $\mathcal{E}_{\mathrm{Haar}(\nu)}$ as follows. We would apply the algorithm to estimate the entanglement of the given state, and if the estimated entanglement were $\Theta(\log^{c'} n)$, we could conclude the state belonged to $\mathcal{E}_{f,S,\nu}$ [since the relative error is $O(1)$] and otherwise infer it belonged to $\mathcal{E}_{\mathrm{Haar}(\nu)}$.

(2) We now show that there is no sample-efficient (hence no efficient) algorithm that distills more than M Bell pairs obeying $M/S_1(\psi_A) = \Omega(1)$. Assume for the sake of contradiction that such an algorithm exists. One could use such an algorithm as a distinguisher, by counting the number of Bell pairs, as for any $|\phi\rangle \in \mathcal{E}_{f,S,\nu}$, this distiller must produce $\Theta(\log^{c'} n)$ Bell pairs, while for $|\psi\rangle \in \mathcal{E}_{\mathrm{Haar}(\nu)}$, it must produce $\Theta(\log^c n)$ Bell pairs. Given that $c' < c$, the distinguisher can actually discern between the two ensembles, thus we have arrived at a contradiction. Therefore, the maximal number M of distillable Bell pairs obeys $M/S_1 = o(1)$. In the extreme case when $\nu = n$, this inequality grows to $M/S_1 = O(\log^c n/n)$ for any $c > 1$.

(3) Finally, to show that any entanglement dilution protocol requires $M/S_1 = \omega(1)$ Bell pairs, assume for the sake of contradiction that there exists a LOCC dilution protocol that requires $M_-/S_1 = O(1)$ Bell pairs to prepare a general MD state ψ. This implies that, if Alice is given access to polynomial copies of a pseudorandom state $\psi \sim \mathcal{E}_{f,S}$ [with $\nu = n$ for simplicity—the general case of $\nu = \Theta(f(n))$ follows similarly], Alice and Bob can together prepare ψ across the bipartition $A|B$ using $M = O(S_1(\psi)) = O(\mathrm{poly}\log n)$ Bell pairs. Similarly, if Alice is given access to polynomial copies of a Haar random state $\psi \sim \mathcal{E}_{\mathrm{Haar}}$, Alice and Bob can prepare ψ across the bipartition $A|B$ using $M = \Theta(n)$ Bell pairs [the lower bound is because ψ has $\Omega(n)$ entanglement with overwhelming probability]. However, this leads to a contradiction; we can construct a distinguisher between $\mathcal{E}_{f,S}$ and $\mathcal{E}_{\mathrm{Haar}}$ as follows. We run the dilution protocol, supplied with $\Theta(n^2)$ Bell pairs. If the dilution protocol is given a pseudorandom state ψ (or polynomially copies thereof), only $O(\mathrm{poly}\log n)$ of the Bell pairs should be consumed, while if it is given a Haar random state, $\Theta(n)$ Bell pairs should be consumed. The number of consumed Bell pairs at the end of the protocol can be found using a simple swap test on each of the supplied Bell pairs. This would then allow us to distinguish whether a pseudorandom state or Haar random state was input, leading to a contradiction.

■

As a consequence of Theorem 10, we can give strong bounds for general entanglement dilution protocols, analogous to the bounds for entanglement distillation given in Ref. [11].

Corollary 5 (No-go for efficient entanglement dilution). Given local query access to polynomially many copies of an unknown state $|\psi\rangle$, the minimum number of Bell pairs needed to dilute $|\psi\rangle$ across $A|B$ is $M_- = \Omega(\exp(S_1(\psi_A)^{1/c}))$ for any $c > 1$.

Proof. In the extreme case of the proof of Item 3 in Theorem 10, when we set $\nu = n$, we must have $M_- = \Theta(n)$, while $S_1(\psi_A) = \Theta(\log^c n)$ with $c > 1$ for pseudoentangled states, so $M_- = \Omega(\exp(S_1(\psi_A)^{1/c}))$ for $c > 1$. ■

D. A computational phase transition between entanglement- and magic-dominated phases

At this stage, we can synthesize our findings to clarify the operational distinction between ED and MD states.

We have shown in Lemmas 13 and 14 that entanglement characterization and distillation are efficient for ED states, while we show the converse in Theorem 10. These two results are summarized in Sec. I A. Another way of understanding this computational phase transition is through the lens of entanglement reversibility. Entanglement distillation allows us to transform (multiple copies of) a state ρ to a number M_+ of Bell pairs, and entanglement dilution allows us to use a number of Bell pairs to prepare the state ρ via LOCC. In particular, the minimum amount of Bell pairs M_- necessary to distill ρ is named as *entanglement cost*. The ratio M_+/M_- quantifies the *reversibility* of entanglement for the state ρ. For pure states, without any restriction to efficient (samplewise and computationally) LOCC protocols, it has been shown that entanglement is always reversible, meaning that the ratio is always one [20]. However, when considering efficient and state-agnostic LOCC protocols, it has been demonstrated in Ref. [11] that the maximal distillable entanglement is in general $M_+ = O(\log^2 S_1(\psi_A))$. Furthermore, in Corollary 5, we demonstrated that $M_- = \Omega\left(\exp\left(S_1(\psi_A)^{1/2}\right)\right)$ Bell pairs are necessary for every efficient state-agnostic dilution protocol. Combining these two results, it follows that, for the efficient and state-agnostic LOCC protocol, the reversibility ratio M_+/M_- for pure states is always asymptotically vanishing in n being $M_+/M_- = o(1)$ [for every bipartition $A|B$ with $\omega(\log n) = n_A, n_B$]. This shows that, even for pure states, entanglement manipulation is not computationally reversible when no further information about the state is available. However, given partial information about the state, one might wonder if entanglement reversibility is somewhat restored. For instance, in the case of ED states, how does the reversibility ratio behave? In stabilizer states, the reversibility ratio is always one [9]: since the entanglement spectrum of stabilizer states is flat, essentially the same one-shot stabilizer LOCC protocol can be used for both distillation and dilution. Below, we put together all our findings and prove that the separation between entanglement- and magic-dominated phases marks a sharp phase transition in entanglement reversibility under efficient and state-agnostic LOCC protocols.

Corollary 6 (Phase transition in entanglement reversibility). Given an unknown state $|\psi\rangle$, let M_+ be the number of Bell pairs distillable by an efficient LOCC protocol, and let M_- be the number of Bell pairs necessary to prepare $|\psi\rangle$ via LOCC. For any ED $|\psi\rangle$,

$$\frac{M_+}{M_-} \geq 1 - o(1), \tag{80}$$

while if $|\psi\rangle$ is MD, no efficient LOCC protocol can do better than

$$\frac{M_+}{M_-} \leq o(1). \tag{81}$$

Proof. From Theorem 6, we know that there exists a state-agnostic one-shot protocol that distills a number M_+ of Bell pairs given by $M_+ \geq S_1(\psi; A|B) - \nu/2$. Conversely, from Theorem 8 we know that $M_- \leq \mathcal{E}(\psi; A|B) + \nu/2$ Bell pairs are sufficient to prepare $|\psi\rangle$ via LOCC. If $|\psi\rangle$ is ED, then

$$\frac{M_+}{M_-} \geq \frac{S_1(\psi_A) - \nu/2}{\mathcal{E}(\psi; A|B) + \nu/2} \geq \frac{S_1(\psi_A) - \nu/2}{S_1(\psi_A) + 2\nu} = 1 - o(1). \tag{82}$$

Conversely, let $|\psi\rangle$ be a MD state. We know that $M_- \geq S_1(\psi; A|B)$ for any dilution protocol [20], and from Theorem 10 we know that for any state-agnostic efficient protocol $M_+/S_1(\psi_A) = o(1)$. Therefore,

$$\frac{M_+}{M_-} \leq \frac{M_+}{S_1(\psi_A)} = o(1), \tag{83}$$

which concludes the proof. ■

Before concluding this section, we mention a recent result that proves the impossibility of reversible entanglement manipulation for general mixed states [71], even under a class of entanglement-free operations that strictly generalize LOCC, and if we do not insist upon state-agnostic or efficient protocols.

E. Robustness of ED-MD phases

In the previous section, we observed the remarkable distinctions between entanglement-dominated and magic-dominated phases. These phases were defined based on the relationship between entanglement entropy and stabilizer nullity. Given the sensitive nature of stabilizer nullity, it is natural to ask whether these phases are robust under perturbations. As discussed in Sec. II A, stabilizer nullity can be extended to a robust form via the smoothed stabilizer nullity (see Definition 5). In this section, we demonstrate that this phase characterization is indeed robust. We begin by extending Definition 1 to define the entanglement-dominated (ED) and magic-dominated (MD) phases in this context. We then proceed to establish the robust versions of Theorems 1 and 2.

Definition 9 (ε-ED and ε-MD phases). Let $|\psi\rangle$ be a pure quantum state and $\varepsilon > 0$. Define

$$|\psi_\varepsilon\rangle := \operatorname{argmin}\{\nu(|\phi\rangle) \,:\, \| \, |\psi\rangle - |\phi\rangle \, \|_1 \leq \varepsilon\}. \tag{84}$$

We say that $|\psi\rangle$ is entanglement dominated if $|\psi_\varepsilon\rangle$ is entanglement dominated. In contrast, $|\psi\rangle$ is magic dominated if $|\psi_\varepsilon\rangle$ is magic dominated.

With this definition at hand, we can now prove the robustness of the computational separation occurring between the two phases.

Theorem 11 (Robust version of Theorem 1). Let $|\psi\rangle$ be a pure quantum state and let $\varepsilon = \Theta(n^{-\alpha})$ for any $\alpha > 1$. If $|\psi\rangle$ is ε-entanglement dominated, then there exist efficient quantum algorithms—specifically those discussed in Secs. IV A and IV B—for performing the following tasks:

(1) estimating the entanglement entropy up to a additive error $o(1)$;
(2) distilling $S_1(\psi_A) - o(1)$ Bell pairs with ε error in trace distance;
(3) diluting $|\psi\rangle$ with ε error in trace distance using $S_1(\psi_A) + o(1)$ many Bell pairs.

Proof. Given $\varepsilon = \Theta(n^{-\alpha})$, it is possible to learn, using a polynomial-time quantum algorithm, the stabilizer group of $|\psi_\varepsilon\rangle$ and thereby measure the stabilizer entanglement $\mathcal{E}(\psi_\varepsilon; A|B)$, as shown in Lemma 16. Recall that Lemma 16 yields the largest stabilizer group (equivalently, the minimal stabilizer nullity) among the states in the ε ball of $|\psi\rangle$, and therefore of $|\psi_\varepsilon\rangle$ by definition (Definition 9).

Since $|\psi_\varepsilon\rangle$ is entanglement dominated by assumption, we have that $S_1(\psi_{\varepsilon A}) = \mathcal{E}(\psi_\varepsilon; A|B)(1 + o(1))$. To complete the proof, we use Fannes' inequality to bound the error in estimating $S_1(\psi_A)$:

$$|S_1(\psi_A) - S_1(\psi_{\varepsilon A})| = o(1), \tag{85}$$

which is ensured by the choice $\varepsilon = o(n^{-1})$. The proofs of (ii) and (iii) follow directly from Remarks 2 and 3—i.e., the robustness of the distillation (respectively, dilution) protocol—and from Eq. (85). ■

Theorem 12. Consider any bipartition $A|B$ with $n_A, n_B = \omega(\log n)$ and let $|\psi\rangle$ a pure quantum state with $S_1(\psi_A) = \omega(\log n)$. Let $\varepsilon = \Theta(n^{-\alpha})$ for any $\alpha \geq 1$. If $|\psi\rangle$ is magic dominated then any efficient protocol

(1) can only estimate $S_1(\psi_A)$ to within a relative error at least $\omega(1)$;
(2) can only distill at most $M_+/S_1(\psi_A) = o(1)$ many Bell pairs;
(3) requires at least $M_+/S_1(\psi_A) = \omega(1)$ many Bell pairs to prepare $|\psi\rangle$ across $A|B$.

Proof. Consider the two families of states in Eqs. (77) and (78) as presented in the proof of Theorem 10. To establish the statement, we first demonstrate that there are no entanglement-dominated states within the ε ball around any state belonging to each family, with high probability. We then proceed with the proof of Theorem 10.

Given $\varepsilon = O(n^{-1})$, thanks to the Fannes inequality, we know that every state $|\phi\rangle$ within the ε ball exhibits the same bipartite entanglement up to an additive error of $O(1)$. Furthermore, since by assumption these states possess entanglement on the order of $\omega(\log n)$, this error corresponds to a relative error of $o(1)$. Now, assuming for the sake of contradiction that an entanglement-dominated state exists within the ε ball, we could distinguish the two families by running the efficient distillation protocol in Theorem 6. To complete the proof, we proceed exactly as in the proof of Theorem 10. ■

V. INFERENCE ON ENTANGLEMENT- VS. MAGIC-DOMINATED STATES

A. Stabilizer group inference

A theme throughout this work is that the stabilizer generators S of the state in Eq. (5) contain most of the relevant information about its entanglement structure, particularly when the state is within the ED phase. However, in the statement of all our results, we assumed *a priori* knowledge of the stabilizer group generators S. This assumption is not necessary: S is readily available in most cases. If we know the unitary that generates $|\psi\rangle$ (i.e., we have an efficient circuit description of the state), we do not even need to run the circuit on real quantum hardware; the generators G can be calculated in polynomial time using the stabilizer formalism, by processing the circuit on classical hardware (see Algorithm 3). Another setting might be where we have black-box access to an unknown state $|\psi\rangle$. In this case, we can efficiently learn the generators G for its stabilizer group using just $O(n)$ copies of $|\psi\rangle$, using the tools of Ref. [48]. For the reader's convenience, we state the result below.

Lemma 16 (Efficient stabilizer group learning [48]). For any $\epsilon, \delta \in (0, 1)$ there exists an algorithm that outputs an Abelian group $\hat{G} \subset \mathbb{P}_n$, that uses only $(16n + 8\log(1/\delta))/\epsilon^2$ 4-copy measurements of an unknown state $|\psi\rangle$. This algorithm takes $O((n^3 + n^2 \log(1/\delta))/\epsilon^2)$ time, and up to a failure probability of at most δ, this Abelian group $\hat{G}$ satisfies two properties:

(1) The true stabilizer group G of $|\psi\rangle$ is a subgroup of $\hat{G}$, which implies $\log_2 |\hat{G}| \geq \log_2 |G| \geq n - \nu$; and
(2) $|\psi\rangle$ is ϵ close in trace distance to some state with a stabilizer group $\hat{G}$.

Proof. This follows from Algorithm 1 and Theorem 5.1 of Ref. [48]. ■

Remark 5. Note that this stabilizer group learning algorithm does not guarantee that we can learn the true stabilizer group of the state $|\psi\rangle$. However, the output group $\hat{G}$ is the largest stabilizer group among the states ϵ close in trace distance to $|\psi\rangle$, thus corresponding to the one in the definition of the smoothed stabilizer nullity in Sec. II A. If $\epsilon = 1/n$, by the Fannes inequality, the entanglement of $|\sigma\rangle$ across any cut $A|B$ differs from the entanglement of $|\psi\rangle$ across $A|B$ by at most 2. If $\epsilon = o(n^{-1})$, this gap is improved to $o(1)$ (i.e., it is vanishing in the limit of large n). This is

all to say that when we do entanglement calculation, when ϵ is small enough, we can pretend $\hat{G}$ is the true stabilizer group of the state. Furthermore, for entanglement manipulation tasks (e.g., Bell pair distillation), we can design protocols that work for $|\sigma\rangle$ (i.e., we will assume that $\hat{G}$ is the true stabilizer group). Although $|\psi\rangle$ may be the true input state, when we run our protocol on $|\psi\rangle$, we are guaranteed that the output will be ϵ close in trace distance to the desired output since the trace distance is monotonically decreasing under all CPTP maps. In summary, for all intents and purposes, when ϵ is very small, we can pretend $\hat{G}$ is the true stabilizer group of the state.

Corollary 7 (Robustness of stabilizer group learning). If an unknown (potentially mixed) state ψ is ϵ_1 close in trace distance to some other unknown state $|\psi'\rangle$ with stabilizer group G', the stabilizer group learning algorithm will still succeed with failure probability at most $(64\epsilon_1/\epsilon^2)\left(2n + \log\left(\epsilon^2/\epsilon_1\right)\right)$. The output stabilizer group $\hat{G}$ will satisfy $G' \subseteq \hat{G}$, and ψ will be $(\epsilon + \epsilon_1)$ close in trace distance to some state with stabilizer group $\hat{G}$.

Proof. The stabilizer group learning algorithm uses exactly $N = (64n + 32\log(1/\delta))/\epsilon^2$ copies of ψ. If ψ is ϵ_1 close in trace distance to some t-doped stabilizer state $|\psi'\rangle$, the maximum probability of distinguishing the two is $N\epsilon_1$ (this is the maximum possible trace distance between $|\psi\rangle^{\otimes N}$ and $|\psi'\rangle^{\otimes N}$). So, we define a distinguisher as follows. If the output of our algorithm is one of the "good" $\hat{G}$ (i.e., $G \subseteq \hat{G}$ and $|\psi'\rangle$ is ϵ close in trace distance to some state with stabilizer group $\hat{G}$) that could be output assuming the input state was $|\psi'\rangle$, we output 1, and 0 otherwise. By definition, this distinguisher outputs 1 with probability at least $1 - \delta$ for an input $|\psi'\rangle$. The properties of the trace distance tell us that the distinguisher will output 1 with probability at least $1 - \delta - 2N\epsilon_1$ when the input is ψ. That is, the failure probability is at most $\delta + 2N\epsilon_1$. If we set $\delta = 64\epsilon_1/\epsilon^2$, the overall failure probability (i.e., the probability that one of the "good" $\hat{G}$ is *not* output) is at most

$$\frac{64\epsilon_1}{\epsilon^2}\left(1 + 2n + \log\left(\frac{\epsilon^2}{64\epsilon_1}\right)\right) \leq \frac{64\epsilon_1}{\epsilon^2}\left(2n + \log\left(\frac{\epsilon^2}{\epsilon_1}\right)\right). \tag{86}$$

■

B. Testing for the entanglement-dominated phase

Having shown the ability to learn a state's stabilizer group, we now apply this to answer even more specific questions about particular states. Given access to an unknown state $|\psi\rangle$, can we tell whether it is ED or MD? In complete generality, this task is not efficiently possible as it would necessitate efficient measurement of entanglement for general states. However, we can reformulate this task as a property testing problem that is efficiently solvable. Recall that a property tester for a class C of quantum states takes copies of a state $|\psi\rangle$ as input and determines whether $|\psi\rangle \in C$ or $|\psi\rangle$ is ϵ far in trace distance from all states in C, *promised that one of these is the case*. Our algorithm efficiently tests whether an input state $|\psi\rangle$ belongs to the ED phase or is ϵ far from all such states and, as such, it is MD.

Theorem 13 (Property testing for entanglement-dominated states). Let $|\psi\rangle$ be an unknown quantum state. For any $\epsilon = \Theta(n^{-\alpha})$ with $\alpha \geq 1$, there is a ED property tester that uses polynomially many copies of $|\psi_n\rangle$, polynomial classical computing time, and has a failure probability of at most $\delta = O(\exp(-n))$ to distinguish between the following cases:

(1) $|\psi\rangle$ is ED;
(2) $|\psi\rangle$ is ε far in trace distance from any ED state, hence $|\psi\rangle$ is MD.

Proof. As shown in Lemma 16, it is possible through $O(\text{poly}\, n)$ queries to $|\psi\rangle$ to determine an Abelian group $\hat{G}$ such that $G \subseteq \hat{G}$ (where G is the true stabilizer group) and $|\psi\rangle$ is ϵ close in trace distance from a state $|\hat{\psi}\rangle$ with stabilizer group $\hat{G}$, with a failure probability that is exponentially small in n.

Assume that $|\psi\rangle$ falls into case a)—that is, it is an ED state. We know that $n - \log_2|G| \leq o(\mathcal{E}(\psi; A|B))$ by definition of the ED phase. Since $|\hat{G}| \geq |G|$ and $|S_1(\psi_A) - S_1(\hat{\psi}_A)| \leq O(1)$ by the Fannes inequality, we must then also have

$$n - \log_2|G| \leq o(\mathcal{E}(\psi; A|B)). \tag{87}$$

This is to say, $|\hat{\psi}\rangle$ is within the ED phase as well.

Take the other possibility, which is that $|\psi\rangle$ falls into case b). Then, $|\hat{\psi}\rangle$ must also be MD by the assumption that $|\psi\rangle$ is ϵ far from all ED states. In this case, we have

$$n - \log_2|\hat{G}| \geq \Omega(\mathcal{E}(\hat{\psi}; A|B)). \tag{88}$$

Since distinguishing between Eqs. (87) and (88) is efficient [calculating both $\log_2|\hat{G}|$ and $\mathcal{E}(\hat{\psi}; A|B)$ is efficient], the property tester is overall efficient. ■

VI. PHYSICAL APPLICATIONS

Entanglement is a fundamental physical quantity, and the tools developed here are therefore expected to have applications across a broad range of physics problems. Of course, for our tools to apply, we need to be able to find entanglement-dominated states in nature. We are able to

identify a number of settings in which these states naturally appear, such as quantum error-correcting codes and many-body systems.

The ubiquity of entanglement-dominated states in these diverse physical settings allows us to leverage the ED-MD framework developed in this work to gain new insights into various subfields of physics. Our tools not only provide a deeper understanding of the role of entanglement in these systems but also offer practical applications, such as efficient entanglement estimation protocols that can be used to study scrambling dynamics and characterize quantum chaos. By bridging the gap between these seemingly disparate fields, our work highlights the fundamental role of the entanglement-magic interplay in understanding the complex behavior of quantum systems. In this section, we showcase the power and versatility of our approach by applying it to three distinct areas: quantum error correction, many-body physics, and the characterization of scrambling dynamics. These applications demonstrate the unifying nature of the ED-MD perspective and its potential to provide fresh insights into seemingly unrelated physical phenomena.

A. Entanglement-dominated states as energy eigenstates

In this section, we show the emergence of ED states in the contexts of many-body Hamiltonians. To establish this connection between energy eigenstates and ED states, it is convenient to begin with stabilizer states. Stabilizer states also find relevance in many-body physics as eigenstates of so-called stabilizer Hamiltonians. A stabilizer Hamiltonian H_G is defined as

$$H_G = \sum_{P\in G} \alpha_P P, \tag{89}$$

where $G \subset \mathbb{P}_n$ is an Abelian subgroup of the Pauli group. It is well known [72] that all the eigenstates of H_G correspond to stabilizer states associated with the stabilizer group G. Recently, we have shown that perturbation to H_G leads to ν-compressible states, see Ref. [26]. In particular, let us introduce a perturbation to H_G formed by k generic Pauli operator $P_1, \ldots, P_k$ as

$$H = H_G + \sum_{i=1}^{k} \gamma_i P_i \tag{90}$$

with γ_i real coefficients. If $K = \langle\{P_1, P_2, \ldots, P_k\}\rangle$, we have shown that there is an Abelian subgroup $J \subset \mathbb{P}_n$ of dimension at least $n - \dim K$ (which is in turn lower bounded by $n - k$), such that there is an eigenbasis for H in which every eigenstate is stabilized by the elements of J. That is to say, they are low-magic states with $\nu \leq \dim K \leq k$. In the following, we analyze the implications of the existence of a separation between ED and MD states in many-body physics, employing the fact that the Hamiltonian (90) features ν-compressible eigenstates with $\nu \leq \dim K$.

Corollary 8 (Near-identical entanglement for all eigenstates). Let $H = H_G + \sum_{i=1}^{k} \gamma_i P_i$ with H_G being a stabilizer Hamiltonian corresponding to the stabilizer group G. In the ν-compressible eigenbasis of H corresponding to the stabilizer group J, the von Neumann entanglement entropy (across any cut) of any two eigenstates differs by at most $\dim K$.

Proof. All of the eigenstates have the same stabilizer group, hence have the same stabilizer entanglement. Since Theorem 4 shows that the stabilizer entanglement differs from the true entanglement by at most $\nu/2$ (which is at most $\dim K/2$), this shows the desired claim. ■

This may seem to contradict physical intuition, as we expect that for local Hamiltonians, generic high-energy states will be volume law entangled while low-energy states are area-law states. However, note that Corollary 8 says that there exists *one particular* eigenbasis for which all the eigenstates have a similar entanglement. It does not rule out the possibility that most of the states in a high-energy subspace have high entanglement; rather, it merely says that a select few of these states have the same entanglement as the ground states.

Let us now consider the Hamiltonian H as a perturbed version of a stabilizer Hamiltonian H_G and analyze the implications for the robustness of the ground-state entanglement, which is frequently employed as a tool to characterize many-body Hamiltonians, as, e.g., distinguishing between phases of matter [28].

Lemma 17 (Robust ground-state entanglement entropy). Consider a Hamiltonian $H = H_G + \sum_{i=1}^{k} \gamma_i P_i$. The ground-state entanglement entropy (across any cut) of H differs from the ground-state entanglement entropy of H_G by at most $\frac{3}{2}\dim K$.

Proof. Let G' be any completion of G (i.e., it will be a size-2^n Abelian subgroup of $\mathbb{P}_n$ such that $G \subseteq G'$). The eigenstates $|\psi_G\rangle$ of H_G are stabilizer states with a stabilizer group G' [72]. Define $K^\perp := \{P \in \mathbb{P}_n \mid [Q, P] = 0, \forall Q \in K\}$. Consider the subgroup $J := K^\perp \cap G'$—this contains all Pauli operators that commute with every element in K and G. In Ref. [26], it was shown that H admits an eigenbasis in which every eigenstate is stabilized by J. The group J has dimension $\dim(J) \geq n - \dim K$. Since $J \subseteq G'$,

$$|\mathcal{E}(\psi; A|B) - \mathcal{E}(\psi_G; A|B)| \leq \dim K. \tag{91}$$

Now, using the fact that ψ_G is an exact stabilizer state and ψ is a ν-compressible state with $\nu \leq \dim K$, we apply

Theorem 4 to conclude

$$|S_1(\psi) - S_1(\psi_G)| \leq \tfrac{3}{2} \dim K. \tag{92}$$

■

Combining Corollary 8 and Lemma 17 is useful for models in which we have knowledge of the entanglement for *just one* of the eigenstates, such as the ground state. In fact, thanks to Lemma 17, we only need to know about the ground state entanglement of the unperturbed (i.e., stabilizer) Hamiltonian, and given the ease with which this ground state can be found analytically, this information can always be found in polynomial classic time. Then, when k is dominated by the ground-state entanglement, we can first use Lemma 17 to estimate the ground-state entanglement of the perturbed model, then use Corollary 8 to estimate the entanglement for *any* of the eigenstates of the perturbed model.

B. A group-theoretic topological-entropy robustness proof in 3D topological models

We now turn our attention to more exotic forms of entanglement, particularly topological entanglement. This measure manifests as a universal correction to area law for the ground-state entanglement, arising from the intrinsic topological properties of the ground state. First identified in the context of the toric code [28] and later extended to a broad class of gapped topological phases [27], topological entanglement entropy has become an essential tool for characterizing such phases. While it provides only a partial characterization of topological order due to the existence of spurious terms [73], it continues to offer valuable insights into the nature of topological models. In this section, we leverage this tool alongside the techniques developed in this paper to explore the robustness of topological order against Hamiltonian perturbations.

For a pure state ψ, the topological entanglement entropy S_{topo} is defined as [27]

$$S_{\text{topo}} = S(\psi_{AB}) + S(\psi_{BC}) - S(\psi_B) - S(\psi_{ABC}), \tag{93}$$

where A, B, and C are appropriately chosen regions [74]. This quantity is also known as the conditional mutual information $I(A:C \mid B)$ [15]. Since each bipartite entanglement entropy in Eq. (93) can be estimated up to an additive error $O(\nu)$, it follows that S_{topo} can be estimated to the same precision. While S_{topo} is traditionally a small constant, certain topologically ordered states of matter can exhibit much larger values. For instance, Ref. [29] showed that ground states of two archetypal models of three-dimensional topological order—the X-cube model and Haah's code—possess topological entanglement entropy that scales linearly with system size.

The existence of models with extensive topological entanglement entropy raises the question: Does this extensive scaling confer greater robustness against Hamiltonian perturbations compared to models with constant topological entropy? While topological entropy is generally robust against weak, local Hamiltonian perturbations due to its nonlocal nature [27], are there specific perturbations where the extensive nature of the topological entropy plays a crucial role? Our tools for estimating entanglement entropy are particularly effective in addressing these questions, as the $O(\nu)$ correction becomes negligible relative to extensive S_{topo}.

To investigate this question, we analyze the behavior of topological entropy under identical perturbations in two distinct topological models. For our first model, we select a stabilizer code with extensive topological entropy, such as Haah's code or the X-cube model (see Ref. [29]), denoting its Hamiltonian as H_{ext} and its ground-state topological entropy as $S^{\text{ext}}_{\text{topo}}$. For our second model, we choose the 2D toric-code Hamiltonian:

$$H_{\text{TC}} = -\sum_v A_v - \sum_p B_p, \tag{94}$$

where A_v and B_p represent the star and plaquette operators, respectively. Formally, $A_v = \prod_{j \in \text{star}(v)} Z_j$ and $B_p = \prod_{j \in \text{bd}(p)} X_j$, with $\text{star}(v)$ indicating the star around vertex v and $\text{bd}(p)$ representing the boundary of plaquette p. As shown in Ref. [28], the ground-state topological entropy $S^{\text{TC}}_{\text{topo}}$ satisfies

$$S^{\text{TC}}_{\text{topo}} = 2. \tag{95}$$

We perturb both models $H \in \{H_{\text{TC}}, H_{\text{ext}}\}$ with a local defect:

$$H \mapsto H(\gamma) = H - \frac{\gamma_1}{3} Z_1 X_2 - \frac{\gamma_2}{3} Z_3 - \frac{\gamma_3}{3} X_4, \tag{96}$$

where $\gamma := \frac{1}{3}(\gamma_1, \gamma_2, \gamma_3)$. We evaluate the topological entanglement entropy in a region ABC surrounding the local defect (involving qubits 1 through 4), see Fig. 8. Leveraging Lemma 17, we can analytically demonstrate the robustness of topological entropy for H_{ext} in the thermodynamic limit.

Corollary 9 (Robust topological entanglement). The topological entanglement entropy of the ground state of H' differs from that of H by at most $O(\dim K)$, where K is defined in Eq. (90). Specifically, for $H_{\text{ext}}(\gamma)$ with the perturbation in Eq. (96), the ratio between the perturbed ground-state topological entropy $S^{\text{ext}}_{\text{topo}}(\gamma)$ and the

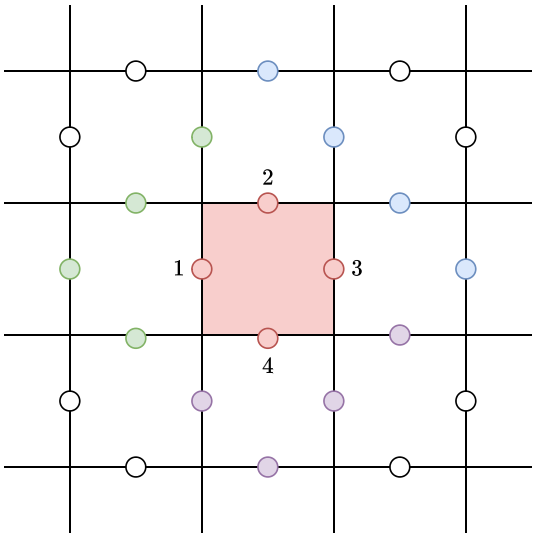


FIG. 8. Topological entanglement entropy evaluated around a defect (shown in red). The regions A, B, and C are chosen to be the qubits in green, purple, and blue, respectively.

unperturbed ground-state topological entropy $S_{\text{topo}}^{\text{ext}}$ satisfies

$$\frac{S_{\text{topo}}^{\text{ext}}(\gamma)}{S_{\text{topo}}^{\text{ext}}} = 1 + O(n^{-1}), \tag{97}$$

for all γ, demonstrating robustness of the topological entropy in the thermodynamic limit ($n \to \infty$).

Proof. The topological entanglement entropy is a linear combination of bipartite entanglement entropies (93). Since each bipartite entanglement entropy differs between ground states by at most $4 \dim K$, the topological entanglement entropies also differ by at most $O(\dim K)$. For $H_{\text{ext}}(\gamma)$, we have $S_{\text{topo}}^{\text{ext}}(\gamma) = S_{\text{topo}}^{\text{ext}} + O(1)$ with $S_{\text{topo}}^{\text{ext}} = \Omega(n)$, from which Eq. (97) follows immediately. ■

Remark 6. We note that Corollary 9 implies robustness of the topological entanglement even against highly nonlocal perturbations that couple the entire system, extending far beyond the local perturbation in Eq. (96). This demonstrates a remarkable robustness property of systems with extensive topological entropy.

To investigate the robustness of $S_{\text{topo}}^{\text{TC}}$ under the same perturbation, we conduct a numerical study on a toric code with $n = 98$ qubits. We remark that for the case of H_{ext}, numerical simulation is unnecessary, as Corollary 9 provides all the relevant information for the comparison of interest. As shown in Ref. [26], the ground state of the perturbed Hamiltonian $H_{\text{TC}}(\gamma)$ exhibits a nullity of $O(1)$ for all values of γ. To find the ground state, we leverage another result from the same reference, which establishes that there exists a classical algorithm running in time $O(n^{\|\gamma\|+3})$ capable of identifying the ground state of $H_{\text{TC}}(\gamma)$ (see Example 1 in Ref. [26]).

After determining the ground state for each value of γ, we compute the topological entropy within a small region ABC surrounding the defect. The results, shown in Fig. 9, indicate that for sufficiently strong perturbations γ, the topological entropy follows the scaling relation:

$$\frac{S_{\text{topo}}^{\text{TC}}(\gamma)}{S_{\text{topo}}^{\text{TC}}} \sim O(\exp(-\gamma/\gamma_0)), \tag{98}$$

for some constant γ_0. This illustrates that the topological entropy of the toric-code ground state can be effectively nullified under such perturbations. However, this finding does not contradict the expected robustness of S_{topo} under weak, local perturbations—while our perturbation is local, it is not weak, with $\|\gamma\| = \Omega(1)$.

Our analysis reveals an unexpected connection between the magnitude of a model's topological entropy (i.e., whether it is extensive or constant) and its robustness against Hamiltonian perturbations. We have rigorously demonstrated that stabilizer codes with extensive topological entropy maintain their robustness against any algebraically weak perturbations [characterized by $\dim K = o(S_{\text{topo}})$], regardless of coupling strength. In contrast, our numerical results show that this robustness does not extend to models with constant topological entropy, such as the 2D toric code.

This leads us to an unexpected conclusion: while topological entanglement is generally considered robust against

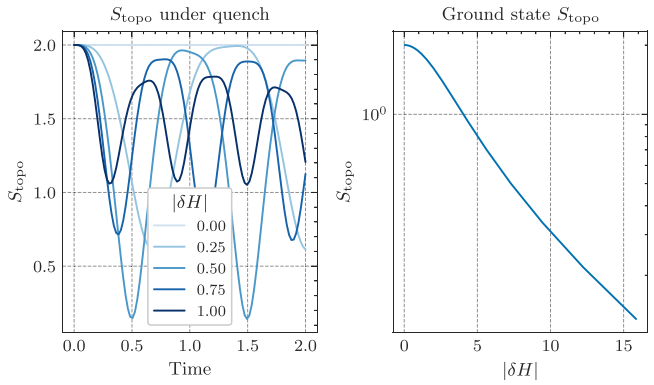


FIG. 9. Behavior of topological entanglement entropy (S_{topo}) in the toric code under local perturbations. Left: time evolution of S_{topo} following a quantum quench with different perturbation strengths $|\delta H|$, showing distinctive oscillatory behavior around $|\delta H| \approx 0.5$. Right: ground state S_{topo} as a function of perturbation strength, demonstrating the decay of topological order with increasing perturbation magnitude. The resonant behavior at intermediate perturbation strengths ($|\delta H| \approx 0.5$) suggests a delicate interplay between topology and local perturbations, where the system neither fully preserves nor completely loses its topological character.

weak, local perturbations, 3D topological models with extensive topological entropy exhibit persistence against *algebraically local* defects, which are, i.e., characterized by $\dim K = o(S_{\text{topo}})$. Notably, algebraically local defects can include highly nonlocal terms and arbitrary perturbation strengths. This distinguishes them from models with $O(1)$ topological entanglement, where similar perturbations can completely eliminate the topological entropy, as illustrated by Fig. 9. Although these results concern topological entropy, which provides an incomplete characterization of topological order, and are applied on stabilizer topological models only, they offer valuable new insights into topological order in 3D systems, an area that remains largely unexplored [29].

1. Robustness of topological entropy under quench

In this section, we further explore topological models through the lens of quantum quenches, complementing our earlier analysis of static perturbations. Our previous results revealed a striking distinction in how topological models respond to Hamiltonian perturbations based on their topological entanglement entropy scaling. Here, we investigate whether this distinction extends to dynamical settings, contributing to the broader understanding of topological order survival following quantum quenches [75].

We study the following quantum quench protocol: Consider a topological system initially prepared in its ground state $|\psi_0\rangle$ under Hamiltonian H. We suddenly apply the perturbation defined in Eq. (96), resulting in a new Hamiltonian $H(\gamma)$. Since $|\psi_0\rangle$ is no longer an eigenstate of $H(\gamma)$, the system evolves as $|\psi_t\rangle = e^{-iH(\gamma)t}|\psi_0\rangle$. Building on our previous findings, we compare the behavior of a topological model with extensive topological entropy (H_{ext}) to one with constant topological entropy (the toric code, H_{TC}).

We begin with analytical results for H_{ext}. The following corollary establishes that the topological entropy $S^{\text{ext}}_{\text{topo}}(t)$ of the state $|\psi_t\rangle$ remains robust for all times t and perturbation strengths $\|\gamma\|$:

Corollary 10 (Robust topological entanglement after quantum quench). The topological entanglement entropy $S^{\text{ext}}_{\text{topo}}(t)$ of $|\psi_t\rangle$ satisfies

$$\frac{S^{\text{ext}}_{\text{topo}}(t)}{S^{\text{ext}}_{\text{topo}}(0)} = 1 + O(n^{-1}) \tag{99}$$

for all $t \in \mathbb{R}$, demonstrating robustness in the thermodynamic limit.

Proof. Our proof relies on Theorem 4 from Ref. [26], which shows that for a stabilizer Hamiltonian H_{ext} perturbed by $O(1)$ Pauli terms, the postquench state takes the form

$$|\psi_t\rangle = C^\dagger(|0\rangle \otimes e^{-i\tilde{H}(\gamma)t}|\phi\rangle), \tag{100}$$

where $C^\dagger$ is the Clifford circuit preparing the ground state $|\psi_0\rangle$ of H_{ext} from $|0\rangle$, i.e., $|\psi_0\rangle = C^\dagger|0\rangle$. The transformed Hamiltonian $\tilde{H}(\gamma)$ acts on only $O(1)$ qubits, though we omit its explicit form here. Crucially, the state in Eq. (100) has stabilizer nullity $O(1)$ and shares the stabilizer group of the ground state $|\psi_0\rangle$. Therefore, by Theorem 4, we have $S^{\text{ext}}_{\text{topo}}(t) = S^{\text{ext}}_{\text{topo}}(0) + O(1)$. Since $S^{\text{ext}}_{\text{topo}}(0) = \Omega(n)$, Eq. (99) follows immediately. ■

We next examine the evolution of $S^{\text{TC}}_{\text{topo}}(t)$ under the same quantum quench. To study the thermodynamic limit while minimizing finite-size effects, we employ the stabilizer formalism. Theorem 4 from Ref. [26] provides an efficient classical algorithm for simulating stabilizer Hamiltonians perturbed by $O(1)$ Pauli terms, with runtime scaling as $O(n^3)$ independent of time t. Using this algorithm, we obtain the numerical results shown in Fig. 9.

In contrast to the model with extensive topological entropy, the toric code's topological entropy vanishes after a short time t. Our results suggest that the early-time decay follows

$$S^{\text{TC}}_{\text{topo}}(t) \sim S^{\text{TC}}_{\text{topo}}(0) \times (1 - (t/t_0)^4), \tag{101}$$

where t_0 depends on the perturbation strength. These findings reinforce our earlier conclusion that the scaling of topological entropy fundamentally impacts the resilience of these models to perturbations, even in dynamical settings.

These results provide fresh insight into 3D topological models like Haah's code and the X-cube code. Their robustness extends far beyond resistance to local, weak perturbations, fundamentally distinguishing them from more fragile 2D models characterized by constant topological entropy. This remarkable stability in both static and dynamic settings suggests these 3D models may be particularly valuable for quantum information storage and processing applications.

C. Entanglement-dominated states as code states: topological error correction

The toric code Hamiltonian, defined in Eq. (94), is one of the most widely studied models of topological order, and as its name suggests, it is connected with error correction. The ground state of this Hamiltonian is fourfold degenerate, which allows these states to be used as logical states. That is, the four energy eigenstates can be identified with $|00\rangle_L$, $|01\rangle_L$, $|10\rangle_L$, and $|11\rangle_L$, where the L subscript denotes the fact that these are logical states, rather than actual physical ones on the lattice. Each of these ground states is the $+1$ eigenstate of $2n-2$ *independent*

stabilizers. The remaining two stabilizers are strings of Z operators that form horizontal or vertical loops around the torus—these are referred to as the logical operators Z_1 and Z_2 of the code. Whether the state is a $+1$ or -1 eigenstate of these two logical operators determines the logical state (e.g., if the state is the $+1$ stabilizer of the two loop operators, it corresponds to $|00\rangle_L$). However, this implies that any state in the ground state space of the toric-code Hamiltonian (i.e., an error-free state, or a *code state*) can be written

$$\rho = \left(\prod_{v=1}^{n-1} \frac{\mathbb{1} + A_v}{2}\right)\left(\prod_{p=1}^{n-1} \frac{\mathbb{1} + B_p}{2}\right)\rho_L, \tag{102}$$

where the portion of the state ρ_L determines the logical state represented by ρ. For instance, if $\rho_L = ((\mathbb{1} + Z_1)(\mathbb{1} + Z_2))/4$, then ρ is identified with $|00\rangle\langle 00|_L$. Of course, in general, if we have a superposition of logical states, ρ_L will not necessarily be a stabilizer state. However, due to the $2n-2$ shared stabilizers, this form makes it obvious that any state in the ground-state space, even if it is a superposition of logical states, is a state with $\nu \leq 2$. We note here that this supplies a new interpretation of the nullity distillation circuit in Fig. 4: the distillation circuit is a *decoding circuit*, since the state left over after the distillation represents the logical content of the code state.

Furthermore, these states fall into the ED phase. Previous work [76] has shown that entanglement for a bipartition $A|B$ in the ground space of the toric-code grows as $S_1(\psi_A) \sim \sqrt{n_A}$, following an area-law state in 2D. Therefore, with $S_1(\psi_A) = \omega(\nu) := \omega(1)$, the ground space of the toric codes is described by ED states, cf. Definition 1.

Another implication of the fact that all code states are ν-compressible states is that, using the same logic as Lemma 17, the entanglement of ρ across any cut can differ from any one of the logical states by at most $4\nu = 8 = O(1)$. Indeed, in general, any stabilizer code Hamiltonian with k logical states will have $\nu \leq k$, and since $k = O(1)$ for most topological codes [77], $\nu = O(1)$ for these codes. An immediate corollary of this means that the entanglement of any code state is equal to the entanglement of the logical states up to an $O(1)$ correction. Why is this important? It tells us that we can reason about the entanglement properties of any code state by knowing the entanglement properties of *just a single logical state*. Furthermore, since the stabilizers of this state are analytically derivable by looking at the Hamiltonian, these entanglement properties are readily available. This is a special property. On the other hand, this reasoning breaks for the recently proposed good qLDPC codes [78–80]. These qLDPC Hamiltonians support $\Theta(n)$ logical qubits [79], hence $\nu = \Theta(n)$, in which case our bounds become too loose to say anything useful.

D. Lyapunov exponent estimation with entanglement-dominated states

Lyapunov exponents were originally introduced in classical mechanics to probe chaos. They measure the rate of divergence of nearby trajectories in phase space, effectively quantifying the sensitivity to initial conditions. This phenomenon is the well-known "butterfly effect," wherein small changes in initial conditions lead to dramatically different outcomes over time. In the context of quantum mechanics, the notion of Lyapunov exponents maintains a similar role as a probe of chaos. However, due to the unitarity of quantum mechanics, they cannot measure the divergence of trajectories in phase space as in classical mechanics. Instead, in quantum mechanics, Lyapunov exponents are related to the degree of noncommutativity between two operators.

Consider two operators V and W supported on nonoverlapping domains such that $[W, V] = 0$ and take a local unitary evolution $U(\tau)$ parametrized by a real parameter τ. Note that τ can be interpreted as the time in Hamiltonian evolution or, as considered below, the number of layers of an evolution from a local quantum circuit. A local operator $W(0) = W$ evolves under the Heisenberg picture as $W(\tau) = U^\dagger(\tau) W U(\tau)$. The commutator between $W(\tau)$ and V will now increase because the domain of W will gradually expand and eventually, it reaches the domain of V. The main difference, in the choice of the unitary evolution, lies in the speed of this expansion. A way to quantify the speed of this expansion is given by the out-of-time-order correlator (OTOC), which is connected to the notion of a group commutator. A way to define it is via the so-called, *entanglement in time* [81]. Let us consider a system made of two bipartitions, an input bipartition $A|B$ and an output bipartition $C|D$, and Hilbert space of $2n$ qubits where $n_A + n_B = n_C + n_D$. The OTOC is then defined as

$$\mathrm{OTOC}\left(U(\tau)\right) := \frac{1}{2^{n+2(n_A+n_D)}} \sum_{P_A, P_D} \mathrm{tr}\left(P_D(\tau) P_A P_D(\tau) P_A\right), \tag{103}$$

where P_X is a local Pauli operator, $P_D(\tau) := U(\tau)^\dagger P_D U(\tau)$. The OTOC is a well-known probe to chaos [82], and as shown in Ref. [81] can be connected to the entanglement entropy of the Choi representation of the unitary $U(t)$ as

$$S_2(\rho_{AC}(\tau)) = n_C - n_A - \log_2\left(\mathrm{OTOC}(U(\tau))\right), \tag{104}$$

where $\rho_{AC}(\tau) := \mathrm{tr}_{BD} |U(\tau)\rangle\langle U(\tau)|$. $|U(\tau)\rangle$ is the Choi representation of the unitary operator $U(\tau)$ and is defined as $|U(\tau)\rangle := 2^{-n/2} \sum_i \mathbb{1} \otimes U(\tau) |ii\rangle$. Note that also $|U(\tau)\rangle$ is a ν-compressible state with stabilizer nullity $\nu \leq 2t$ since it corresponds to the action of a t-doped Clifford unitary on the stabilizer state $|\mathbb{1}\rangle = 2^{-n/2} \sum_i |ii\rangle$ (see Definition 3).

We now show the connection between OTOC and entanglement entropy with the Lyapunov exponent, as done in

Ref. [31]. Below, we will focus on the class of unitary operators $U(\tau)$ generated by a local quantum circuit. Following Ref. [31], we consider a discrete-time evolution, made of layers of two-site random unitary gates, acting on pairs of neighboring sites. Each time step consists of two layers, an odd layer acting on all odd neighboring bonds, and an even layer acting on all the even neighboring bonds. In a clear analogy to what was done in Ref. [31] for the entanglement, one can show that the average OTOC for two-qubit gates drawn according to a 2-design distribution in a brickwork fashion behaves as

$$\langle \mathrm{OTOC}(U(\tau))\rangle = 2^{-2n_A} + 2^{-2n_D} - 2^{-2(n_A+n_D)} + \left(1 - 2^{-2n_A} - 2^{-2n_D} + 2^{-2(n_A+n_D)}\right) \times e^{-\frac{\tau}{2}\log\left(\frac{1}{1-\lambda_L^2}\right)} \tag{105}$$

where the average is denoted by $\langle\cdot\rangle$, and λ_L represents the Lyapunov exponent. Here, we adopt the notion of the Lyapunov exponent as given in Ref. [82], where it quantifies the exponential divergence rate in operator evolution, similarly to its classical counterpart. Therefore, one has the following scaling for

$$\langle \mathrm{OTOC}(U(\tau))\rangle = \exp\left(-\frac{\tau}{2}\log\left(1-\lambda_L^2\right)^{-1}\right) + O(\exp(\tau)/2^{2n_A}). \tag{106}$$

If we use the connection between OTOC(U) and entanglement entropy (103) we obtain

$$\langle S_2(\rho_{AC}(\tau))\rangle = n_C - n_A + c(\lambda_L)\tau + O(\exp(\tau)/2^{2n_A}) \tag{107}$$

for $c(\lambda_L) = (1/2\log 2)\log\left(1/(1-\lambda_L^2)\right)$. The result shows how in a 2-design random circuit model, the scrambling velocity is connected to the Lyapunov exponent. Notably, Eq. (107) holds for local random Clifford circuits (being unitary 2-design), as well as for t-doped local random Clifford circuits for any value of the doping.

Motivated by Eq. (107), we hypothesize that for a single instance of a typical random local t-doped Clifford circuit, the behavior of $S_2(\rho_{AC}(\tau))$ is approximately the one given in Eq. (107). Therefore, we assume that for $\tau = o(n_A)$ one has $S_2(\rho_{AC}(t)) = n_c - n_A + c(\lambda_L)\tau$ with the constant $c(\lambda_L) = (1/2\log 2)\log\left(1/(1-\lambda_L^2)\right)$ and some Lyapunov exponent λ_L to be determined. Then, one can estimate the Lyapunov exponent of the circuit by performing a linear regression of the entanglement entropy of $\rho_{AC}(\tau)$ versus the number of layers τ. In the following, we will consider the case $n_A = n_C - \omega(\log n)$, with $n_C > n_A > 1$. In such a case, it is clear that a simple swap test cannot be used to estimate the Lyapunov exponent because the number of measurements required to estimate $S_2(\rho_{AC})$ will be superpolynomial. However, using our results on ED phase, from Lemma 13, we are able to estimate the entanglement entropy S_1, or the 2-Rényi entropy S_2, up to a $o(1)$ relative error in the ED phase. Assuming the entanglement growth to be linear in τ, for any $\tau \in [\omega(\max(\log n, \nu)), o(n_A)]$ the state $\rho(\tau)$ is ED in the bipartition $AB|CD$. Therefore, as long as the doping $\nu = o(n_A)$, one can always find a useful time window to perform a linear regression and estimate the Lyapunov exponent λ_L. The only aspect that changes with the nullity, ν, of $|U(\tau)\rangle$ is the useful time window during which one can estimate the OTOC using stabilizer entanglement. In particular, as shown in Lemma 13, in the ED phase, we can estimate $c(\lambda_L)$ with an asymptotically vanishing error $o(1)$, which, via standard error propagation, transfers to an asymptotic vanishing error on the Lyapunov exponent λ_L.

In summary, given query access to a unitary $U(\tau)$, an experimenter can construct its Choi representation by applying it to a maximally entangled state $2^{-n/2}\sum_i |ii\rangle$. Then, they can apply Lemma 16 to approximate the stabilizer entanglement, hence allowing them to estimate the Lyapunov exponent of the circuit. Alternatively, if the experimenter has access to a gate set, it is possible to use Algorithm 3 to directly monitor the entanglement of the Choi representation $U(\tau)$ and obtain an estimate of the Lyapunov exponent efficiently through classical simulation.

VII. UNDERSTANDING PHENOMENOLOGY WITH ED-MD PHASES

In addition to the physical applications discussed in the previous section, the tools and techniques developed in this work can also provide theoretical insights into various phenomena observed in numerical simulations of quantum circuits. In this section, we showcase the utility of the separation between ED and MD phases for gaining a theoretical understanding of previously observed phenomena in numerical simulations.

A. Entanglement cooling

The first phenomenon we investigate is known as *entanglement cooling* [83,84]. The core idea is that we can quantify a state's entanglement complexity by measuring how difficult it is to disentangle across a given bipartition, without knowledge of the dynamics that generated the state. While this task was initially studied in the context of quantum circuits [83], it was soon recognized that similar analyses could be extended to many-body Hamiltonian evolutions [85]. Specifically, in the context of many-body Hamiltonians, the success or failure of the entanglement cooling task was found to be closely related to the integrability-chaos transition.

Integrable dynamics, such as those produced by the transverse-field Ising model (TFIM), generate entangled

states that can be efficiently disentangled using an unstructured Monte Carlo algorithm. In contrast, the presence of a longitudinal field, which breaks the integrability of the TFIM, completely inhibits the success of this disentangling algorithm. This parallel between quantum circuits and many-body Hamiltonian evolution is natural, as similar connections appear in various contexts. Indeed, a central theme of this paper is the modeling of many-body unitary dynamics using quantum circuits (see, e.g., Sec. VI D). Specifically, we use k-designs with sufficiently large k to represent chaotic quantum evolution, such as Clifford circuits doped with an adequate number of non-Clifford gates. For further details on the strong relationship between Clifford + T circuits (with high T count) and unitary k-designs, we refer the reader to the discussion in the overview (Sec. I A).

The relationship between entanglement cooling failure and the onset of quantum chaos provides insight into quantum chaotic evolution, particularly regarding the complex entanglement structures that emerge. In this section, we rigorously analyze the entanglement cooling task within the framework of Clifford + T circuits, providing analytical proofs that support the numerical findings of Ref. [33], and draw conclusions regarding the presence of ED-MD phases in this integrability-chaos separation.

To formalize the entanglement cooling task, we consider the following setup. Fix $A|B$ and a entangling unitary U that produces $|\psi\rangle = U|0\rangle^{\otimes n}$. The entanglement cooling task is the following: given only query access to $|\psi\rangle$ (i.e., no knowledge of U), design a unitary operator U' such that the state $|\psi'\rangle := U'|\psi\rangle$ is minimally entangled across $A|B$. The success of the entanglement cooling is quantified by the ratio R between the entanglement entropy before and after the cooling:

$$R = \frac{S_1(\psi'_A)}{S_1(\psi_A)}. \tag{108}$$

The lower the ratio R, the more successful the protocol. In Ref. [33], the authors explore the entanglement cooling problem concerning states formed by random Clifford+T circuits. They restrict themselves to Clifford operations in the cooling process and aim to identify good unitaries via a local greedy search. Importantly, they find that the efficacy of this entanglement cooling method decreases as the number of T gates used to prepare the state increases. In particular, their numerical results indicate that Clifford unitaries suffice to cool the entanglement of a random t-doped state as long as $t \lesssim n$, but fail otherwise. Notably, the tools and methods developed in this paper give us an analytical way of understanding these results.

Lemma 18 (Entanglement cooling via Clifford operations). For any bipartition $A|B$, let $|\psi\rangle$ be a state generated by a Clifford + T circuit with $t \leq n/2$ T gates. There exists a Clifford circuit C' such that, when applied on the state $|\psi'\rangle = C'|\psi\rangle$, the cooling ratio is $R = 0$. That is, $|\psi'\rangle$ is a product state with respect to the bipartition $A|B$. Moreover, it is possible to find C' using just $O(n)$ queries to $|\psi\rangle$.

Proof. Given query access to some state $|\psi\rangle$, we first learn the stabilizer generators of the state using $O(n)$ queries. We note that this state must have at least $n - t$ stabilizer generators (as opposed to the general bound $n - 2lt$, which comes from the fact that the T gate is diagonal [45]). Then, using nullity distillation (Algorithm 1), we can find a Clifford unitary that maps $|\psi\rangle$ to $|\psi'\rangle = |0\rangle^{\otimes(n-t)} \otimes |\psi''\rangle$, where $|\psi''\rangle$ is supported on $t \leq \frac{n}{2}$ qubits. Also, at least one of the bipartitions has $\geq n/2$ qubits. By applying swaps appropriately, we can move $|\psi'\rangle$ so that it is supported only on this bipartition, with the rest of the system in a product of $|0\rangle$ states. Now, the resulting state is completely unentangled across the bipartition $A|B$. ■

While Lemma 18 provides an analytical validation of the numerical results reported in Ref. [33], we now explore the performance of the cooling algorithm in the context of the ED-MD phases introduced in this paper, interpreting the integrability-chaos transition through the more general ED-MD states, and not limiting ourselves solely to Clifford + T circuits.

First, we remark that there are two important aspects of the results presented in Ref. [85]. First, the study employed an algorithm that disentangles states produced by many-body Hamiltonians exclusively using Clifford circuits. Interestingly, when the authors examined a similar algorithm that outputs general unitaries drawn from the full unitary group, they found a noticeably poorer performance in the cooling process. Second, the main conclusion of their work indicates that the failure of the disentangling algorithm is linked to the emergence of quantum chaos in many-body Hamiltonians, as the algorithm succeeds for the integrable transverse field Ising model (TFIM). In the following theorem, we extend Lemma 18 to encompass any cooling algorithm that outputs a Clifford circuit, as in the approach taken by Ref. [85], for attempting to disentangle a given state $|\psi\rangle$ in light of the ED-MD phases.

Theorem 14 (Entanglement cooling and ED-MD separation). Consider a bipartition $A|B$ and an unknown state $|\psi\rangle$. Let R denote the ratio defined in Eq. (108), representing the entanglement of the cooled state $|\psi'\rangle = C|\psi\rangle$, where C is a Clifford circuit, relative to the entanglement of the original state $|\psi\rangle$. Then if $|\psi\rangle \in$ ED, there exists a Clifford circuit C such that $R = o(1)$. Conversely, if no Clifford circuit can disentangle the state, resulting in $R = \Omega(1)$, then $|\psi\rangle \in$ MD necessarily.

Proof. Similarly to Eq. (18), the proof proceeds as follows: if $|\psi\rangle$ is entanglement dominated, then $S_1(\psi_A) =$

$\omega(\nu)$, implying that the stabilizer nullity ν of $|\psi\rangle$ satisfies $\nu = o(S_1(\psi_A))$. By applying the nullity distillation procedure (see Algorithm 1), we know that there exists a Clifford circuit C such that $|\psi'\rangle = |0\rangle^{\otimes(n-\nu)} \otimes |\psi''\rangle$, where $|\psi''\rangle$ is supported on ν qubits. In any bipartition $A|B$, the entanglement of $|\psi'\rangle$ is trivially bounded above by ν, thus we obtain $S_1(\psi'_A) \leq \nu$. Evaluating the ratio, we find

$$R = \frac{S_1(\psi'_A)}{S_1(\psi_A)} \leq \frac{\nu}{S_1(\psi_A)} = o(1), \tag{109}$$

which completes the proof.

For the converse, note that Theorem 14 shows that any entanglement-dominated state can be cooled down, yielding $R = o(1)$. Therefore, if $R = \Omega(1)$ for any Clifford circuit, it necessarily implies that $|\psi\rangle$ is magic dominated. ■

These results complement the findings of Ref. [85]. While we cannot definitively state that TFIM evolution results in ED phases, we observe a clear connection between the integrability-chaos transition and the ED-MD phase transition. Specifically, while the cooling algorithm always succeeds in the ED phase (as seen in integrable dynamics), its failure implies the state must be MD—indicating that chaotic dynamics necessarily lead to MD states. This suggests that the separation between ED and MD phases can serve as an effective model for distinguishing the entanglement structure (or entanglement complexity, as described in Ref. [85]) in integrable versus chaotic dynamics.

B. Entanglement and magic phases in hybrid quantum circuits

The second phenomenon we study focuses on hybrid quantum circuits, which consist of random circuits interspersed with measurements. These constructions have gained significant attention in recent years [42,86,87], as the concept of *quantum trajectories* has allowed for the study of quantum information dynamics in the context of open quantum systems. Quantum systems subjected to both continuous measurements and unitary dynamics present another class of quantum dynamical behavior, described through "quantum trajectories," which have been explored in few-qubit systems [88], quantum spin systems [89], and trapped ultracold atoms [90]. Under strong and continuous measurements, the state vector can localize in the Hilbert space, exemplifying the quantum Zeno effect [91].

Although many-body open systems are traditionally studied using mixed-state density matrices and Lindblad equations, this formalism provides limited insight into quantum entanglement. Since entanglement serves as a crucial tool for detecting phase transitions in quantum many-body physics, an alternative approach is needed. Modeling many-body quantum dynamics with unitary circuits interspersed with measurements offers a more manageable setup for numerical simulations that mimic open many-body dynamics and allows for the investigation of entanglement through extensive simulations.

A series of works has revealed the critical role of entanglement in quantum trajectories induced by unitary circuits interspersed with measurements, demonstrating a phase transition from volume-law entanglement to area-law entanglement in the resulting states [92–94]. This behavior resembles a quantum Zeno effect, where entanglement is suppressed, resulting in a near-product state, and transitions to an error-corrected phase with long-range entanglement. This phase transition can be detected in Clifford unitary circuits interspersed with measurements in the computational basis, where stabilizer formalism techniques enable the simulation of systems with up to, and beyond, approximately 500 qubits [95].

The classical simulability of Clifford circuits has motivated an investigation into whether a similar phase transition occurs for magic. In Ref. [34], the authors numerically demonstrated an "entanglement-magic separation" in hybrid quantum circuits constructed using random local Clifford gates interspersed with both measurements and T gates. Although one might expect four distinct phases corresponding to each combination of (volume law, area law) $\times$ (entanglement, magic), only three phases were observed: the *volume-law entanglement* and *area-law magic* phases were not realized. Notably, this missing phase corresponds to entanglement-dominated (ED) states, where magic scales subextensively with entanglement.

In our work, using the toolkit developed in this paper, we provide an analytical explanation for this phenomenon. Specifically, we show that for a hybrid quantum circuit to exhibit entanglement-dominated states, the probability of applying T gates must decrease exponentially with n. This finding quantitatively explains the absence of this phase in the numerical results of Ref. [34], which used only a polynomially vanishing probability of applying T gates relative to n, insufficient to suppress magic to reach the ED phase.

Since our focus lies in the transition ED-MD, we design a simplified version of hybrid quantum circuits characterized solely by volume-law entangled states. Typically, random hybrid circuits consist of random local gates arranged in a brickwork structure. However, in our case, we employ global random Clifford circuits, pumping up entanglement to its maximum value while leaving magic invariant. This design choice allows us to specifically address the "ratio" between magic and entanglement. In particular, we consider a hybrid quantum circuit composed of layers of *random global* Clifford circuits interleaving the application of $R_z(\theta)$ gates (with random θ) and measurements in the computational basis (see Fig. 10). While we assume that the Clifford layers are applied with unit probability, at each layer we consider a probability p_t for applying

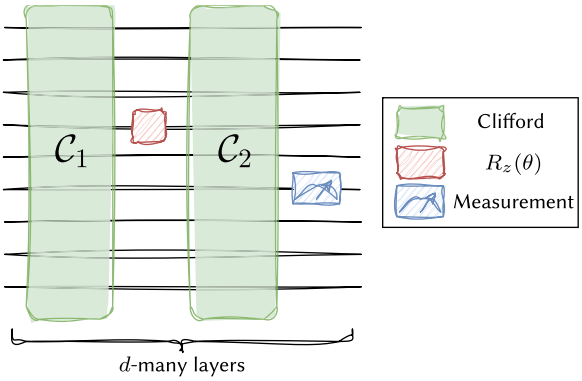


FIG. 10. A schematic representation of a t-doped random circuit consists of alternating layers of random Clifford circuits, random diagonal gates $R_z(\theta)$ and measurements in the local Z basis.

a $R_z(\theta)$ gate and p_m for performing a measurement with $p_t + p_m \leq 1$. On the other hand, we consider the probability of applying both at the same time to be zero. Thanks to the Clifford invariance of the problem, the Markov chain governing the evolution of the stabilizer nullity ν of the resultant state is simple and can be mapped into a simple birth-death process.

Before stating our result, we lay down some preliminary considerations. Consider a state $|\psi\rangle$ with a stabilizer group G_ψ. We apply a $R_z(\theta)$ gate on $|\psi\rangle$ sandwiched by a global *random* Clifford circuit $\mathcal{C}_1$, so we define $|\phi\rangle = \mathcal{C}_1^\dagger R_z(\theta)\mathcal{C}_1 |\psi\rangle$. This sandwiching can always be done by mapping $\mathcal{C}_2 \to \mathcal{C}_2\mathcal{C}_1$, which does not change the distribution of $\mathcal{C}_2$ because both $\mathcal{C}_1, \mathcal{C}_2$ are uniformly sampled from the Clifford group. We know that $|G_\phi| \geq |G_\psi|/2$. However, since θ is chosen at random $\theta \in [0, 2\pi]$, up to a set of measure zero, it holds that $|G_\phi| \leq |G_\psi|$. Therefore, we have $|G_\phi| \in \{|G_\psi|, |G_\psi|/2\}$ with probability 1. For a random Clifford $\mathcal{C}_1$, $\mathcal{C}_1^\dagger R_z(\theta)\mathcal{C}_1 = e^{i\theta P}$ where P is a random nonidentity Pauli operator. Now, there are two cases: either P commutes with all the elements in G_ψ, or P anticommutes with $|G_\psi|/2$ elements of G_ψ. In the first case, one has $|G_\phi| = |G_\psi|$, while in the latter case one has $|G_\phi| = |G_\psi|/2$. It is quite easy to be convinced that the probability that a random Pauli operator anticommutes with $|G_\psi|/2$ elements of G_ψ is $f(\nu) := 1 - (2^{n+\nu} - 1)/(4^n - 1)$. Now, we analyze measurement in the computational basis. Consider a measurement sandwiched by a random Clifford $\mathcal{C}_2$ applied to a state $|\psi\rangle$ with stabilizer group G_ψ. The resultant states are $|\phi\rangle \propto (I + P)|\phi\rangle$ with P being a random Pauli operator. There are two situations that can arise, P commutes with all elements of G_ψ or P anticommutes with $|G_\psi|/2$ elements of G_ψ. In the latter case, it is easy to see that $|G_\phi| = |G_\psi|$ [4]. In the former case, we have not yet ensured that the dimension of the stabilizer group of $|\phi\rangle$ doubles. In order for the stabilizer group to double, on top of commuting with all the elements in G_ψ, we must also have $P \notin G_\psi$. The probability that P commutes with G_ψ is given by $1 - f(\nu)$ as before, and if P commutes with G_ψ and given that both P and G_ψ are sandwiched by random Clifford operations, the probability that P is generated by G_ψ is given by $|G_\psi|/d = 2^{-\nu}$. Therefore, putting all together, the probability that a measurement doubles the size of G_ψ is given by $(1 - f(\nu))(1 - 2^{-\nu})$. From this analysis, it becomes clear that the localizing power of the measurements, is much less than the spreading of magic given by the application of $R_z(\theta)$ gates. We make this intuition rigorous in the following theorem.

Theorem 15 (Absence of entanglement-dominated states in hybrid quantum circuits). Consider infinite layers of the hybrid quantum circuit in Fig. 10 with p_t the probability of applying a $R_z(\theta)$ and p_m being the probability of applying a measurement in the computational basis. Define $r = p_t/p_m$, then for any $r = \omega(\exp(-n))$ the resulting state is in the MD phase.

Proof. As anticipated above, the stabilizer nullity at each layer can be modeled as a simple Markov chain—specifically, a birth-death process. The birth probability $p_+(\nu)$ refers to the probability of an increasing stabilizer nullity, while the death probability $p_-(\nu)$ refers to the case where nullity decreases:

$$\begin{aligned} p_-(\nu) &= p_m \times (1 - f(\nu))(1 - 2^{-\nu}), \\ p_+(\nu) &= p_t \times f(\nu). \end{aligned} \tag{110}$$

Note that $p_-(0) = p_+(n) = 0$, while the probability that the nullity stays the same is $1 - p_+(\nu) - p_-(\nu)$. In the limit of infinite layers, we converge to the stationary distribution of a birth-death Markov process with birth probability p_+ and death probability p_-. The steady-state distribution of such a process is [96]

$$\pi(\nu) = \begin{cases} \dfrac{p_+(0)}{Z} & \nu = 0 \\ \dfrac{1}{Z}\dfrac{p_+(0)\cdots p_+(\nu-1)}{p_-(1)\cdots p_-(\nu)} & \nu > 0 \end{cases}; \tag{111}$$
$$Z = \sum_{\nu=0}^{n} \frac{p_+(0)\cdots p_+(\nu-1)}{p_-(1)\cdots p_-(\nu)}.$$

Recalling the definition of $f(\nu) = 1 - (2^{n+\nu} - 1)/(4^n - 1)$, we approximate $f(\nu) \approx 1 - 2^{\nu-n}$ for simplicity. Therefore, we have $p_-(\nu) = (p_m/2^n)(2^\nu - 1)$ and $p_+(\nu) =$

$p_t(1-2^{\nu-n})$. Plugging the probabilities, the steady probability distribution $\pi(\nu)$ reads

$$\begin{aligned}\pi(\nu) &= \frac{r^\nu}{Z}\prod_{i=1}^{\nu}\frac{(2^n-2^{i-1})}{(2^i-1)}\\ &\geq \frac{r^\nu 2^{n\nu}}{Z\cdot 2^{(\nu+1)/2}}\prod_{i=1}^{\nu}(1-2^{i-n-1})\\ &\geq \frac{r^\nu 2^{n\nu}}{4Z\cdot 2^{(\nu+1)/2}}\\ &\geq \frac{1}{6Z}\left(\frac{2^n r}{\sqrt{2}}\right)^\nu. \end{aligned} \tag{112}$$

Since we aim to lower bound the expectation $\langle\nu\rangle$, it suffices to upper bound Z.

$$\begin{aligned}Z &= p_t(1-2^{-n}) + \sum_{\nu=1}^{n} r^\nu \prod_{i=1}^{\nu}\frac{2^n-2^{i-1}}{2^i-1}\\ &\leq p_t + \sum_{\nu=1}^{n} r^\nu \prod_{i=1}^{\nu}\frac{2^n-2^{i-1}}{2^i-1}\\ &\leq p_t + \sum_{\nu=1}^{n} r^\nu \prod_{i=0}^{\nu-1}\left(2^{n-i}-1\right)\\ &\leq p_t + \sum_{\nu=1}^{n} r^\nu 2^{n\nu-\nu^2/2^2/2}\\ &\leq \sum_{\nu=0}^{n} r^\nu 2^{\nu(n-\nu/2)\nu^2/2} \leq O(2^{n^2/2}), \end{aligned} \tag{113}$$

where we assume, without loss of generality, that $r \leq 1$ in the last line (if $r > 1$, the expected stabilizer nullity can only decrease). Now, observe that we can combine Eqs. (112) and (113) to find $\pi(n) \geq \Omega(r^n 2^{n^2/2})$, so $\langle\nu\rangle \geq n\pi(n) \geq \Omega(n)$ when $r = \omega(\exp(-n))$. ■

The implications of Theorem 15 are twofold. Firstly, it provides an analytical explanation for the empirical observation in Ref. [34] that the fourth phase—namely, the entanglement-dominated one—is absent from random hybrid quantum circuits. Secondly, it firmly establishes that hybrid quantum circuits cannot generate ED states within a reasonable timeframe, since Theorem 15 says that the probability ratio must decrease exponentially with n, consequently the convergence time of the process scales at least exponentially with n. A natural question is whether it is possible to improve upon this result. The answer is yes, but slightly. In the following, we design a slightly more optimistic scenario for which hybrid quantum circuits create ED states. This is achieved without an exponentially small probability of applying $R_z(\theta)$ gates, but rather by adapting probabilities to the stabilizer nullity of the state at a given layer, with the ratio being $r(\nu)$ a function of ν.

Theorem 16 (Producing entanglement-dominated states via adaptive measurements). Consider infinite layers of the hybrid quantum circuit in Fig. 10 with $p_t(\nu)$ the probability of applying a $R_z(\theta)$ and $p_m(\nu)$ being the probability of applying a measurement in the computational basis, both depending on the nullity ν of the state at each layer. Defining $r(\nu) = p_t(\nu-1)/p_m(\nu)$ for $\nu > 0$ and $r(\nu) = p_t(0)$, we obtain the following:

(1) For $r(\nu) = \omega(\exp(-k))$ for $k = n-\nu$, then the state is within the MD phase.
(2) For $r(\nu) = O(\exp(-k))$ for $k = n-\nu$ then the resulting state is within the ED phase.

Proof. The proof uses the same notation and techniques of Theorem 15. The steady-state distribution $\pi(\nu)$ is

$$\begin{aligned}\pi(\nu) &= \frac{1}{Z}\left(\prod_{i=1}^{\nu} r(i)\right)\frac{\prod_{i=0}^{\nu-1}(2^n-2^i)}{\prod_{i=1}^{\nu}(2^i-1)}\\ &= \frac{1}{Z}\prod_{i=1}^{\nu} r(i)\frac{(2^n-2^{i-1})}{(2^i-1)}, \end{aligned} \tag{114}$$

where we defined $r(i) = p_t(i-1)/p_m(i)$ for $i \geq 1$. We only need to establish that for $r(\nu) = O(\exp(-(\nu-n))$, ED states are realized. We will proceed with a constructive proof as follows. Starting from Eq. (114), we choose

$$r(i) = r_0\frac{(2^i-1)}{(2^n-2^{i-1})}, \quad i \in [1,n] \tag{115}$$

with some $r_0 \neq 1$ to be fixed later, so we have $\pi(\nu) = r_0^\nu/Z$. With such a choice, we compute Z:

$$\begin{aligned}Z = Z\pi(0) + \sum_{\nu=1}^{n} r_0^\nu &= Z\pi(0) + r_0\frac{r_0^n-1}{r_0-1}\\ \implies Z &= \frac{1}{1-\pi(0)}r_0\frac{r_0^n-1}{r_0-1}. \end{aligned} \tag{116}$$

We have the following consistency condition

$$1 = (1-\pi(0))\frac{1}{r_0}\frac{r_0-1}{r_0^n-1} \implies \pi(0) = 1 - r_0\frac{1-r_0^n}{1-r_0}. \tag{117}$$

If we fix $r_0 = 1/2$, we get $\pi(0) = 1/2^n$. We can thus express the steady probability as

$$\pi_n(\nu) = \begin{cases} 2^{-n}, & \nu = 0\\ 2^{-\nu}, & \nu \in [1,n] \end{cases} \tag{118}$$

from which we can bound the average stabilizer nullity $\langle\nu\rangle_{\nu\sim\pi_n} \leq \sum_{\nu=1}^{\infty}\nu 2^{-\nu} = 2$, which realizes the ED

phase. In this case, recall that through Eq. (115), we chose $r(\nu) = \frac{1}{2}(2^{\nu} - 1)/(2^n - 2^{n-\nu}) = \Theta(2^{-(n-\nu)})$. For the case where $r(\nu) = \omega(\exp(-(n-\nu)))$ and defining $r_0 = \min_{\nu \in [1,n]} r(\nu)$, we have $r_0 = \omega(\exp(-n))$, so a proof similar to the one presented in Theorem 15 follows. ■

In Theorem 16, we demonstrated that by adapting the probabilities p_t and p_m based on the outcome of the stabilizer nullity measurement (which can be conducted efficiently, see Sec. V), the outlook for the presence of an ED phase is more optimistic compared to the result in Theorem 15. Specifically, by tuning the ratio between the probability of applying T gates and measurements to decrease exponentially in the size of the stabilizer group $n - \nu$ rather than in n, hybrid quantum circuits can indeed generate ED states. While this scaling of the ratio is significantly better than $\exp(-n)$, particularly in the high stabilizer nullity regime, achieving convergence to the ED phase, i.e., in the regime of small nullity ν, still requires the ratio to be exponentially small for assurance of convergence. Thus, although this approach slightly accelerates the convergence to ED states, the convergence time remains exponential in n.

Despite these findings, the emergence of ED states in hybrid quantum circuits appears unlikely. However, Ref. [97] introduces another scenario involving correlated monitoring, where measurements are conducted immediately after a non-Clifford gate with a certain probability. This technique can evidently result in ED states. In fact, with perfect correlation (i.e., a measurement following every non-Clifford gate application), the entire circuit effectively behaves as a Clifford circuit. While this scenario provides valuable insights, it assumes perfect knowledge of the applied gates from the observer's perspective, which may not always be realistic. In contrast, our adaptive measurement strategy allows us to determine nullity in polynomial time (see Sec. V), but identifying the occurrence and location of a T gate remains challenging [98]. However, investigating the transition from MD to ED phases based on the observer's monitoring knowledge is an intriguing avenue for future research, as discussed in Ref. [97].

VIII. PSEUDOCODE

Here we outline algorithms based on the main results of this work. These algorithms are designed for three tasks: exact calculation of the 2-Rényi entropy, entanglement monitoring, and efficient evaluation of a multipartite entanglement witness.

A. 2-Rényi entropy estimation for t-doped stabilizer states

Algorithm 2 exactly calculates the 2-Rényi entanglement entropy of a state ψ. We assume we know the stabilizer tableau $\mathcal{T}_S$ of the state ψ as well as its bad generators $(h_i, \mathrm{tr}(h_i\psi))$. We first apply the map $\mathcal{P}_B : g_A \otimes g_B \to \mathbb{1} \otimes g_B$ to each row of the tableau. Then, since $G_A = \ker(\mathcal{P}_B(\mathcal{T}_S))$, to estimate $|S_A|$, one computes $\dim(\ker(\mathcal{P}_B(\mathcal{T}_S)))$. Then, the map $\mathcal{P}_B(\mathcal{T}_S)$ is applied to the bad generators h_i, and if they are within $\mathrm{rowsp}(\mathcal{P}_B(\mathcal{T}_S))$, their corresponding values $\mathrm{tr}(h_i\psi)$ are summed. This yields $r = \sum_{\mathcal{P}_B(h_i) \in \mathrm{rowsp}(\mathcal{P}_B(\mathcal{T}_S))} \mathrm{tr}(h_i\psi)$. Finally, the 2-Rényi entropy is $S_2(\psi_A) = n_A - |S_A| - \log(r)$. The overall computational complexity is $O(n^3 2^{4lt})$, because the linear algebraic operations each have a runtime $O(n^3)$, and we loop over $i = 0, \ldots, k$, where k can be as large as $k = 2^{4lt}$.

ALGORITHM 2. Exact calculation of the 2-Rényi entanglement entropy across a bipartition $A|B$.

Input: A representation of ψ in terms of its stabilizer tableau $\mathcal{T}_S$ and bad generators $\{(h_i, \mathrm{tr}(h_i\psi)) \mid i = 0, \ldots, k\}$.

function 2RÉNYIENTROPY(ψ)
  $|S_A| \leftarrow \dim(\ker(\mathcal{P}_B(\mathcal{T}_S)))$ ▷ $\mathcal{P}_B$ is the map that takes $g_A \otimes g_B \to \mathbb{1} \otimes g_B$
  $r \leftarrow 0$
  for $i \leftarrow 0$ to k **do**
    $\vec{y}_i \in \mathbb{F}_2^{2n} \leftarrow$ symplectic representation of h_i
    if $\mathcal{P}_B(\vec{y}_i) \in \mathrm{rowsp}(\mathcal{P}_B(\mathcal{T}_S))$ **then** ▷ $\mathcal{P}_B(\vec{y}_i)$ is in the row space of $\mathcal{P}_B(\mathcal{T}_G) \iff \delta_i = 1$
      $r \leftarrow r + \mathrm{tr}^2(h_i\psi)$
    end if
  end for
  return $n_A - |S_A| - \log(r)$
end function

B. Efficient entanglement monitoring in t-doped Clifford circuits

Here, we present an algorithm for classically monitoring the entanglement of t-doped Clifford circuits, *without running anything on quantum hardware*. The task is to compute the entanglement entropy across a bipartition $A|B$ of a state generated by a sequence of gates $\{U_1, U_2, \ldots U_{|C|}\}$, where $|C|$ is the number of gates in the circuit. To start, we initialize the stabilizer tableau $\mathcal{T}_S$ corresponding to the stabilizer set $S := \{Z_1, Z_2, \ldots, Z_n\}$ of the state $|0\rangle\langle 0|\, 0^{\otimes n}$, to which we will sequentially apply the gates U_i. If U_i is a Clifford gate, we update the tableau $\mathcal{T}_S$ using the stabilizer formalism. Otherwise, letting X_i be the sites on which the non-Clifford gate acts, we perform a row reduction of $\mathcal{T}_S$ to reformat the tableau such that all generators except the first $2|X_i|$ act trivially on the sites X_i, then we remove the first $2|X_i|$ generators from S. After applying every gate, we compute $|S_A| = \dim(\ker(\mathcal{P}_B(\mathcal{T}_S)))$, and finally return $n_A - |S_A| + 2lt$ as our entanglement estimate. The algorithm has computational complexity $O(|C|n^3)$, as the linear algebraic manipulations each have runtime $O(n^3)$ and they are repeated $|C|$ times.

ALGORITHM 3. Entanglement monitoring.

Input: A t-doped circuit C in terms of a sequence of gates $\{U_1, U_2, \ldots, U_{|C|}\}$.
Input: A bipartition $A|B$
Output: $\tilde{S}_\alpha(\psi_A)$: an estimate of the entanglement entropy of the state ψ generated by C across $A|B$, such that $|\tilde{S}_\alpha(\psi_A) - S_\alpha(\psi_A)| \leq 2lt$

function EstimateEntropy(C, $A|B$)
  $S \leftarrow \{Z_1, Z_2, \ldots, Z_n\}$ ▷ Initial state is $|0\rangle\langle 0|^{\otimes n}$
  for $i \leftarrow 1, \ldots, |C|$ **do**
    if U_i is a Clifford gate **then**
      for $g_j \in S$ **do**
        $g_j \leftarrow U_i g_j U_i^\dagger$
      end for
    else
      $X_i \leftarrow$ the sites on which U_i acts (note that $l := \max_i |X_i|$)
      Using row reduction on $\mathcal{T}_S$, reformat the stabilizer generators such that all except the first $2|X_i|$ act as the identity on X_i
      Remove the first $2|X_i|$ generators from S
    end if
  end for
  $|S_A| \leftarrow \dim(\ker(\mathcal{P}_B(\mathcal{T}_S)))$ ▷ $\mathcal{P}_B$ is the map that takes $g_A \otimes g_B \to \mathbb{1} \otimes g_B$
  return $n_A - |S_A| + 2lt$ ▷ See Lemma 11.
end function

C. Efficient estimation for multipartite entanglement witness

To efficiently estimate a multipartite entanglement witness, we have developed Algorithm 4. In words, the algorithm takes as input the stabilizer tableau $\mathcal{T}_S$ associated with the target state $|\psi\rangle\langle\psi|$, a list of bipartitions $\mathcal{B} = \{(A_i, B_i) \mid i = 1, \ldots, |\mathcal{B}|\}$, a threshold error ϵ, and a maximum failure probability δ. The algorithm then proceeds as follows: It computes the minimum bound on the entropy across the bipartitions, $M = \min_i \lfloor \mathcal{E}(\psi; A_i|B_i) - lt \rfloor$, then samples a random stabilizer $P \in G$, and measures the expectation value of the observable P over ψ. This process is repeated

$$N = \left\lceil \frac{2\log(2/\delta)}{(1-\epsilon-2^{-M})^2} \right\rceil$$

times, and one computes $\tilde{\Pi} = \sum_i b_i/N$. At the end of the computation, the algorithm returns $2^{-M} - \tilde{\Pi}$. The algorithm has computational complexity

$$O\left(n^3 \left\lceil \frac{2\log(2/\delta)}{(1-\epsilon-2^{-M})^2} \right\rceil + \mathcal{B}n^3\right).$$

Estimating $\mathcal{E}(\psi; A_i|B_i)$ for each partition requires at most $O(n^3)$ resources, and such a task has to be repeated $|\mathcal{B}|$ times, then the runtime to find the minimum value has runtime $O(|\mathcal{B}|)$. The loop, on the other hand, has computational complexity

$$O\left(n^3 \left\lceil \frac{2\log(2/\delta)}{(1-\epsilon-2^{-M})^2} \right\rceil\right)$$

due to the repeated sampling of Pauli operators.

ALGORITHM 4. Efficient estimation protocol for a multipartite entanglement witness.

Input: A representation of the target state $|\psi\rangle$ in terms of its stabilizer tableau $\mathcal{T}_S \in \mathbb{F}_2^{|S|\times 2n}$
Input: A list of bipartitions $\mathcal{B} = \{(A_i, B_i) \mid i = 1, \ldots, |\mathcal{B}|\}$
Input: An upper bound on the state preparation error $\epsilon \geq \frac{1}{2}\|\psi - |\psi\rangle\langle\psi|\|_1$ and a maximum failure probability δ

function MultipartiteWitness($\psi, \mathcal{B}, \epsilon, \delta$)
  $M \leftarrow \min_i \lfloor \mathcal{E}(\psi; A_i|B_i) - lt \rfloor$
  $\tilde{\Pi} \leftarrow 0$ ▷ Running estimate of $\mathrm{tr}(\Pi\rho)$
  $N \leftarrow \left\lceil \frac{2\log(2/\delta)}{(1-\epsilon-2^{-M})^2} \right\rceil$
  for $i \leftarrow 1, \ldots, N$ **do**
    $\vec{x} \leftarrow$ a random binary vector $\mathbb{F}_2^{|S|}$
    $P \in \mathbb{P}_n \leftarrow \vec{x}^T \mathcal{T}_S$ ▷ Select a random stabilizer $P \in G$
    $b \in \{-1, 1\} \leftarrow$ Measure(ψ, P) ▷ Measure the two-outcome Pauli observable P
    $\tilde{\Pi} \leftarrow \tilde{\Pi} + b/N$
  end for
  return $2^{-M} - \tilde{\Pi}$
end function

ACKNOWLEDGMENTS

Special thanks to Ludovico Lami for inspiring our exploration of entanglement dilution, suggesting delving into entanglement witnessing, and providing guidance through the literature of entanglement theory. We thank Sumeet Khatri for insightful comments on birth and death chains. We express our appreciation to Jacopo Rizzo for recommending a perspective on our results through the lens of entanglement irreversibility. Additionally, we acknowledge Antonio and Francesco A. Mele for their valuable comments and engaging discussions. We also thank Susanne Yelin, Pablo Bonilla, Nazlı Uğur Köylüoğlu, and Varun Menon for helpful discussions and suggestions on potential applications of the ED-MD framework. We thank the Unitary Fund for their support. A.G. acknowledges support from the NSF through the Q-IDEAS HDR, as well as the AWS Generation Q Fund at the Harvard Quantum Initiative. S.F.E.O. acknowledges support from PNRR MUR project PE0000023-NQSTI. L.L. is funded through the Munich Quantum Valley project (MQV-K8) by Bayerisches Staatsministerium für Wissenschaft und Kunst.

[1] D. Gottesman, The Heisenberg representation of quantum computers, arxiv:quant-ph/9807006.

[2] E. T. Campbell, Catalysis and activation of magic states in fault-tolerant architectures, Phys. Rev. A **83**, 032317 (2011).

[3] S. Bravyi, G. Smith, and J. A. Smolin, Trading classical and quantum computational resources, Phys. Rev. X **6**, 021043 (2016).

[4] M. Beverland, E. Campbell, M. Howard, and V. Kliuchnikov, Lower bounds on the non-Clifford resources for quantum computations, Quantum Sci. Technol. **5**, 035009 (2020).

[5] L. Leone, S. F. E. Oliviero, and A. Hamma, Stabilizer Rényi entropy, Phys. Rev. Lett. **128**, 050402 (2022).

[6] L. Leone, S. F. E. Oliviero, and A. Hamma, Nonstabilizerness determining the hardness of direct fidelity estimation, Phys. Rev. A **107**, 022429 (2023).

[7] K. Goto, T. Nosaka, and M. Nozaki, Probing chaos by magic monotones, Phys. Rev. D **106**, 126009 (2022).

[8] R. J. Garcia, K. Bu, and A. Jaffe, Resource theory of quantum scrambling, Proc. Natl. Acad. Sci. U.S.A. **120**, e2217031120 (2023).

[9] D. Fattal, T. S. Cubitt, Y. Yamamoto, S. Bravyi, and I. L. Chuang, Entanglement in the stabilizer formalism, arxiv:quant-ph/0406168.

[10] E. Tirrito, P. S. Tarabunga, G. Lami, T. Chanda, L. Leone, S. F. E. Oliviero, M. Dalmonte, M. Collura, and A. Hamma, Quantifying nonstabilizerness through entanglement spectrum flatness, Phys. Rev. A **109**, L040401 (2024).

[11] S. Aaronson, A. Bouland, B. Fefferman, S. Ghosh, U. Vazirani, C. Zhang, and Z. Zhou, In *15th Innovations in Theoretical Computer Science Conference (ITCS 2024). Leibniz International Proceedings in Informatics (LIPIcs)*, (Dagstuhl – Leibniz-Zentrum für Informatik, 2024), Vol. 287, pp. 2:1–2:21.

[12] R. Arnon-Friedman, Z. Brakerski, and T. Vidick, Computational entanglement theory, arxiv:2310.02783 [quant-ph].

[13] D. Poulin, A. Qarry, R. Somma, and F. Verstraete, Quantum simulation of time-dependent hamiltonians and the convenient illusion of Hilbert space, Phys. Rev. Lett. **106**, 170501 (2011).

[14] E. Knill, Approximation by quantum circuits, arXiv:quant-ph/9508006 [quant-ph].

[15] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information: 10th Anniversary Edition* (Cambridge University Press, Cambridge, 2000).

[16] S. Aaronson and D. Gottesman, Improved simulation of stabilizer circuits, Phys. Rev. A **70**, 052328 (2004).

[17] S. Bravyi, D. Browne, P. Calpin, E. Campbell, D. Gosset, and M. Howard, Simulation of quantum circuits by low-rank stabilizer decompositions, Quantum **3**, 181 (2019).

[18] R. Horodecki, P. Horodecki, M. Horodecki, and K. Horodecki, Quantum entanglement, Rev. Mod. Phys. **81**, 865 (2009).

[19] To remind the reader, $f = o(g(n)) \Rightarrow \lim_{n\to\infty} f(n)/g(n) = 0$, $f = \omega(g(n)) \Rightarrow \lim f(n)/g(n) = \infty$, $f = O(g(n)) \Rightarrow \exists C_2 > 0$ such that $\lim f(n)/g(n) \leq C_2$, $f = \Omega(g(n)) \Rightarrow \exists C_1 > 0$ such that $\lim f(n)/g(n) \geq C_1$ and finally $f = \Theta(g(n))$ means $f = \Omega(g(n))$ and $f = O(g(n))$.

[20] C. H. Bennett, H. J. Bernstein, S. Popescu, and B. Schumacher, Concentrating partial entanglement by local operations, Phys. Rev. A **53**, 2046 (1996).

[21] L. Leone, J. Rizzo, J. Eisert, and S. Jerbi, Entanglement theory with limited computational resources, arXiv:2502.12284.

[22] J. I. de Vicente, C. Spee, and B. Kraus, Maximally entangled set of multipartite quantum states, Phys. Rev. Lett. **111**, 110502 (2013).

[23] D. Sauerwein, N. R. Wallach, G. Gour, and B. Kraus, Transformations among pure multipartite entangled states via local operations are almost never possible, Phys. Rev. X **8**, 031020 (2018).

[24] B. M. Terhal and K. G. H. Vollbrecht, Entanglement of formation for isotropic states, Phys. Rev. Lett. **85**, 2625 (2000).

[25] O. Gühne and G. Tóth, Entanglement detection, Phys. Rep. **474**, 1 (2009).

[26] A. Gu, S. F. E. Oliviero, and L. Leone, Doped stabilizer states in many-body physics and where to find them, arXiv:2403.14912 [quant-ph].

[27] A. Kitaev and J. Preskill, Topological entanglement entropy, Phys. Rev. Lett. **96**, 110404 (2006).

[28] A. Hamma, R. Ionicioiu, and P. Zanardi, Ground state entanglement and geometric entropy in the Kitaev model, Phys. Lett. A **337**, 22 (2005).

[29] H. Ma, A. T. Schmitz, S. A. Parameswaran, M. Hermele, and R. M. Nandkishore, Topological entanglement entropy of fracton stabilizer codes, Phys. Rev. B **97**, 125101 (2018).

[30] J. Haferkamp, F. Montealegre-Mora, M. Heinrich, J. Eisert, D. Gross, and I. Roth, Efficient unitary designs with a system-size independent number of non-Clifford gates, Commun. Math. Phys. **397**, 995 (2023).

[31] C. W. von Keyserlingk, T. Rakovszky, F. Pollmann, and S. L. Sondhi, Operator hydrodynamics, OTOCs, and entanglement growth in systems without conservation laws, Phys. Rev. X **8**, 021013 (2018).

[32] D. A. Roberts and B. Yoshida, Chaos and complexity by design, J. High Energy Phys. **2017**, 121 (2017).

[33] S. True and A. Hamma, Transitions in entanglement complexity in random circuits, Quantum **6**, 818 (2022).

[34] G. E. Fux, E. Tirrito, M. Dalmonte, and R. Fazio, Entanglement – nonstabilizerness separation in hybrid quantum circuits, Phys. Rev. Res. **6**, L042030 (2024).

[35] A. Gu, L. Leone, S. Ghosh, J. Eisert, S. F. Yelin, and Y. Quek, Pseudomagic quantum states, Phys. Rev. Lett. **132**, 210602 (2024).

[36] M. M. Wilde, *Quantum Information Theory*, 1st edn (Cambridge University Press, Cambridge, 2013).

[37] S. Bravyi and D. Gosset, Improved classical simulation of quantum circuits dominated by Clifford gates, Phys. Rev. Lett. **116**, 250501 (2016).

[38] C. Huang, M. Newman, and M. Szegedy, Explicit lower bounds on strong quantum simulation, IEEE Trans. Inf. Theory **66**, 5585 (2020).

[39] Z. Webb, The Clifford group forms a unitary 3-design, QIC **16**, 1379 (2016).

[40] H. Zhu, R. Kueng, M. Grassl, and D. Gross, The Clifford group fails gracefully to be a unitary 4-design, arXiv:1609.08172 [quant-ph].

[41] L. Leone, S. F. E. Oliviero, Y. Zhou, and A. Hamma, Quantum chaos is quantum, Quantum **5**, 453 (2021).

[42] S. F. E. Oliviero, L. Leone, and A. Hamma, Transitions in entanglement complexity in random quantum circuits by measurements, Phys. Lett. A **418**, 127721 (2021).

[43] H.-Y. Huang, Y. Liu, M. Broughton, I. Kim, A. Anshu, Z. Landau, and J. R. McClean, in *Proceedings of the 56th Annual ACM Symposium on Theory of Computing*, STOC 2024 (Association for Computing Machinery, New York, USA, 2024), pp. 1343–1351.

[44] M. Cramer, M. B. Plenio, S. T. Flammia, R. Somma, D. Gross, S. D. Bartlett, O. Landon-Cardinal, D. Poulin, and Y.-K. Liu, Efficient quantum state tomography, Nat. Commun. **1**, 149 (2010).

[45] L. Leone, S. F. E. Oliviero, S. Lloyd, and A. Hamma, Learning efficient decoders for quasichaotic quantum scramblers, Phys. Rev. A **109**, 022429 (2024).

[46] S. F. E. Oliviero, L. Leone, S. Lloyd, and A. Hamma, Unscrambling quantum information with Clifford decoders, Phys. Rev. Lett. **132**, 080402 (2024).

[47] L. Leone, S. F. E. Oliviero, and A. Hamma, Learning t-doped stabilizer states, Quantum **8**, 1361 (2024).

[48] S. Grewal, V. Iyer, W. Kretschmer, and D. Liang, Efficient learning of quantum states prepared with few Non-Clifford gates, arXiv:2305.13409 [quant-ph].

[49] D. Hangleiter and M. J. Gullans, Bell sampling from quantum circuits, Phys. Rev. Lett. **133**, 020601 (2024).

[50] N.-H. Chia, C.-Y. Lai, and H.-H. Lin, Efficient learning of *t*-doped stabilizer states with single-copy measurements, Quantum **8**, 1250 (2024).

[51] O. Landon-Cardinal, Y.-K. Liu, and D. Poulin, Efficient direct tomography for matrix product states, arXiv:1002.4632 [quant-ph].

[52] S. Grewal, V. Iyer, W. Kretschmer, and D. Liang, Pseudoentanglement ain't cheap, arXiv:2404.00126 [quant-ph].

[53] L. Leone, S. F. E. Oliviero, G. Esposito, and A. Hamma, Phase transition in stabilizer entropy and efficient purity estimation, Phys. Rev. A **109**, 032403 (2024).

[54] V. Veitch, S. A. H. Mousavian, D. Gottesman, and J. Emerson, The resource theory of stabilizer quantum computation, New J. Phys. **16**, 013009 (2014).

[55] J. Eisert, D. E. Browne, S. Scheel, and M. B. Plenio, Distillation of continuous-variable entanglement with optical means, Ann. Phys. **311**, 431 (2004).

[56] D. E. Browne, J. Eisert, S. Scheel, and M. B. Plenio, Driving non-Gaussian to Gaussian states with linear optics, Phys. Rev. A **67**, 062320 (2003).

[57] J. Eisert, M. B. Plenio, D. E. Browne, S. Scheel, and A. Feito, On the experimental feasibility of continuous-variable optical entanglement distillation, Opt. Spectrosc. **103**, 173 (2007).

[58] E. T. Campbell and J. Eisert, Gaussification and entanglement distillation of continuous-variable systems: A unifying picture, Phys. Rev. Lett. **108**, 020501 (2012).

[59] M. Cramer, M. B. Plenio, S. T. Flammia, R. Somma, D. Gross, S. D. Bartlett, O. Landon-Cardinal, D. Poulin, and Y.-K. Liu, Efficient quantum state tomography, Nat. Commun. **1**, 149 (2010).

[60] M. Fanizza, N. Galke, J. Lumbreras, C. Rouzé, and A. Winter, Learning finitely correlated states: Stability of the spectral reconstruction, arXiv:2312.07516 [quant-ph].

[61] E. Chitambar, D. Leung, L. Mančinska, M. Ozols, and A. Winter, Everything you always wanted to know about LOCC (but were afraid to ask), Commun. Math. Phys. **328**, 303 (2014).

[62] Z.-W. Liu and A. Winter, Many-body quantum magic, PRX Quantum **3**, 020333 (2022).

[63] G. Saxena and G. Gour, Quantifying multiqubit magic channels with completely stabilizer-preserving operations, Phys. Rev. A **106**, 042422 (2022).

[64] K. Bu, W. Gu, and A. Jaffe, Quantum entropy and central limit theorem, Proc. Natl. Acad. Sci. U.S.A. **120**, e2304589120 (2023).

[65] J. Jiang and X. Wang, Lower bound for the *t* count via unitary stabilizer nullity, Phys. Rev. Appl. **19**, 034052 (2023).

[66] L. Leone and L. Bittel, Stabilizer entropies are monotones for magic-state resource theory, Phys. Rev. A **110**, L040403 (2024).

[67] E. Chitambar and G. Gour, Quantum resource theories, Rev. Mod. Phys. **91**, 025001 (2019).

[68] S. Bravyi, D. Gosset, and R. König, Quantum advantage with shallow circuits, Science **362**, 308 (2018).

[69] S. Khatri and M. M. Wilde, Principles of quantum communication theory: A modern approach, arxiv:2011.04672 [cond-mat, physics:hep-th, physics:math-ph, physics:quant-ph].

[70] S. Bravyi, D. Fattal, and D. Gottesman, GHZ extraction yield for multipartite stabilizer states, J. Math. Phys **47**, 062106 (2006).

[71] L. Lami and B. Regula, No second law of entanglement manipulation after all, Nat. Phys. **19**, 184 (2023).

[72] N. J. Coble, M. Coudron, J. Nelson, and S. S. Nezhadi, in *18th Conference on the Theory of Quantum Computation, Communication and Cryptography (TQC 2023), Leibniz International Proceedings in Informatics (LIPIcs)*, edited by O. Fawzi and M. Walter (Schloss Dagstuhl – Leibniz-Zentrum für Informatik, Dagstuhl, Germany, 2023), Vol. 266, pp. 14:1–14:21.

[73] D. J. Williamson, A. Dua, and M. Cheng, Spurious topological entanglement entropy from subsystem symmetries, Phys. Rev. Lett. **122**, 140506 (2019).

[74] B. Zeng, X. Chen, D.-L. Zhou, and X.-G. Wen, *Quantum Information Meets Quantum Matter: From Quantum Entanglement to Topological Phases of Many-Body Systems*, Quantum Science and Technology (Springer, New York, 2019).

[75] D. I. Tsomokos, A. Hamma, W. Zhang, S. Haas, and R. Fazio, Topological order following a quantum quench, Phys. Rev. A **80**, 060302 (2009).

[76] A. Hamma, R. Ionicioiu, and P. Zanardi, Bipartite entanglement and entropic boundary law in lattice spin systems, Phys. Rev. A **71**, 022315 (2005).

[77] S. Bravyi, D. Poulin, and B. Terhal, Tradeoffs for reliable quantum information storage in 2D systems, Phys. Rev. Lett. **104**, 050503 (2010).

[78] N. P. Breuckmann and J. N. Eberhardt, Quantum low-density parity-check codes, PRX Quantum **2**, 040101 (2021).

[79] P. Panteleev and G. Kalachev, Degenerate quantum LDPC codes with good finite length performance, Quantum **5**, 585 (2021).

[80] Q. Xu, J. P. Bonilla Ataides, C. A. Pattison, N. Raveendran, D. Bluvstein, J. Wurtz, B. Vasić, M. D. Lukin, L. Jiang, and H. Zhou, Constant-overhead fault-tolerant quantum computation with reconfigurable atom arrays, Nat. Phys. **20**, 1084 (2024).

[81] P. Hosur, X.-L. Qi, D. A. Roberts, and B. Yoshida, Chaos in quantum channels, J. High Energy Phys. **2016**, 4 (2016).

[82] A. Kitaev, in *Talk Given at the Fundamental Physics Prize Symposium* (2014), Vol. 10.

[83] C. Chamon, A. Hamma, and E. R. Mucciolo, Emergent irreversibility and entanglement spectrum statistics, Phys. Rev. Lett. **112**, 240501 (2014).

[84] D. Shaffer, C. Chamon, A. Hamma, and E. R. Mucciolo, Irreversibility and entanglement spectrum statistics in quantum circuits, J. Statist. Mech.: Theory Exp. **2014**, P12007 (2014).

[85] Z.-C. Yang, A. Hamma, S. M. Giampaolo, E. R. Mucciolo, and C. Chamon, Entanglement complexity in quantum many-body dynamics, thermalization, and localization, Phys. Rev. B **96**, 020408 (2017).

[86] A. Zabalo, M. J. Gullans, J. H. Wilson, S. Gopalakrishnan, D. A. Huse, and J. H. Pixley, Critical properties of the measurement-induced transition in random quantum circuits, Phys. Rev. B **101**, 060301 (2020).

[87] M. J. Gullans and D. A. Huse, Dynamical purification phase transition induced by quantum measurements, Phys. Rev. X **10**, 041020 (2020).

[88] H. M. Wiseman, Quantum trajectories and quantum measurement theory, Quantum Semiclassical Opt.: J. Eur. Opt. Soc. Part B **8**, 205 (1996).

[89] S. Dhar and S. Dasgupta, Measurement-induced phase transition in a quantum spin system, Phys. Rev. A **93**, 050103 (2016).

[90] T. Elliott, W. Kozlowski, S. Caballero-Benitez, and I. Mekhov, Multipartite entangled spatial modes of ultracold atoms generated and controlled by quantum measurement, Phys. Rev. Lett. **114**, 113604 (2015).

[91] B. Misra and E. C. G. Sudarshan, The Zeno's paradox in quantum theory, J. Math. Phys **18**, 756 (1977).

[92] B. Skinner, J. Ruhman, and A. Nahum, Measurement-induced phase transitions in the dynamics of entanglement, Phys. Rev. X **9**, 031009 (2019).

[93] Y. Li, X. Chen, and M. P. A. Fisher, Measurement-driven entanglement transition in hybrid quantum circuits, Phys. Rev. B **100**, 134306 (2019).

[94] Y. Bao, S. Choi, and E. Altman, Theory of the phase transition in random unitary circuits with measurements, Phys. Rev. B **101**, 104301 (2020).

[95] Y. Li, X. Chen, and M. P. A. Fisher, Quantum Zeno effect and the many-body entanglement transition, Phys. Rev. B **98**, 205136 (2018).

[96] S. Karlin and J. McGregor, The classification of birth and death processes, Trans. Am. Math. Soc. **86**, 366 (1957).

[97] M. Bejan, C. McLauchlan, and B. Béri, Dynamical magic transitions in monitored Clifford + T circuits, PRX Quantum **5**, 030332 (2024).

[98] J. van de Wetering and M. Amy, Optimising quantum circuits is generally hard, arXiv:2310.05958.